

Territory Analyses with Year as Factor

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Warning: package 'ggplot2' was built under R version 4.4.3
Warning: package 'tibble' was built under R version 4.4.3
Warning: package 'tidyr' was built under R version 4.4.3
Warning: package 'readr' was built under R version 4.4.3
Warning: package 'purrr' was built under R version 4.4.3

-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.6
v forcats    1.0.0      v stringr    1.6.0
v lubridate  1.9.4      v tibble     3.3.1
v purrr      1.2.1      v tidyr      1.3.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

Warning in check_dep_version(dep_pkg = "TMB"): package version mismatch:
glmmTMB was built with TMB package version 1.9.17
Current TMB package version is 1.9.19
Please re-install glmmTMB from source or restore original 'TMB' package (see '?reinstalling' for more info)

Loading required package: nlme

Attaching package: 'nlme'

The following object is masked from 'package:dplyr':

    collapse

This is mgcv 1.9-3. For overview type 'help("mgcv-package")'.

Warning: package 'emmeans' was built under R version 4.4.3

Welcome to emmeans.
Caution: You lose important information if you filter this package's results.
See '? untidy'
This is DHARMA 0.4.7. For overview type '?DHARMA'. For recent changes, type news(package = 'DHARMA')
Loading required package: lattice

Attaching package: 'caret'

The following object is masked from 'package:purrr':
```

```
lift
```

```
Loading required package: carData
```

```
Use the command
```

```
  lattice::trellis.par.set(effectsTheme())
```

```
  to customize lattice options for effects plots.
```

```
See ?effectsTheme for details.
```

```
Welcome! Want to learn more? See two factoextra-related books at https://goo.gl/ve3WBa
```

```
Attaching package: 'gridExtra'
```

```
The following object is masked from 'package:dplyr':
```

```
  combine
```

```
corrplot 0.95 loaded
```

```
Attaching package: 'kableExtra'
```

```
The following object is masked from 'package:dplyr':
```

```
  group_rows
```

```
Attaching package: 'gratia'
```

```
The following object is masked from 'package:stringr':
```

```
  boundary
```

For this report, we generated 20 different simulated data sets varying probability of mate fidelity, a pair's territory fidelity, a single male's territory fidelity, and a single female's territory fidelity. Latin hypercube sampling (LHS) was used to generate these probabilities for the 20 different runs. In addition to the 20 runs generated from the LHS, we also generated two simulated datasets, where probabilities were chosen to either show the territory effect, or to make the territory effect more difficult to detect.

Year is being modeled as a factor instead of with a smoother. The reason being for this is that we did try modelling year with both cyclic and cubic regression splines. However, edfs on the year smoother were large and the k-indices were often too small, meaning our basis dimension was inadequate. Since we only have 20 years in the dataset, we cannot get the adequate dimension for our basis. A year term is not included in the models with z-score as a response, as the construction of the z-score removes the year effect.

This report does not include diagnostic plots. However, a few of the models had diagnostics to warrant concern about the validity of model assumptions. Most of the residuals in the z-score models had a slight left skew, with a select few having a notable left skew. The z-score models should be interpreted a bit more cautiously.

The probability of fledgling were randomly drawn from a $\text{beta}(14, (14/0.65 - 14))$ distribution for good territories, a $\text{beta}(9, (9/0.55 - 9))$ distribution for mediocre territories, and a $\text{beta}(14, (5/0.45 - 5))$ distribution for bad territories. Note that maybe playing with priors could be interesting. I initially ran the simulation where priors were very different and the models seemed to have an easier time picking up on the territory effect.

Simulation 1

Number of fledgling as the response (Poisson)

Table 1: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npars	Family	RE
54.10890	-495.3884	1102.784	1323.446	432.8877	325.8911	380	0.2099745	128	Poisson	Terr
56.29970	-493.0690	1116.182	1372.380	428.2488	323.7003	380	0.2168068	254	Poisson	BirdID+Terr
30.40247	-528.9373	1129.318	1270.068	499.9853	349.5975	380	0.1069678	146	Poisson	BirdID

Table 2: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	10.402475	125	11.80533	0.1898715	Poisson	BirdID
s(Terrs)	34.108902	107	55.26825	0.0001767	Poisson	Terr
s(birdID)	2.440038	125	2.52828	0.4129371	Poisson	BirdID+Terr
s(Terrs)	33.859661	107	55.03280	0.0002091	Poisson	BirdID+Terr

Table 3: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.3737769	0.2339301	1.5978145	0.1100843	Poisson	BirdID
factor(fac.yr)2	0.0950657	0.2984007	0.3185841	0.7500419	Poisson	BirdID
factor(fac.yr)3	-0.8047759	0.3485338	-2.3090330	0.0209418	Poisson	BirdID
factor(fac.yr)4	0.0935880	0.2844975	0.3289588	0.7421868	Poisson	BirdID
factor(fac.yr)5	-0.2862026	0.2925978	-0.9781434	0.3280034	Poisson	BirdID
factor(fac.yr)6	-0.0741514	0.2847640	-0.2603959	0.7945584	Poisson	BirdID
factor(fac.yr)7	-1.5988208	0.4082503	-3.9162760	0.0000899	Poisson	BirdID
factor(fac.yr)8	-0.2415985	0.2886457	-0.8370070	0.4025886	Poisson	BirdID
factor(fac.yr)9	0.2862208	0.2751881	1.0400917	0.2982973	Poisson	BirdID
factor(fac.yr)10	0.0687893	0.2878257	0.2389965	0.8111083	Poisson	BirdID
factor(fac.yr)11	-0.3190489	0.3122863	-1.0216553	0.3069441	Poisson	BirdID
factor(fac.yr)12	-0.2812180	0.3153448	-0.8917793	0.3725112	Poisson	BirdID
factor(fac.yr)13	-0.0470649	0.3028974	-0.1553822	0.8765200	Poisson	BirdID
factor(fac.yr)14	0.0718622	0.3229980	0.2224851	0.8239363	Poisson	BirdID
factor(fac.yr)15	-0.5705759	0.3842100	-1.4850629	0.1375272	Poisson	BirdID
factor(fac.yr)16	-0.2517480	0.3582057	-0.7028028	0.4821786	Poisson	BirdID
factor(fac.yr)17	0.2051473	0.3678679	0.5576657	0.5770727	Poisson	BirdID
factor(fac.yr)18	-0.9765319	0.5553323	-1.7584640	0.0786686	Poisson	BirdID
factor(fac.yr)19	-1.3886909	0.6255504	-2.2199503	0.0264221	Poisson	BirdID
factor(fac.yr)20	-0.5961762	0.5088424	-1.1716322	0.2413448	Poisson	BirdID
(Intercept)1	0.4325881	0.2456391	1.7610720	0.0782262	Poisson	Terr
factor(fac.yr)21	0.0440217	0.3099395	0.1420332	0.8870538	Poisson	Terr
factor(fac.yr)31	-0.9022384	0.3607360	-2.5011046	0.0123807	Poisson	Terr
factor(fac.yr)41	0.0697470	0.2979168	0.2341158	0.8148951	Poisson	Terr
factor(fac.yr)51	-0.3447337	0.3036389	-1.1353408	0.2562326	Poisson	Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)61	-0.1366417	0.2948333	-0.4634539	0.6430390	Poisson	Terr
factor(fac.yr)71	-1.6274894	0.4148923	-3.9226796	0.0000876	Poisson	Terr
factor(fac.yr)81	-0.3560882	0.2973459	-1.1975557	0.2310900	Poisson	Terr
factor(fac.yr)91	0.1775688	0.2879191	0.6167314	0.5374119	Poisson	Terr
factor(fac.yr)101	-0.0202419	0.2999938	-0.0674744	0.9462040	Poisson	Terr
factor(fac.yr)111	-0.4725065	0.3198543	-1.4772554	0.1396072	Poisson	Terr
factor(fac.yr)121	-0.2978849	0.3238682	-0.9197719	0.3576920	Poisson	Terr
factor(fac.yr)131	-0.1031416	0.3131033	-0.3294170	0.7418405	Poisson	Terr
factor(fac.yr)141	0.0817752	0.3365538	0.2429782	0.8080223	Poisson	Terr
factor(fac.yr)151	-0.6202976	0.3967345	-1.5635082	0.1179331	Poisson	Terr
factor(fac.yr)161	-0.2877129	0.3702707	-0.7770338	0.4371388	Poisson	Terr
factor(fac.yr)171	0.0970172	0.3894690	0.2491012	0.8032825	Poisson	Terr
factor(fac.yr)181	-1.0307229	0.5694446	-1.8100496	0.0702881	Poisson	Terr
factor(fac.yr)191	-1.3850356	0.6347303	-2.1820852	0.0291032	Poisson	Terr
factor(fac.yr)201	-0.6771297	0.5266598	-1.2857061	0.1985457	Poisson	Terr
(Intercept)2	0.4303541	0.2466804	1.7445812	0.0810578	Poisson	BirdID+Terr
factor(fac.yr)22	0.0440680	0.3107546	0.1418098	0.8872303	Poisson	BirdID+Terr
factor(fac.yr)32	-0.8998042	0.3614684	-2.4893021	0.0127994	Poisson	BirdID+Terr
factor(fac.yr)42	0.0698431	0.2989796	0.2336049	0.8152917	Poisson	BirdID+Terr
factor(fac.yr)52	-0.3453692	0.3047405	-1.1333223	0.2570789	Poisson	BirdID+Terr
factor(fac.yr)62	-0.1330499	0.2960693	-0.4493876	0.6531521	Poisson	BirdID+Terr
factor(fac.yr)72	-1.6223301	0.4157952	-3.9017533	0.0000955	Poisson	BirdID+Terr
factor(fac.yr)82	-0.3509383	0.2985761	-1.1753733	0.2398454	Poisson	BirdID+Terr
factor(fac.yr)92	0.1810974	0.2892641	0.6260625	0.5312740	Poisson	BirdID+Terr
factor(fac.yr)102	-0.0169553	0.3013948	-0.0562561	0.9551378	Poisson	BirdID+Terr
factor(fac.yr)112	-0.4681923	0.3211447	-1.4578858	0.1448720	Poisson	BirdID+Terr
factor(fac.yr)122	-0.2960656	0.3252011	-0.9104076	0.3626076	Poisson	BirdID+Terr
factor(fac.yr)132	-0.1025106	0.3145382	-0.3259083	0.7444938	Poisson	BirdID+Terr
factor(fac.yr)142	0.0754018	0.3380746	0.2230332	0.8235097	Poisson	BirdID+Terr
factor(fac.yr)152	-0.6245553	0.3980786	-1.5689245	0.1166655	Poisson	BirdID+Terr
factor(fac.yr)162	-0.2917781	0.3717050	-0.7849721	0.4324699	Poisson	BirdID+Terr
factor(fac.yr)172	0.0899421	0.3913948	0.2297990	0.8182480	Poisson	BirdID+Terr
factor(fac.yr)182	-1.0373659	0.5707069	-1.8176859	0.0691122	Poisson	BirdID+Terr
factor(fac.yr)192	-1.3891733	0.6358199	-2.1848536	0.0288996	Poisson	BirdID+Terr
factor(fac.yr)202	-0.6840720	0.5281347	-1.2952605	0.1952304	Poisson	BirdID+Terr

The territory only model performed best on AIC and had an adjusted R^2 of 20%, which is close to the best adjusted R^2 seen out of these three models. The band ID only model did perform the best on BIC. However, its adjusted R^2 was about half of the other 2 models adjusted R^2 . In addition, the territory smoothers were the only significant smoothers

Z-score number of fledgling (Gaussian)

Table 4: Z-score GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npars	Family	RE
31.98857	-490.0768	1050.913	1190.316	293.4319	348.0114	380	0.1123321	109	Gaussian	Terr
32.54085	-489.4595	1058.417	1215.033	292.4802	347.4591	380	0.1138047	235	Gaussian	BandID+Terr

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
1.00402	-528.9199	1061.856	1069.768	359.9926	378.9960	380	0.0000101	127	Gaussian	BandID

Table 5: Z-score GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	0.0040201	125	0.0000314	0.5330127	Gaussian	BandID
s(Terrs)	30.9885660	107	0.4482386	0.0026014	Gaussian	Terr
s(birdID)	0.5185845	125	0.0041858	0.4622962	Gaussian	BandID+Terr
s(Terrs)	31.0222684	107	0.4494781	0.0025891	Gaussian	BandID+Terr

Table 6: Z-score GAMMs linear terms table.

	Estimate	Std. Error	t value	Pr(> t)	Family	RE
(Intercept)	-0.0000015	0.0499977	-0.0000293	0.9999767	Gaussian	BandID
(Intercept)1	0.0108211	0.0578865	0.1869363	0.8518195	Gaussian	Terr
(Intercept)2	0.0106362	0.0580219	0.1833141	0.8546585	Gaussian	BandID+Terr

The model diagnostics on these three models were severe enough to warrant concern. The residuals seemed as though the left skew was more notable, and the residuals vs fitted plots seemed to funnel out. I would not recommened working with these models.

The QQ plot does suggest violation in the model assumptions.

Eggs (Binomial)

Table 7: No. fledged GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
53.19068	-190.9750	493.9359	714.5577	381.9500	326.8093	380	0.2146451	128	Binomial	Terr
71.36412	-171.2458	501.0728	813.4914	342.4915	308.6359	380	0.2830034	254	Binomial	BirdID+Terr
45.02733	-201.8515	501.1914	693.2520	403.7029	334.9727	380	0.1772003	146	Binomial	BirdID

Table 8: No. fledged GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	25.02733	124	32.73317	0.0269031	Binomial	BirdID
s(Terrs)	33.19068	107	47.08774	0.0048875	Binomial	Terr
s(birdID)	19.99139	124	25.11433	0.1388589	Binomial	BirdID+Terr
s(Terrs)	31.37273	107	43.90992	0.0244160	Binomial	BirdID+Terr

Table 9: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.4016516	6.017405e-01	0.6674832	0.5044636	Binomial	BirdID
factor(fac.yr)2	0.5244897	8.001079e-01	0.6555237	0.5121306	Binomial	BirdID
factor(fac.yr)3	-0.5111355	7.346270e-01	-0.6957755	0.4865694	Binomial	BirdID
factor(fac.yr)4	0.5894941	7.720034e-01	0.7635900	0.4451116	Binomial	BirdID
factor(fac.yr)5	-0.1909280	7.141276e-01	-0.2673584	0.7891933	Binomial	BirdID
factor(fac.yr)6	0.0012349	7.215767e-01	0.0017114	0.9986345	Binomial	BirdID
factor(fac.yr)7	-1.5925736	7.509901e-01	-2.1206319	0.0339528	Binomial	BirdID
factor(fac.yr)8	-0.4287605	7.100056e-01	-0.6038832	0.5459213	Binomial	BirdID
factor(fac.yr)9	0.8721148	7.737130e-01	1.1271813	0.2596658	Binomial	BirdID
factor(fac.yr)10	0.5495289	7.665463e-01	0.7168894	0.4734423	Binomial	BirdID
factor(fac.yr)11	-0.1064362	7.507002e-01	-0.1417826	0.8872517	Binomial	BirdID
factor(fac.yr)12	-0.0890853	7.641089e-01	-0.1165872	0.9071872	Binomial	BirdID
factor(fac.yr)13	0.2295780	7.798429e-01	0.2943901	0.7684598	Binomial	BirdID
factor(fac.yr)14	42.9366894	1.861265e+07	0.0000023	0.9999982	Binomial	BirdID
factor(fac.yr)15	-1.0104573	8.526935e-01	-1.1850182	0.2360102	Binomial	BirdID
factor(fac.yr)16	0.5304737	9.229219e-01	0.5747764	0.5654425	Binomial	BirdID
factor(fac.yr)17	0.3644860	1.065204e+00	0.3421750	0.7322192	Binomial	BirdID
factor(fac.yr)18	-1.5337592	1.065060e+00	-1.4400680	0.1498482	Binomial	BirdID
factor(fac.yr)19	-2.5289671	1.254495e+00	-2.0159239	0.0438079	Binomial	BirdID
factor(fac.yr)20	-0.5383313	1.053531e+00	-0.5109782	0.6093663	Binomial	BirdID
(Intercept)1	0.6981032	6.168791e-01	1.1316694	0.2577734	Binomial	Terr
factor(fac.yr)21	0.3268518	8.240105e-01	0.3966598	0.6916184	Binomial	Terr
factor(fac.yr)31	-0.8915133	7.617296e-01	-1.1703803	0.2418480	Binomial	Terr
factor(fac.yr)41	0.5779732	7.902424e-01	0.7313873	0.4645426	Binomial	Terr
factor(fac.yr)51	-0.3473526	7.290335e-01	-0.4764564	0.6337493	Binomial	Terr
factor(fac.yr)61	-0.2545973	7.307596e-01	-0.3484009	0.7275391	Binomial	Terr
factor(fac.yr)71	-1.8341017	7.581270e-01	-2.4192540	0.0155524	Binomial	Terr
factor(fac.yr)81	-0.8701220	7.178513e-01	-1.2121201	0.2254664	Binomial	Terr
factor(fac.yr)91	0.3801849	7.859184e-01	0.4837460	0.6285661	Binomial	Terr
factor(fac.yr)101	0.1521355	7.774535e-01	0.1956844	0.8448572	Binomial	Terr
factor(fac.yr)111	-0.4753812	7.549250e-01	-0.6297065	0.5288866	Binomial	Terr
factor(fac.yr)121	-0.2294843	7.697427e-01	-0.2981312	0.7656031	Binomial	Terr
factor(fac.yr)131	-0.0456840	7.882885e-01	-0.0579534	0.9537858	Binomial	Terr
factor(fac.yr)141	43.0172383	1.861265e+07	0.0000023	0.9999982	Binomial	Terr
factor(fac.yr)151	-1.1183779	8.693685e-01	-1.2864256	0.1982946	Binomial	Terr
factor(fac.yr)161	0.4236411	9.372506e-01	0.4520041	0.6512660	Binomial	Terr
factor(fac.yr)171	0.2131770	1.101271e+00	0.1935737	0.8465097	Binomial	Terr
factor(fac.yr)181	-1.5887712	1.085738e+00	-1.4633102	0.1433825	Binomial	Terr
factor(fac.yr)191	-2.5593975	1.274612e+00	-2.0079812	0.0446453	Binomial	Terr
factor(fac.yr)201	-0.7357349	1.081954e+00	-0.6800055	0.4965010	Binomial	Terr
(Intercept)2	0.6210348	6.434380e-01	0.9651820	0.3344536	Binomial	BirdID+Terr
factor(fac.yr)22	0.3578230	8.432673e-01	0.4243293	0.6713257	Binomial	BirdID+Terr
factor(fac.yr)32	-0.8164565	7.858660e-01	-1.0389257	0.2988393	Binomial	BirdID+Terr
factor(fac.yr)42	0.5522474	8.162505e-01	0.6765660	0.4986814	Binomial	BirdID+Terr
factor(fac.yr)52	-0.3091601	7.601257e-01	-0.4067223	0.6842120	Binomial	BirdID+Terr
factor(fac.yr)62	-0.1173882	7.635569e-01	-0.1537386	0.8778158	Binomial	BirdID+Terr
factor(fac.yr)72	-1.7168013	7.914199e-01	-2.1692672	0.0300624	Binomial	BirdID+Terr
factor(fac.yr)82	-0.7261626	7.515898e-01	-0.9661688	0.3339597	Binomial	BirdID+Terr
factor(fac.yr)92	0.5500640	8.214257e-01	0.6696455	0.5030838	Binomial	BirdID+Terr
factor(fac.yr)102	0.2935222	8.157599e-01	0.3598145	0.7189859	Binomial	BirdID+Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)112	-0.3855318	7.918446e-01	-0.4868782	0.6263447	Binomial	BirdID+Terr
factor(fac.yr)122	-0.1385441	8.107687e-01	-0.1708800	0.8643181	Binomial	BirdID+Terr
factor(fac.yr)132	0.0712434	8.244707e-01	0.0864111	0.9311396	Binomial	BirdID+Terr
factor(fac.yr)142	41.8854907	1.861265e+07	0.0000023	0.9999982	Binomial	BirdID+Terr
factor(fac.yr)152	-1.1523766	9.107846e-01	-1.2652570	0.2057792	Binomial	BirdID+Terr
factor(fac.yr)162	0.3819553	9.826345e-01	0.3887054	0.6974941	Binomial	BirdID+Terr
factor(fac.yr)172	0.1713200	1.146696e+00	0.1494031	0.8812356	Binomial	BirdID+Terr
factor(fac.yr)182	-1.6721975	1.132815e+00	-1.4761428	0.1399056	Binomial	BirdID+Terr
factor(fac.yr)192	-2.5713950	1.306828e+00	-1.9676619	0.0491070	Binomial	BirdID+Terr
factor(fac.yr)202	-0.7863259	1.134891e+00	-0.6928650	0.4883943	Binomial	BirdID+Terr

Each of these models is the best on one of the 3 selection criteria. The smoothers in the models with only one random effect are significant. However, the model that includes both smoothers only saw significance with the territory smoother.

Probabilites used to generate simulation 1

Simulation 1

Probability of mate fidelity: 0.029

Probability of pair dispersal: 0.149

Probability of single male dispersal: 0.575

Probability of single female dispersal: 0.342

Simulation 2

Number of fledgling as the response (Poisson)

Table 10: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
49.79766	-382.2875	867.7066	1061.700	294.0521	268.2023	318	0.2536522	131	Poisson	Terr
55.50406	-376.1305	878.2445	1115.223	281.7380	262.4959	318	0.2808516	247	Poisson	BirdID+Terr
42.09184	-394.9720	879.5722	1048.165	319.4210	275.9082	318	0.1849786	136	Poisson	BirdID

Table 11: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	22.091840	115	30.356665	0.0150737	Poisson	BirdID
s(Terrs)	29.797660	110	48.217001	0.0002527	Poisson	Terr
s(birdID)	8.116753	115	9.520933	0.2075024	Poisson	BirdID+Terr
s(Terrs)	27.387304	110	43.788040	0.0008197	Poisson	BirdID+Terr

Table 12: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.3831109	0.2955041	1.2964654	0.1948152	Poisson	BirdID
factor(fac.yr)2	-0.0383600	0.4193112	-0.0914834	0.9271085	Poisson	BirdID
factor(fac.yr)3	-0.8221996	0.4254702	-1.9324491	0.0533041	Poisson	BirdID
factor(fac.yr)4	-0.3043641	0.3907202	-0.7789821	0.4359902	Poisson	BirdID
factor(fac.yr)5	0.0986797	0.3554011	0.2776572	0.7812755	Poisson	BirdID
factor(fac.yr)6	0.0527096	0.3399684	0.1550427	0.8767876	Poisson	BirdID
factor(fac.yr)7	-0.6520410	0.3825139	-1.7046205	0.0882652	Poisson	BirdID
factor(fac.yr)8	-0.6493017	0.3826165	-1.6970041	0.0896959	Poisson	BirdID
factor(fac.yr)9	-0.0882325	0.3494771	-0.2524699	0.8006778	Poisson	BirdID
factor(fac.yr)10	-0.4686836	0.3663577	-1.2793061	0.2007893	Poisson	BirdID
factor(fac.yr)11	-1.5285169	0.4642858	-3.2921894	0.0009941	Poisson	BirdID
factor(fac.yr)12	-0.3089556	0.3730180	-0.8282592	0.4075237	Poisson	BirdID
factor(fac.yr)13	0.1375377	0.3415704	0.4026628	0.6871963	Poisson	BirdID
factor(fac.yr)14	-0.0885647	0.3631527	-0.2438773	0.8073258	Poisson	BirdID
factor(fac.yr)15	-1.2049659	0.5099349	-2.3629799	0.0181287	Poisson	BirdID
factor(fac.yr)16	-0.5366463	0.4409758	-1.2169516	0.2236226	Poisson	BirdID
factor(fac.yr)17	-0.0444553	0.4244452	-0.1047374	0.9165842	Poisson	BirdID
factor(fac.yr)18	0.1212931	0.4488086	0.2702558	0.7869635	Poisson	BirdID
factor(fac.yr)19	-0.0774282	0.6026311	-0.1284836	0.8977662	Poisson	BirdID
factor(fac.yr)20	0.1330955	0.5625835	0.2365790	0.8129834	Poisson	BirdID
(Intercept)1	0.3828493	0.2981391	1.2841297	0.1990966	Poisson	Terr
factor(fac.yr)21	-0.1155495	0.4265452	-0.2708962	0.7864708	Poisson	Terr
factor(fac.yr)31	-0.7818073	0.4277451	-1.8277411	0.0675884	Poisson	Terr
factor(fac.yr)41	-0.2821581	0.3968935	-0.7109165	0.4771360	Poisson	Terr
factor(fac.yr)51	0.1535122	0.3523698	0.4356564	0.6630861	Poisson	Terr
factor(fac.yr)61	0.0847453	0.3369561	0.2515024	0.8014257	Poisson	Terr
factor(fac.yr)71	-0.6879961	0.3832953	-1.7949506	0.0726616	Poisson	Terr
factor(fac.yr)81	-0.6948960	0.3805027	-1.8262575	0.0678115	Poisson	Terr
factor(fac.yr)91	-0.0679291	0.3528601	-0.1925101	0.8473426	Poisson	Terr
factor(fac.yr)101	-0.4546447	0.3679353	-1.2356647	0.2165832	Poisson	Terr
factor(fac.yr)111	-1.5142802	0.4653671	-3.2539478	0.0011381	Poisson	Terr
factor(fac.yr)121	-0.3408498	0.3723123	-0.9154943	0.3599322	Poisson	Terr
factor(fac.yr)131	0.0993354	0.3435691	0.2891278	0.7724836	Poisson	Terr
factor(fac.yr)141	-0.0144490	0.3647115	-0.0396176	0.9683980	Poisson	Terr
factor(fac.yr)151	-1.1576016	0.5104078	-2.2679934	0.0233296	Poisson	Terr
factor(fac.yr)161	-0.4442085	0.4448974	-0.9984515	0.3180604	Poisson	Terr
factor(fac.yr)171	-0.0852222	0.4307070	-0.1978659	0.8431500	Poisson	Terr
factor(fac.yr)181	0.1190523	0.4570498	0.2604800	0.7944935	Poisson	Terr
factor(fac.yr)191	-0.0362378	0.6073642	-0.0596640	0.9524233	Poisson	Terr
factor(fac.yr)201	0.0660228	0.5740991	0.1150025	0.9084431	Poisson	Terr
(Intercept)2	0.3739555	0.3032768	1.2330499	0.2175572	Poisson	BirdID+Terr
factor(fac.yr)22	-0.0946260	0.4320930	-0.2189944	0.8266544	Poisson	BirdID+Terr
factor(fac.yr)32	-0.7778735	0.4327339	-1.7975795	0.0722437	Poisson	BirdID+Terr
factor(fac.yr)42	-0.2762459	0.4024418	-0.6864244	0.4924455	Poisson	BirdID+Terr
factor(fac.yr)52	0.1626811	0.3592223	0.4528703	0.6506421	Poisson	BirdID+Terr
factor(fac.yr)62	0.0902913	0.3435489	0.2628194	0.7926898	Poisson	BirdID+Terr
factor(fac.yr)72	-0.6692713	0.3890848	-1.7201169	0.0854112	Poisson	BirdID+Terr
factor(fac.yr)82	-0.6738368	0.3863958	-1.7439029	0.0811760	Poisson	BirdID+Terr
factor(fac.yr)92	-0.0719027	0.3589312	-0.2003244	0.8412269	Poisson	BirdID+Terr
factor(fac.yr)102	-0.4531846	0.3738070	-1.2123492	0.2253787	Poisson	BirdID+Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)112	-1.5140930	0.4700404	-3.2211980	0.0012766	Poisson	BirdID+Terr
factor(fac.yr)122	-0.3428200	0.3789860	-0.9045717	0.3656923	Poisson	BirdID+Terr
factor(fac.yr)132	0.1075937	0.3500798	0.3073406	0.7585842	Poisson	BirdID+Terr
factor(fac.yr)142	-0.0212948	0.3711671	-0.0573724	0.9542485	Poisson	BirdID+Terr
factor(fac.yr)152	-1.1620259	0.5154179	-2.2545315	0.0241628	Poisson	BirdID+Terr
factor(fac.yr)162	-0.4478044	0.4510546	-0.9927942	0.3208103	Poisson	BirdID+Terr
factor(fac.yr)172	-0.0909859	0.4381052	-0.2076804	0.8354785	Poisson	BirdID+Terr
factor(fac.yr)182	0.1257504	0.4643820	0.2707909	0.7865519	Poisson	BirdID+Terr
factor(fac.yr)192	-0.0243941	0.6173163	-0.0395163	0.9684787	Poisson	BirdID+Terr
factor(fac.yr)202	0.0761712	0.5836642	0.1305052	0.8961667	Poisson	BirdID+Terr

The territory only model is performing the best on AIC, and has an adjusted R^2 close to the best adjusted R^2 seen out of these three models. The band ID only model is performing the best on BIC and its adjusted R^2 is a bit worse compared to the other two models. Both models that only included one random effect saw those random effects to be significant. However, the model fitting both random effects only saw the smoother for territory as significant.

Z-score number of fledgling (Gaussian)

Table 13: Z-score GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npars	Family	RE
63.80818	-	847.7483	1110.940	172.4403	254.1918	318	0.2783607	228	Gaussian	BandID+Terr
	353.9144									
41.14550	-	857.6132	1022.967	209.4817	276.8545	318	0.1951083	112	Gaussian	Terr
	384.8534									
32.39456	-	873.9034	1009.349	231.7975	285.6054	318	0.1366533	117	Gaussian	BandID
	400.9487									

Table 14: Z-score GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	31.39456	115	0.4363111	0.0021839	Gaussian	BandID
s(Terrs)	40.14550	110	0.6985619	0.0000252	Gaussian	Terr
s(birdID)	25.19950	115	0.3798684	0.0096707	Gaussian	BandID+Terr
s(Terrs)	37.60868	110	0.7105244	0.0001259	Gaussian	BandID+Terr

Table 15: Z-score GAMMs linear terms table.

	Estimate	Std. Error	t value	Pr(> t)	Family	RE
(Intercept)	0.0058010	0.0642355	0.0903082	0.9281056	Gaussian	BandID
(Intercept)1	0.0257052	0.0655248	0.3922965	0.6951410	Gaussian	Terr
(Intercept)2	0.0226367	0.0708870	0.3193355	0.7497344	Gaussian	BandID+Terr

The model fitting both random effects is performing the best on AIC and adjusted R^2 . However, it is performing the worst on BIC, with the band ID only model performing the best on BIC. All smoothers were significant.

Eggs (Binomial)

Table 16: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
58.29239	-143.0502	407.7701	636.6339	286.1004	259.7076	318	0.2683491	131	Binomial	Terr
56.77574	-145.2109	410.0171	634.9789	290.4218	261.2243	318	0.2627205	136	Binomial	BirdID
77.39405	-122.1032	412.9864	730.4658	244.2064	240.6059	318	0.3673096	247	Binomial	BirdID+Terr

Table 17: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	36.77574	113	54.49025	0.0024213	Binomial	BirdID
s(Terrs)	38.29239	109	58.32967	0.0009728	Binomial	Terr
s(birdID)	26.28167	113	36.19856	0.1126207	Binomial	BirdID+Terr
s(Terrs)	31.11238	109	44.06327	0.0670208	Binomial	BirdID+Terr

Table 18: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.3447456	7.824289e-01	0.4406095	0.6594957	Binomial	BirdID
factor(fac.yr)2	0.1229443	1.072964e+00	0.1145838	0.9087750	Binomial	BirdID
factor(fac.yr)3	-0.6953503	9.519783e-01	-0.7304266	0.4651295	Binomial	BirdID
factor(fac.yr)4	1.0064404	9.920815e-01	1.0144735	0.3103569	Binomial	BirdID
factor(fac.yr)5	1.5043301	1.034902e+00	1.4535965	0.1460582	Binomial	BirdID
factor(fac.yr)6	1.2749696	9.548408e-01	1.3352693	0.1817882	Binomial	BirdID
factor(fac.yr)7	-0.2491368	9.139029e-01	-0.2726075	0.7851549	Binomial	BirdID
factor(fac.yr)8	-0.6356901	9.173559e-01	-0.6929591	0.4883352	Binomial	BirdID
factor(fac.yr)9	0.2831706	9.295309e-01	0.3046381	0.7606418	Binomial	BirdID
factor(fac.yr)10	0.0892880	9.200676e-01	0.0970451	0.9226906	Binomial	BirdID
factor(fac.yr)11	-1.5387936	9.439553e-01	-1.6301552	0.1030687	Binomial	BirdID
factor(fac.yr)12	0.6060616	9.646975e-01	0.6282400	0.5298467	Binomial	BirdID
factor(fac.yr)13	0.6507657	9.384725e-01	0.6934308	0.4880393	Binomial	BirdID
factor(fac.yr)14	0.9991941	1.007614e+00	0.9916438	0.3213713	Binomial	BirdID
factor(fac.yr)15	-0.9945043	1.017330e+00	-0.9775626	0.3282907	Binomial	BirdID
factor(fac.yr)16	-0.3318353	1.044472e+00	-0.3177064	0.7507077	Binomial	BirdID
factor(fac.yr)17	0.5147921	1.195337e+00	0.4306668	0.6667106	Binomial	BirdID
factor(fac.yr)18	1.1782921	1.406708e+00	0.8376239	0.4022420	Binomial	BirdID
factor(fac.yr)19	42.1912992	3.874532e+07	0.0000011	0.9999991	Binomial	BirdID
factor(fac.yr)20	42.2990597	3.874532e+07	0.0000011	0.9999991	Binomial	BirdID
(Intercept)1	0.3790256	7.623664e-01	0.4971698	0.6190693	Binomial	Terr
factor(fac.yr)21	-0.0109190	1.074865e+00	-0.0101585	0.9918949	Binomial	Terr
factor(fac.yr)31	-0.5855208	9.341417e-01	-0.6268008	0.5307898	Binomial	Terr
factor(fac.yr)41	0.9674961	9.930584e-01	0.9742591	0.3299279	Binomial	Terr
factor(fac.yr)51	1.5437878	1.011135e+00	1.5267867	0.1268141	Binomial	Terr
factor(fac.yr)61	1.3569506	9.233441e-01	1.4696045	0.1416689	Binomial	Terr
factor(fac.yr)71	-0.3188155	8.863546e-01	-0.3596929	0.7190768	Binomial	Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)81	-0.7772568	8.847134e-01	-0.8785409	0.3796503	Binomial	Terr
factor(fac.yr)91	0.2972002	9.086630e-01	0.3270742	0.7436118	Binomial	Terr
factor(fac.yr)101	0.2206038	8.929367e-01	0.2470543	0.8048662	Binomial	Terr
factor(fac.yr)111	-1.4650698	9.193790e-01	-1.5935428	0.1110385	Binomial	Terr
factor(fac.yr)121	0.4560056	9.373474e-01	0.4864852	0.6266232	Binomial	Terr
factor(fac.yr)131	0.4828341	9.125384e-01	0.5291110	0.5967285	Binomial	Terr
factor(fac.yr)141	1.2430051	9.817794e-01	1.2660737	0.2054867	Binomial	Terr
factor(fac.yr)151	-0.7365076	9.862246e-01	-0.7467951	0.4551873	Binomial	Terr
factor(fac.yr)161	0.0483649	1.024657e+00	0.0472011	0.9623530	Binomial	Terr
factor(fac.yr)171	0.5834293	1.170837e+00	0.4983010	0.6182719	Binomial	Terr
factor(fac.yr)181	1.1219899	1.403294e+00	0.7995401	0.4239773	Binomial	Terr
factor(fac.yr)191	42.7525617	3.874532e+07	0.0000011	0.9999991	Binomial	Terr
factor(fac.yr)201	42.1206433	3.874532e+07	0.0000011	0.9999991	Binomial	Terr
(Intercept)2	0.3751265	8.302925e-01	0.4518005	0.6514127	Binomial	BirdID+Terr
factor(fac.yr)22	0.0031516	1.154961e+00	0.0027287	0.9978228	Binomial	BirdID+Terr
factor(fac.yr)32	-0.6285043	1.011597e+00	-0.6212993	0.5344027	Binomial	BirdID+Terr
factor(fac.yr)42	1.1027625	1.056200e+00	1.0440849	0.2964461	Binomial	BirdID+Terr
factor(fac.yr)52	1.6237941	1.086347e+00	1.4947290	0.1349852	Binomial	BirdID+Terr
factor(fac.yr)62	1.3850968	9.953749e-01	1.3915329	0.1640639	Binomial	BirdID+Terr
factor(fac.yr)72	-0.2495842	9.702780e-01	-0.2572296	0.7970016	Binomial	BirdID+Terr
factor(fac.yr)82	-0.7130038	9.685712e-01	-0.7361398	0.4616456	Binomial	BirdID+Terr
factor(fac.yr)92	0.3050250	9.920645e-01	0.3074649	0.7584896	Binomial	BirdID+Terr
factor(fac.yr)102	0.1912286	9.788351e-01	0.1953635	0.8451084	Binomial	BirdID+Terr
factor(fac.yr)112	-1.5330528	1.006309e+00	-1.5234422	0.1276481	Binomial	BirdID+Terr
factor(fac.yr)122	0.5836397	1.021730e+00	0.5712271	0.5678457	Binomial	BirdID+Terr
factor(fac.yr)132	0.5731347	9.926478e-01	0.5773797	0.5636830	Binomial	BirdID+Terr
factor(fac.yr)142	1.2041558	1.064902e+00	1.1307666	0.2581533	Binomial	BirdID+Terr
factor(fac.yr)152	-0.8966963	1.073820e+00	-0.8350530	0.4036879	Binomial	BirdID+Terr
factor(fac.yr)162	-0.0911063	1.107840e+00	-0.0822378	0.9344576	Binomial	BirdID+Terr
factor(fac.yr)172	0.4756127	1.270599e+00	0.3743217	0.7081650	Binomial	BirdID+Terr
factor(fac.yr)182	1.0734711	1.484656e+00	0.7230438	0.4696530	Binomial	BirdID+Terr
factor(fac.yr)192	39.6046270	3.874532e+07	0.0000010	0.9999992	Binomial	BirdID+Terr
factor(fac.yr)202	39.1446932	3.874532e+07	0.0000010	0.9999992	Binomial	BirdID+Terr

The single random effects models are both performing similarly on the model selection criteria. The model fitting both random effects is performing poorly, comparatively, on AIC and BIC, but has an adjusted R^2 that is about 10% higher. all smoothers were significant, except for the band ID smoother in the model fitting both random effects.

Probabilites used to generate simulation 2

Simulation 2

Probability of mate fidelity: 0.159

Probability of pair dispersal: 0.568

Probability of single male dispersal: 0.79

Probability of single female dispersal: 0.085

NULL

Simulation 3

Number of fledgling as the response (Poisson)

Table 19: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
50.27942	-359.2667	823.5984	1016.9270	287.1940	242.7206	293	0.2094825	129	Poisson	Terr
40.49636	-373.3262	833.7734	994.0834	315.3130	252.5036	293	0.1434583	142	Poisson	BirdID
55.34128	-353.8546	835.0908	1069.4840	276.3697	237.6587	293	0.2368882	251	Poisson	BirdID+Terr

Table 20: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	20.49636	121	26.489530	0.0318031	Poisson	BirdID
s(Terr)	30.27942	108	44.720504	0.0019697	Poisson	Terr
s(birdID)	7.90765	121	8.970129	0.2465400	Poisson	BirdID+Terr
s(Terr)	27.43363	108	39.839112	0.0057106	Poisson	BirdID+Terr

Table 21: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.4170611	0.2260293	1.8451642	0.0650136	Poisson	BirdID
factor(fac.yr)2	0.0450327	0.2938019	0.1532756	0.8781809	Poisson	BirdID
factor(fac.yr)3	-0.4123395	0.3262677	-1.2638072	0.2062992	Poisson	BirdID
factor(fac.yr)4	0.0047095	0.3080292	0.0152890	0.9878016	Poisson	BirdID
factor(fac.yr)5	-0.0346154	0.2966324	-0.1166947	0.9071020	Poisson	BirdID
factor(fac.yr)6	-0.2547561	0.3140254	-0.8112597	0.4172165	Poisson	BirdID
factor(fac.yr)7	-0.7308264	0.3484917	-2.0971132	0.0359836	Poisson	BirdID
factor(fac.yr)8	-0.1744360	0.3113026	-0.5603421	0.5752461	Poisson	BirdID
factor(fac.yr)9	0.0649317	0.3133352	0.2072277	0.8358321	Poisson	BirdID
factor(fac.yr)10	-0.5054352	0.3656188	-1.3824103	0.1668457	Poisson	BirdID
factor(fac.yr)11	-0.9177190	0.3629929	-2.5282009	0.0114649	Poisson	BirdID
factor(fac.yr)12	-0.2393176	0.3296124	-0.7260574	0.4678036	Poisson	BirdID
factor(fac.yr)13	-0.0855947	0.3311904	-0.2584457	0.7960629	Poisson	BirdID
factor(fac.yr)14	-0.2563085	0.3303164	-0.7759486	0.4377793	Poisson	BirdID
factor(fac.yr)15	-1.0963121	0.4472513	-2.4512217	0.0142372	Poisson	BirdID
factor(fac.yr)16	-0.6676359	0.4289674	-1.5563790	0.1196180	Poisson	BirdID
factor(fac.yr)17	-0.7526888	0.4765297	-1.5795211	0.1142166	Poisson	BirdID
factor(fac.yr)18	-0.7655244	0.5118748	-1.4955306	0.1347760	Poisson	BirdID
factor(fac.yr)19	-0.7894786	0.5119269	-1.5421706	0.1230321	Poisson	BirdID
factor(fac.yr)20	-1.5068110	0.7509144	-2.0066349	0.0447885	Poisson	BirdID
(Intercept)1	0.3546384	0.2316579	1.5308709	0.1258013	Poisson	Terr
factor(fac.yr)21	0.0646923	0.3092305	0.2092042	0.8342888	Poisson	Terr
factor(fac.yr)31	-0.3536454	0.3318333	-1.0657319	0.2865448	Poisson	Terr
factor(fac.yr)41	0.0245742	0.3143774	0.0781677	0.9376946	Poisson	Terr
factor(fac.yr)51	-0.0040305	0.3036400	-0.0132738	0.9894093	Poisson	Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)61	-0.1942254	0.3166579	-0.6133605	0.5396380	Poisson	Terr
factor(fac.yr)71	-0.7069877	0.3498701	-2.0207153	0.0433093	Poisson	Terr
factor(fac.yr)81	-0.0991206	0.3154032	-0.3142664	0.7533187	Poisson	Terr
factor(fac.yr)91	0.1077101	0.3196886	0.3369219	0.7361758	Poisson	Terr
factor(fac.yr)101	-0.3719808	0.3727680	-0.9978882	0.3183336	Poisson	Terr
factor(fac.yr)111	-0.8556384	0.3650041	-2.3441887	0.0190685	Poisson	Terr
factor(fac.yr)121	-0.1805835	0.3319025	-0.5440861	0.5863823	Poisson	Terr
factor(fac.yr)131	0.0220209	0.3365163	0.0654380	0.9478253	Poisson	Terr
factor(fac.yr)141	-0.0877934	0.3338927	-0.2629390	0.7925976	Poisson	Terr
factor(fac.yr)151	-0.9331749	0.4490621	-2.0780529	0.0377045	Poisson	Terr
factor(fac.yr)161	-0.5970783	0.4334178	-1.3776044	0.1683255	Poisson	Terr
factor(fac.yr)171	-0.5762669	0.4850357	-1.1880916	0.2347973	Poisson	Terr
factor(fac.yr)181	-0.6538662	0.5166486	-1.2655917	0.2056593	Poisson	Terr
factor(fac.yr)191	-0.6837277	0.5169856	-1.3225275	0.1859926	Poisson	Terr
factor(fac.yr)201	-1.5793484	0.7549968	-2.0918612	0.0364509	Poisson	Terr
(Intercept)2	0.3687988	0.2351395	1.5684253	0.1167819	Poisson	BirdID+Terr
factor(fac.yr)22	0.0612870	0.3110903	0.1970070	0.8438220	Poisson	BirdID+Terr
factor(fac.yr)32	-0.3634754	0.3358182	-1.0823576	0.2790937	Poisson	BirdID+Terr
factor(fac.yr)42	0.0079507	0.3192437	0.0249049	0.9801308	Poisson	BirdID+Terr
factor(fac.yr)52	-0.0242152	0.3080567	-0.0786065	0.9373457	Poisson	BirdID+Terr
factor(fac.yr)62	-0.2109196	0.3217620	-0.6555143	0.5121367	Poisson	BirdID+Terr
factor(fac.yr)72	-0.7077213	0.3540489	-1.9989363	0.0456152	Poisson	BirdID+Terr
factor(fac.yr)82	-0.1118512	0.3202314	-0.3492826	0.7268772	Poisson	BirdID+Terr
factor(fac.yr)92	0.0969235	0.3249241	0.2982957	0.7654775	Poisson	BirdID+Terr
factor(fac.yr)102	-0.3900199	0.3771568	-1.0341053	0.3010869	Poisson	BirdID+Terr
factor(fac.yr)112	-0.8614337	0.3692951	-2.3326431	0.0196669	Poisson	BirdID+Terr
factor(fac.yr)122	-0.1985256	0.3373471	-0.5884906	0.5562030	Poisson	BirdID+Terr
factor(fac.yr)132	-0.0067002	0.3420239	-0.0195897	0.9843707	Poisson	BirdID+Terr
factor(fac.yr)142	-0.1331028	0.3389764	-0.3926610	0.6945699	Poisson	BirdID+Terr
factor(fac.yr)152	-0.9759586	0.4533897	-2.1525821	0.0313515	Poisson	BirdID+Terr
factor(fac.yr)162	-0.6173118	0.4381046	-1.4090513	0.1588200	Poisson	BirdID+Terr
factor(fac.yr)172	-0.6218705	0.4902241	-1.2685434	0.2046040	Poisson	BirdID+Terr
factor(fac.yr)182	-0.6991609	0.5228472	-1.3372186	0.1811513	Poisson	BirdID+Terr
factor(fac.yr)192	-0.7016270	0.5223218	-1.3432849	0.1791797	Poisson	BirdID+Terr
factor(fac.yr)202	-1.5819988	0.7592011	-2.0837677	0.0371813	Poisson	BirdID+Terr

Each model performs the best on a different criteria. However, the criteria that the territory only model isn't the best on it is instead the second best. All smoothers were significant except for the smoother on band ID in the model fitting both random effects.

Z-score number of fledgling (Gaussian)

Table 22: Z-score GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npars	Family	RE
38.31367	-	797.0366	950.6556	195.9043	254.6863	293	0.1772679	110	Gaussian	Terr
	356.7759									
53.27205	-	797.5548	1022.9539	171.7874	239.7279	293	0.2335340	232	Gaussian	BandID+Terr
	337.5305									

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
18.63816	-	818.3799	905.7092	238.2746	274.3618	293	0.0710887	123	Gaussian	BandID
	385.4603									

Table 23: Z-score GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	17.63816	121	0.1846833	0.0740391	Gaussian	BandID
s(Terrs)	37.31367	108	0.5825464	0.0006399	Gaussian	Terr
s(birdID)	15.37963	121	0.1682174	0.1180305	Gaussian	BandID+Terr
s(Terrs)	36.89242	108	0.6115938	0.0009188	Gaussian	BandID+Terr

Table 24: Z-score GAMMs linear terms table.

	Estimate	Std. Error	t value	Pr(> t)	Family	RE
(Intercept)	0.0012245	0.0608429	0.0201252	0.9839581	Gaussian	BandID
(Intercept)1	0.0199768	0.0661215	0.3021228	0.7628050	Gaussian	Terr
(Intercept)2	0.0183788	0.0692300	0.2654744	0.7908722	Gaussian	BandID+Terr

A similar response as the previous batch of models.

Eggs (Binomial)

Table 25: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
63.67659	-	388.3243	633.6472	255.0028	229.3234	293	0.2780774	129	Binomial	Terr
	127.5014									
49.72291	-	401.3688	599.4277	293.7332	243.2771	293	0.1832232	142	Binomial	BirdID
	146.8666									
76.92935	-	403.6288	725.6125	228.6459	216.0707	293	0.3441954	251	Binomial	BirdID+Terr
	114.3229									

Table 26: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	29.72291	121	38.90936	0.0232127	Binomial	BirdID
s(Terrs)	43.67659	108	64.68863	0.0015515	Binomial	Terr
s(birdID)	15.85202	121	18.01485	0.4349326	Binomial	BirdID+Terr
s(Terrs)	41.07733	108	59.13619	0.0225978	Binomial	BirdID+Terr

Table 27: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.5333873	0.5861984	0.9099092	0.3628704	Binomial	BirdID
factor(fac.yr)2	0.8472008	0.8161052	1.0381025	0.2992223	Binomial	BirdID
factor(fac.yr)3	0.3153610	0.7846062	0.4019354	0.6877316	Binomial	BirdID
factor(fac.yr)4	-0.0041150	0.8051839	-0.0051106	0.9959224	Binomial	BirdID
factor(fac.yr)5	-0.1453363	0.7670402	-0.1894767	0.8497192	Binomial	BirdID
factor(fac.yr)6	0.7646681	0.8369531	0.9136331	0.3609097	Binomial	BirdID
factor(fac.yr)7	-0.7233791	0.7584524	-0.9537568	0.3402068	Binomial	BirdID
factor(fac.yr)8	0.8323743	0.8378354	0.9934819	0.3204752	Binomial	BirdID
factor(fac.yr)9	0.4594728	0.8593297	0.5346874	0.5928660	Binomial	BirdID
factor(fac.yr)10	0.0151401	0.8438482	0.0179418	0.9856853	Binomial	BirdID
factor(fac.yr)11	-1.0000418	0.7624858	-1.3115547	0.1896704	Binomial	BirdID
factor(fac.yr)12	-0.1917254	0.8085367	-0.2371264	0.8125588	Binomial	BirdID
factor(fac.yr)13	1.1554466	1.0073027	1.1470698	0.2513528	Binomial	BirdID
factor(fac.yr)14	-0.0286310	0.8399172	-0.0340879	0.9728070	Binomial	BirdID
factor(fac.yr)15	-1.5263902	0.8792313	-1.7360507	0.0825549	Binomial	BirdID
factor(fac.yr)16	-0.6286531	0.9240968	-0.6802893	0.4963213	Binomial	BirdID
factor(fac.yr)17	-0.6479713	0.9966162	-0.6501713	0.5155816	Binomial	BirdID
factor(fac.yr)18	-1.5687846	1.0978870	-1.4289127	0.1530293	Binomial	BirdID
factor(fac.yr)19	-0.3415175	1.0374996	-0.3291736	0.7420245	Binomial	BirdID
factor(fac.yr)20	-2.0998570	1.3032164	-1.6112881	0.1071170	Binomial	BirdID
(Intercept)1	0.4254428	0.6147746	0.6920306	0.4889181	Binomial	Terr
factor(fac.yr)21	0.9897248	0.8932818	1.1079648	0.2678770	Binomial	Terr
factor(fac.yr)31	0.5125652	0.8221706	0.6234293	0.5330025	Binomial	Terr
factor(fac.yr)41	-0.0568619	0.8586927	-0.0662192	0.9472033	Binomial	Terr
factor(fac.yr)51	-0.1312031	0.8133815	-0.1613057	0.8718527	Binomial	Terr
factor(fac.yr)61	0.9860454	0.8768279	1.1245597	0.2607757	Binomial	Terr
factor(fac.yr)71	-0.8838425	0.7828549	-1.1289991	0.2588982	Binomial	Terr
factor(fac.yr)81	1.1344674	0.8689275	1.3055950	0.1916903	Binomial	Terr
factor(fac.yr)91	0.7899275	0.8960076	0.8816080	0.3779888	Binomial	Terr
factor(fac.yr)101	0.3907523	0.8949603	0.4366141	0.6623913	Binomial	Terr
factor(fac.yr)111	-0.9686835	0.7859403	-1.2325154	0.2177566	Binomial	Terr
factor(fac.yr)121	0.0236588	0.8497489	0.0278422	0.9777880	Binomial	Terr
factor(fac.yr)131	1.4215824	1.0509426	1.3526737	0.1761599	Binomial	Terr
factor(fac.yr)141	0.5590372	0.8672880	0.6445808	0.5191989	Binomial	Terr
factor(fac.yr)151	-1.0739614	0.9170510	-1.1711031	0.2415573	Binomial	Terr
factor(fac.yr)161	-0.3486769	0.9396659	-0.3710647	0.7105893	Binomial	Terr
factor(fac.yr)171	-0.1240590	1.0561113	-0.1174678	0.9064894	Binomial	Terr
factor(fac.yr)181	-1.4307544	1.1825173	-1.2099226	0.2263086	Binomial	Terr
factor(fac.yr)191	0.0204606	1.1062091	0.0184961	0.9852431	Binomial	Terr
factor(fac.yr)201	-2.2023801	1.3786610	-1.5974777	0.1101593	Binomial	Terr
(Intercept)2	0.4812982	0.6509815	0.7393424	0.4596991	Binomial	BirdID+Terr
factor(fac.yr)22	1.0037531	0.9252248	1.0848748	0.2779771	Binomial	BirdID+Terr
factor(fac.yr)32	0.4984897	0.8629409	0.5776638	0.5634912	Binomial	BirdID+Terr
factor(fac.yr)42	-0.0749290	0.9041470	-0.0828726	0.9339528	Binomial	BirdID+Terr
factor(fac.yr)52	-0.2135787	0.8596178	-0.2484577	0.8037803	Binomial	BirdID+Terr
factor(fac.yr)62	0.8863912	0.9151302	0.9685957	0.3327470	Binomial	BirdID+Terr
factor(fac.yr)72	-0.8669838	0.8255977	-1.0501286	0.2936590	Binomial	BirdID+Terr
factor(fac.yr)82	1.0784831	0.9131624	1.1810420	0.2375861	Binomial	BirdID+Terr
factor(fac.yr)92	0.7279845	0.9434047	0.7716566	0.4403178	Binomial	BirdID+Terr
factor(fac.yr)102	0.3293017	0.9431837	0.3491385	0.7269854	Binomial	BirdID+Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)112	-0.9616825	0.8331094	-1.1543293	0.2483652	Binomial	BirdID+Terr
factor(fac.yr)122	0.0164498	0.9037089	0.0182026	0.9854773	Binomial	BirdID+Terr
factor(fac.yr)132	1.3591026	1.1026226	1.2326091	0.2177217	Binomial	BirdID+Terr
factor(fac.yr)142	0.3716378	0.9157438	0.4058317	0.6848663	Binomial	BirdID+Terr
factor(fac.yr)152	-1.2497987	0.9734121	-1.2839359	0.1991644	Binomial	BirdID+Terr
factor(fac.yr)162	-0.4759268	1.0030924	-0.4744596	0.6351722	Binomial	BirdID+Terr
factor(fac.yr)172	-0.2766113	1.1292407	-0.2449534	0.8064925	Binomial	BirdID+Terr
factor(fac.yr)182	-1.6718898	1.2746003	-1.3116973	0.1896223	Binomial	BirdID+Terr
factor(fac.yr)192	-0.1020496	1.1612556	-0.0878787	0.9299731	Binomial	BirdID+Terr
factor(fac.yr)202	-2.3014722	1.4632453	-1.5728546	0.1157525	Binomial	BirdID+Terr

A similar response as for the previous batch of models.

Probabilites used to generate simulation 3

Simulation 1

Probability of mate fidelity: 0.251

Probability of pair dispersal: 0.793

Probability of single male dispersal: 0.388

Probability of single female dispersal: 0.885

NULL

Here, we should notice the low mate fidelity rate and the fact that dispersal probabilities aren't terribly low.

Simulation 4

Number of fledgling as the response (Poisson)

Table 28: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
52.83835	-	952.1909	1163.852	327.7745	303.1617	356	0.3121997	129	Poisson	Terr
	421.4723									
55.37711	-	967.0012	1217.452	322.5636	300.6229	356	0.3220840	247	Poisson	BirdID+Terr
	418.8669									
38.94084	-	969.6726	1132.921	370.2438	317.0592	356	0.2169497	138	Poisson	BirdID
	442.7070									

Table 29: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	18.940837	117	24.223850	0.0433657	Poisson	BirdID
s(Terrs)	32.838347	108	53.430502	0.0001597	Poisson	Terr
s(birdID)	3.489294	117	3.697705	0.3705005	Poisson	BirdID+Terr
s(Terrs)	31.887814	108	51.657930	0.0002669	Poisson	BirdID+Terr

Table 30: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.0162574	0.4209611	0.0386197	0.9691936	Poisson	BirdID
factor(fac.yr)2	0.3329679	0.4809464	0.6923182	0.4887375	Poisson	BirdID
factor(fac.yr)3	-0.1650850	0.4896952	-0.3371178	0.7360281	Poisson	BirdID
factor(fac.yr)4	0.4235109	0.4588459	0.9229916	0.3560116	Poisson	BirdID
factor(fac.yr)5	0.5127053	0.4548828	1.1271153	0.2596937	Poisson	BirdID
factor(fac.yr)6	0.4313752	0.4507990	0.9569126	0.3386114	Poisson	BirdID
factor(fac.yr)7	-0.4782329	0.4880277	-0.9799298	0.3271208	Poisson	BirdID
factor(fac.yr)8	0.0557508	0.4598601	0.1212342	0.9035055	Poisson	BirdID
factor(fac.yr)9	0.5342103	0.4518160	1.1823624	0.2370619	Poisson	BirdID
factor(fac.yr)10	0.0164702	0.4797970	0.0343275	0.9726160	Poisson	BirdID
factor(fac.yr)11	-0.1358989	0.4822622	-0.2817947	0.7781009	Poisson	BirdID
factor(fac.yr)12	0.4758525	0.4539987	1.0481363	0.2945758	Poisson	BirdID
factor(fac.yr)13	0.5114149	0.4588996	1.1144374	0.2650916	Poisson	BirdID
factor(fac.yr)14	-0.3618792	0.5074179	-0.7131778	0.4757358	Poisson	BirdID
factor(fac.yr)15	-0.5800208	0.5535859	-1.0477521	0.2947528	Poisson	BirdID
factor(fac.yr)16	-1.1008469	0.6574721	-1.6743629	0.0940593	Poisson	BirdID
factor(fac.yr)17	-0.8458175	0.5901595	-1.4332015	0.1518003	Poisson	BirdID
factor(fac.yr)18	-1.2106444	0.7187133	-1.6844608	0.0920926	Poisson	BirdID
factor(fac.yr)19	-2.4007842	1.0875551	-2.2075058	0.0272787	Poisson	BirdID
factor(fac.yr)20	0.0319158	0.5566601	0.0573345	0.9542787	Poisson	BirdID
(Intercept)1	0.1882697	0.4316033	0.4362100	0.6626843	Poisson	Terr
factor(fac.yr)21	0.1780383	0.4965353	0.3585613	0.7199233	Poisson	Terr
factor(fac.yr)31	-0.3039054	0.5033703	-0.6037412	0.5460157	Poisson	Terr
factor(fac.yr)41	0.2280797	0.4681994	0.4871423	0.6261575	Poisson	Terr
factor(fac.yr)51	0.3544720	0.4659742	0.7607116	0.4468293	Poisson	Terr
factor(fac.yr)61	0.2306161	0.4591707	0.5022448	0.6154954	Poisson	Terr
factor(fac.yr)71	-0.6351084	0.4984686	-1.2741193	0.2026211	Poisson	Terr
factor(fac.yr)81	-0.0756832	0.4690144	-0.1613664	0.8718048	Poisson	Terr
factor(fac.yr)91	0.3072477	0.4627389	0.6639765	0.5067054	Poisson	Terr
factor(fac.yr)101	-0.1477842	0.4902048	-0.3014744	0.7630528	Poisson	Terr
factor(fac.yr)111	-0.2962299	0.4937998	-0.5998988	0.5485737	Poisson	Terr
factor(fac.yr)121	0.2917160	0.4638300	0.6289288	0.5293957	Poisson	Terr
factor(fac.yr)131	0.3192055	0.4708445	0.6779425	0.4978082	Poisson	Terr
factor(fac.yr)141	-0.5423637	0.5179935	-1.0470472	0.2950778	Poisson	Terr
factor(fac.yr)151	-0.8262782	0.5640196	-1.4649813	0.1429260	Poisson	Terr
factor(fac.yr)161	-1.3475459	0.6632003	-2.0318836	0.0421654	Poisson	Terr
factor(fac.yr)171	-1.0053397	0.5942007	-1.6919195	0.0906613	Poisson	Terr
factor(fac.yr)181	-1.3435093	0.7262721	-1.8498703	0.0643322	Poisson	Terr
factor(fac.yr)191	-2.4601113	1.0925111	-2.2517953	0.0243352	Poisson	Terr
factor(fac.yr)201	-0.1102789	0.5681893	-0.1940883	0.8461067	Poisson	Terr
(Intercept)2	0.1833550	0.4332749	0.4231840	0.6721610	Poisson	BirdID+Terr
factor(fac.yr)22	0.1816460	0.4980543	0.3647113	0.7153269	Poisson	BirdID+Terr
factor(fac.yr)32	-0.2971591	0.5049314	-0.5885139	0.5561874	Poisson	BirdID+Terr
factor(fac.yr)42	0.2365758	0.4701779	0.5031622	0.6148502	Poisson	BirdID+Terr
factor(fac.yr)52	0.3586708	0.4680090	0.7663759	0.4434527	Poisson	BirdID+Terr
factor(fac.yr)62	0.2356010	0.4611859	0.5108590	0.6094498	Poisson	BirdID+Terr
factor(fac.yr)72	-0.6300313	0.5002647	-1.2593958	0.2078874	Poisson	BirdID+Terr
factor(fac.yr)82	-0.0721261	0.4709590	-0.1531473	0.8782821	Poisson	BirdID+Terr
factor(fac.yr)92	0.3107930	0.4647925	0.6686705	0.5037057	Poisson	BirdID+Terr
factor(fac.yr)102	-0.1451703	0.4921922	-0.2949464	0.7680348	Poisson	BirdID+Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)112	-0.2954204	0.4957711	-0.5958806	0.5512550	Poisson	BirdID+Terr
factor(fac.yr)122	0.2966220	0.4658321	0.6367574	0.5242829	Poisson	BirdID+Terr
factor(fac.yr)132	0.3198218	0.4728862	0.6763188	0.4988383	Poisson	BirdID+Terr
factor(fac.yr)142	-0.5401190	0.5199593	-1.0387716	0.2989110	Poisson	BirdID+Terr
factor(fac.yr)152	-0.8195641	0.5659315	-1.4481683	0.1475700	Poisson	BirdID+Terr
factor(fac.yr)162	-1.3396320	0.6650522	-2.0143261	0.0439753	Poisson	BirdID+Terr
factor(fac.yr)172	-0.9983207	0.5961101	-1.6747255	0.0939881	Poisson	BirdID+Terr
factor(fac.yr)182	-1.3422655	0.7280160	-1.8437307	0.0652224	Poisson	BirdID+Terr
factor(fac.yr)192	-2.4549719	1.0935588	-2.2449382	0.0247721	Poisson	BirdID+Terr
factor(fac.yr)202	-0.0991734	0.5705651	-0.1738160	0.8620100	Poisson	BirdID+Terr

The territory only model is performing well on AIC and has an adjusted R^2 close to that of the highest adjusted R^2 . The band ID only model has the lowest BIC. All smoothers are significant except for the band ID smoother in the model that fits both.

Z-score number of fledgling (Gaussian)

Table 31: Z-score GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npars	Family	RE
81.03942	-379.6564	934.1624	1272.927	175.9069	274.9606	356	0.3240704	228	Gaussian	BandID+Terr
49.98886	-423.9525	953.4324	1157.888	225.6103	306.0111	356	0.2210478	110	Gaussian	Terr
28.54096	-461.7730	989.5048	1117.298	279.0201	327.4590	356	0.0997406	119	Gaussian	BandID

Table 32: Z-score GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	27.54096	117	0.3361654	0.0133849	Gaussian	BandID
s(Terrs)	48.98886	108	0.9327819	0.0000018	Gaussian	Terr
s(birdID)	30.59196	117	0.4742447	0.0140523	Gaussian	BandID+Terr
s(Terrs)	49.44746	108	1.1220589	0.0000055	Gaussian	BandID+Terr

Table 33: Z-score GAMMs linear terms table.

	Estimate	Std. Error	t value	Pr(> t)	Family	RE
(Intercept)	0.0136900	0.0595269	0.2299793	0.8182515	Gaussian	BandID
(Intercept)1	0.0439841	0.0660062	0.6663622	0.5056816	Gaussian	Terr
(Intercept)2	0.0506930	0.0728135	0.6962036	0.4868894	Gaussian	BandID+Terr

This time, the model fitting both random effects is performing the best on AIC and has the highest adjusted R^2 , while the band ID only model had the lowest BIC. All smoothers were significant.

Eggs (Binomial)

Table 34: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npars	Family	RE
61.19477	-154.8537	437.9334	686.3667	309.7075	294.8052	356	0.3143651	129	Binomial	Terr
89.49117	-123.3954	440.5196	815.8626	246.7908	266.5088	356	0.4452068	247	Binomial	BirdID+Terr
52.51146	-166.6792	444.7271	660.4999	333.3585	303.4885	356	0.2727431	138	Binomial	BirdID

Table 35: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	32.51146	117	46.03729	0.0059847	Binomial	BirdID
s(Terrs)	41.19477	108	62.14628	0.0011350	Binomial	Terr
s(birdID)	29.77849	117	42.19562	0.1272888	Binomial	BirdID+Terr
s(Terrs)	39.71267	108	59.63466	0.0465599	Binomial	BirdID+Terr

Table 36: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.8479675	0.9562002	0.8868096	0.3751814	Binomial	BirdID
factor(fac.yr)2	0.1573283	1.1231541	0.1400772	0.8885990	Binomial	BirdID
factor(fac.yr)3	-0.5732446	1.0657855	-0.5378611	0.5906729	Binomial	BirdID
factor(fac.yr)4	0.7811229	1.1176307	0.6989096	0.4846085	Binomial	BirdID
factor(fac.yr)5	0.5183641	1.1023521	0.4702346	0.6381874	Binomial	BirdID
factor(fac.yr)6	-0.0799390	1.0560517	-0.0756961	0.9396608	Binomial	BirdID
factor(fac.yr)7	-1.2452193	1.0497723	-1.1861804	0.2355510	Binomial	BirdID
factor(fac.yr)8	-0.6122844	1.0420228	-0.5875922	0.5568061	Binomial	BirdID
factor(fac.yr)9	0.6450020	1.1190585	0.5763792	0.5643589	Binomial	BirdID
factor(fac.yr)10	-0.6454077	1.0850807	-0.5948016	0.5519761	Binomial	BirdID
factor(fac.yr)11	0.0509930	1.0929742	0.0466553	0.9627880	Binomial	BirdID
factor(fac.yr)12	1.5806317	1.2283976	1.2867428	0.1981839	Binomial	BirdID
factor(fac.yr)13	0.1540155	1.1141773	0.1382325	0.8900567	Binomial	BirdID
factor(fac.yr)14	-0.7118897	1.0942509	-0.6505726	0.5153224	Binomial	BirdID
factor(fac.yr)15	-1.5149636	1.1409349	-1.3278265	0.1842354	Binomial	BirdID
factor(fac.yr)16	-1.8803885	1.2011127	-1.5655387	0.1174566	Binomial	BirdID
factor(fac.yr)17	-1.6588777	1.1556669	-1.4354289	0.1511649	Binomial	BirdID
factor(fac.yr)18	-3.0284932	1.4580182	-2.0771300	0.0377896	Binomial	BirdID
factor(fac.yr)19	-3.1285964	1.4524556	-2.1540049	0.0312398	Binomial	BirdID
factor(fac.yr)20	-1.0457604	1.2406300	-0.8429269	0.3992693	Binomial	BirdID
(Intercept)1	1.5483473	0.9686236	1.5985026	0.1099312	Binomial	Terr
factor(fac.yr)21	-0.4163334	1.1567600	-0.3599133	0.7189119	Binomial	Terr
factor(fac.yr)31	-1.2583312	1.0974727	-1.1465718	0.2515586	Binomial	Terr
factor(fac.yr)41	0.0891211	1.1408886	0.0781155	0.9377362	Binomial	Terr
factor(fac.yr)51	-0.1354973	1.1124548	-0.1218002	0.9030572	Binomial	Terr
factor(fac.yr)61	-0.7386080	1.0505062	-0.7030973	0.4819951	Binomial	Terr
factor(fac.yr)71	-1.9714412	1.0637007	-1.8533796	0.0638280	Binomial	Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)81	-1.0482303	1.0460560	-1.0020786	0.3163056	Binomial	Terr
factor(fac.yr)91	0.0769584	1.1299446	0.0681081	0.9456996	Binomial	Terr
factor(fac.yr)101	-1.0583270	1.0946856	-0.9667863	0.3336509	Binomial	Terr
factor(fac.yr)111	-0.6154344	1.1080259	-0.5554333	0.5785984	Binomial	Terr
factor(fac.yr)121	0.9580300	1.2228693	0.7834279	0.4333759	Binomial	Terr
factor(fac.yr)131	-0.4016822	1.1236808	-0.3574701	0.7207399	Binomial	Terr
factor(fac.yr)141	-1.3206400	1.1031264	-1.1971792	0.2312367	Binomial	Terr
factor(fac.yr)151	-2.2410082	1.1561904	-1.9382692	0.0525904	Binomial	Terr
factor(fac.yr)161	-2.7413314	1.2093870	-2.2667116	0.0234079	Binomial	Terr
factor(fac.yr)171	-2.2997023	1.1513262	-1.9974376	0.0457777	Binomial	Terr
factor(fac.yr)181	-3.6238956	1.5011130	-2.4141391	0.0157724	Binomial	Terr
factor(fac.yr)191	-3.5531053	1.4774852	-2.4048331	0.0161799	Binomial	Terr
factor(fac.yr)201	-1.8761199	1.2601714	-1.4887815	0.1365449	Binomial	Terr
(Intercept)2	1.5292536	1.0696797	1.4296369	0.1528212	Binomial	BirdID+Terr
factor(fac.yr)22	-0.2837915	1.2230478	-0.2320363	0.8165098	Binomial	BirdID+Terr
factor(fac.yr)32	-1.1241825	1.1999621	-0.9368483	0.3488366	Binomial	BirdID+Terr
factor(fac.yr)42	0.2118689	1.2433164	0.1704063	0.8646907	Binomial	BirdID+Terr
factor(fac.yr)52	0.0423118	1.2320983	0.0343412	0.9726050	Binomial	BirdID+Terr
factor(fac.yr)62	-0.6814240	1.1667644	-0.5840288	0.5592009	Binomial	BirdID+Terr
factor(fac.yr)72	-1.9389576	1.1785700	-1.6451781	0.0999331	Binomial	BirdID+Terr
factor(fac.yr)82	-0.9810667	1.1611449	-0.8449133	0.3981593	Binomial	BirdID+Terr
factor(fac.yr)92	0.1021759	1.2450026	0.0820689	0.9345920	Binomial	BirdID+Terr
factor(fac.yr)102	-1.1076000	1.2191885	-0.9084731	0.3636283	Binomial	BirdID+Terr
factor(fac.yr)112	-0.6142429	1.2339200	-0.4977980	0.6186264	Binomial	BirdID+Terr
factor(fac.yr)122	1.0599616	1.3254352	0.7997083	0.4238798	Binomial	BirdID+Terr
factor(fac.yr)132	-0.4280419	1.2434793	-0.3442292	0.7306739	Binomial	BirdID+Terr
factor(fac.yr)142	-1.4001825	1.2317535	-1.1367392	0.2556473	Binomial	BirdID+Terr
factor(fac.yr)152	-2.3011965	1.2929142	-1.7798525	0.0751001	Binomial	BirdID+Terr
factor(fac.yr)162	-2.7779240	1.3445223	-2.0661048	0.0388186	Binomial	BirdID+Terr
factor(fac.yr)172	-2.2434147	1.2759181	-1.7582748	0.0787008	Binomial	BirdID+Terr
factor(fac.yr)182	-3.7402570	1.6607229	-2.2521862	0.0243105	Binomial	BirdID+Terr
factor(fac.yr)192	-3.5958229	1.6294330	-2.2067939	0.0273285	Binomial	BirdID+Terr
factor(fac.yr)202	-1.5514288	1.3808182	-1.1235577	0.2612007	Binomial	BirdID+Terr

We notice that the territory only model is performing the best on AIC, the band ID model is performing the best on BIC, and the model fitting both random effects has a notably higher adjusted R^2 .

Probabilites used to generate simulation 4

Simulation 1

Probability of mate fidelity: 0.919

Probability of pair dispersal: 0.547

Probability of single male dispersal: 0.743

Probability of single female dispersal: 0.976

NULL

Simulation 5

Number of fledgling as the response (Poisson)

Table 37: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npars	Family	RE
58.38266	-	972.6485	1204.889	339.5054	292.6173	351	0.2437012	130	Poisson	Terr
	426.1706									
64.73475	-	983.8059	1263.038	326.3197	286.2652	351	0.2659123	253	Poisson	BirdID+Terr
	419.5778									
47.21381	-	988.9202	1180.133	377.0304	303.7862	351	0.1542558	143	Poisson	BirdID
	444.9332									

Table 38: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	27.213809	122	39.23953	0.0035441	Poisson	BirdID
s(Terrs)	38.382664	109	66.35079	0.0000224	Poisson	Terr
s(birdID)	9.952611	122	12.14562	0.1414338	Poisson	BirdID+Terr
s(Terrs)	34.782142	109	58.61634	0.0002109	Poisson	BirdID+Terr

Table 39: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.7768595	0.2651364	2.9300373	0.0033892	Poisson	BirdID
factor(fac.yr)2	-0.2986973	0.3163621	-0.9441627	0.3450865	Poisson	BirdID
factor(fac.yr)3	-0.8110366	0.3279979	-2.4726886	0.0134101	Poisson	BirdID
factor(fac.yr)4	-0.6773779	0.3227086	-2.0990388	0.0358135	Poisson	BirdID
factor(fac.yr)5	-0.1831584	0.3120760	-0.5869031	0.5572688	Poisson	BirdID
factor(fac.yr)6	-0.6820000	0.3286415	-2.0752096	0.0379671	Poisson	BirdID
factor(fac.yr)7	-1.0675554	0.3312703	-3.2226111	0.0012703	Poisson	BirdID
factor(fac.yr)8	-0.5574343	0.3225185	-1.7283791	0.0839203	Poisson	BirdID
factor(fac.yr)9	-0.7382497	0.3403806	-2.1688945	0.0300907	Poisson	BirdID
factor(fac.yr)10	-0.6399507	0.3383513	-1.8913798	0.0585737	Poisson	BirdID
factor(fac.yr)11	-1.3441095	0.3978073	-3.3787956	0.0007280	Poisson	BirdID
factor(fac.yr)12	-1.0681743	0.4097056	-2.6071753	0.0091293	Poisson	BirdID
factor(fac.yr)13	-0.5415557	0.3450552	-1.5694754	0.1165372	Poisson	BirdID
factor(fac.yr)14	-0.8528272	0.4012542	-2.1254039	0.0335529	Poisson	BirdID
factor(fac.yr)15	-1.3947594	0.5291461	-2.6358681	0.0083922	Poisson	BirdID
factor(fac.yr)16	-0.6370240	0.4139094	-1.5390421	0.1237940	Poisson	BirdID
factor(fac.yr)17	-0.5931105	0.4147338	-1.4300993	0.1526885	Poisson	BirdID
factor(fac.yr)18	-0.6550160	0.4407139	-1.4862613	0.1372100	Poisson	BirdID
factor(fac.yr)19	-1.3467272	0.5309406	-2.5364930	0.0111969	Poisson	BirdID
factor(fac.yr)20	-0.9047025	0.5012758	-1.8047997	0.0711060	Poisson	BirdID
(Intercept)1	0.4397026	0.2704686	1.6257062	0.1040122	Poisson	Terr
factor(fac.yr)21	0.0815378	0.3326454	0.2451193	0.8063641	Poisson	Terr
factor(fac.yr)31	-0.5313763	0.3354236	-1.5841950	0.1131493	Poisson	Terr
factor(fac.yr)41	-0.3422679	0.3325397	-1.0292541	0.3033603	Poisson	Terr
factor(fac.yr)51	0.1771198	0.3177091	0.5574906	0.5771923	Poisson	Terr
factor(fac.yr)61	-0.3340549	0.3310475	-1.0090845	0.3129341	Poisson	Terr
factor(fac.yr)71	-0.8274865	0.3278851	-2.5237086	0.0116124	Poisson	Terr
factor(fac.yr)81	-0.2619842	0.3216238	-0.8145672	0.4153201	Poisson	Terr
factor(fac.yr)91	-0.3873030	0.3373861	-1.1479519	0.2509884	Poisson	Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)101	-0.2924428	0.3407456	-0.8582437	0.3907579	Poisson	Terr
factor(fac.yr)111	-0.8636788	0.3998849	-2.1598182	0.0307867	Poisson	Terr
factor(fac.yr)121	-0.7370349	0.4144529	-1.7783320	0.0753493	Poisson	Terr
factor(fac.yr)131	-0.2231209	0.3423265	-0.6517780	0.5145444	Poisson	Terr
factor(fac.yr)141	-0.5523423	0.4072626	-1.3562314	0.1750256	Poisson	Terr
factor(fac.yr)151	-1.0729659	0.5315000	-2.0187505	0.0435132	Poisson	Terr
factor(fac.yr)161	-0.2455997	0.4209207	-0.5834822	0.5595688	Poisson	Terr
factor(fac.yr)171	-0.4365896	0.4090749	-1.0672608	0.2858541	Poisson	Terr
factor(fac.yr)181	-0.4097564	0.4300112	-0.9528971	0.3406422	Poisson	Terr
factor(fac.yr)191	-0.9994316	0.5291341	-1.8888058	0.0589179	Poisson	Terr
factor(fac.yr)201	-0.5479618	0.5049218	-1.0852408	0.2778150	Poisson	Terr
(Intercept)2	0.4802444	0.2771663	1.7326938	0.0831501	Poisson	BirdID+Terr
factor(fac.yr)22	0.0321408	0.3364183	0.0955381	0.9238874	Poisson	BirdID+Terr
factor(fac.yr)32	-0.5626868	0.3406974	-1.6515734	0.0986215	Poisson	BirdID+Terr
factor(fac.yr)42	-0.3827125	0.3385223	-1.1305385	0.2582494	Poisson	BirdID+Terr
factor(fac.yr)52	0.1271717	0.3249281	0.3913839	0.6955135	Poisson	BirdID+Terr
factor(fac.yr)62	-0.3771354	0.3383209	-1.1147268	0.2649675	Poisson	BirdID+Terr
factor(fac.yr)72	-0.8521288	0.3357406	-2.5380570	0.0111470	Poisson	BirdID+Terr
factor(fac.yr)82	-0.2934638	0.3295643	-0.8904598	0.3732191	Poisson	BirdID+Terr
factor(fac.yr)92	-0.4393039	0.3452856	-1.2722912	0.2032697	Poisson	BirdID+Terr
factor(fac.yr)102	-0.3542483	0.3482962	-1.0170891	0.3091111	Poisson	BirdID+Terr
factor(fac.yr)112	-0.9347493	0.4062770	-2.3007685	0.0214047	Poisson	BirdID+Terr
factor(fac.yr)122	-0.7882810	0.4219123	-1.8683529	0.0617129	Poisson	BirdID+Terr
factor(fac.yr)132	-0.2773236	0.3508532	-0.7904263	0.4292789	Poisson	BirdID+Terr
factor(fac.yr)142	-0.6010396	0.4148527	-1.4488023	0.1473928	Poisson	BirdID+Terr
factor(fac.yr)152	-1.1242854	0.5380888	-2.0894050	0.0366713	Poisson	BirdID+Terr
factor(fac.yr)162	-0.3064353	0.4289268	-0.7144234	0.4749654	Poisson	BirdID+Terr
factor(fac.yr)172	-0.4746702	0.4190529	-1.1327215	0.2573312	Poisson	BirdID+Terr
factor(fac.yr)182	-0.4552975	0.4403769	-1.0338815	0.3011916	Poisson	BirdID+Terr
factor(fac.yr)192	-1.0326659	0.5366631	-1.9242350	0.0543251	Poisson	BirdID+Terr
factor(fac.yr)202	-0.5849549	0.5131629	-1.1399010	0.2543275	Poisson	BirdID+Terr

The territory only model has the best AIC score and an adjusted R^2 comparable to the best adjusted R^2 . The band ID only model has the best BIC score. All smoothers are significant, except for the band ID smoother that is present in the model fitting both random effects.

Z-score number of fledgling (Gaussian)

Table 40: Z-score GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
64.62452	-	945.2321	1221.330	202.0255	286.3755	351	0.2540488	234	Gaussian	BandID+Terr
	401.1026									
45.35775	-	949.2361	1135.826	233.2018	305.6422	351	0.1932136	111	Gaussian	Terr
	426.2886									
24.11554	-	977.4652	1086.422	283.4128	326.8845	351	0.0832199	124	Gaussian	BandID
	460.5112									

Table 41: Z-score GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	23.11554	122	0.2604178	0.0217333	Gaussian	BandID
s(Terrs)	44.35775	109	0.7689898	0.0000239	Gaussian	Terr
s(birdID)	20.54922	122	0.2634710	0.0219588	Gaussian	BandID+Terr
s(Terrs)	43.07530	109	0.8008559	0.0000822	Gaussian	BandID+Terr

Table 42: Z-score GAMMs linear terms table.

	Estimate	Std. Error	t value	Pr(> t)	Family	RE
(Intercept)	0.0039059	0.0574024	0.0680444	0.9457919	Gaussian	BandID
(Intercept)1	0.0059716	0.0649021	0.0920090	0.9267512	Gaussian	Terr
(Intercept)2	0.0009972	0.0681694	0.0146278	0.9883393	Gaussian	BandID+Terr

I would say something similar as to what I said for the last batch of models with z-score as the response.

Eggs (Binomial)

```
Warning in newton(lsp = lsp, X = G$X, y = G$y, Eb = G$Eb, UrS = G$UrS, L = G$L,
: Fitting terminated with step failure - check results carefully
Warning in newton(lsp = lsp, X = G$X, y = G$y, Eb = G$Eb, UrS = G$UrS, L = G$L,
: Fitting terminated with step failure - check results carefully
Warning in newton(lsp = lsp, X = G$X, y = G$y, Eb = G$Eb, UrS = G$UrS, L = G$L,
: Fitting terminated with step failure - check results carefully
```

Table 43: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
60.85736	-	441.2686	659.2648	328.3401	290.1426	351	0.2508526	143	Binomial	BirdID
	164.1701									
76.24202	-	448.4681	759.0265	287.5898	274.7580	351	0.3295388	130	Binomial	Terr
	143.7949									
106.83444	-	450.4481	901.3777	216.8534	244.1656	351	0.4940652	253	Binomial	BirdID+Terr
	108.4267									

Table 44: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	40.85736	121	61.83766	0.0002296	Binomial	BirdID
s(Terrs)	56.24202	109	97.13006	0.0000878	Binomial	Terr
s(birdID)	36.97383	121	68.70320	0.0208491	Binomial	BirdID+Terr
s(Terrs)	49.86061	109	84.21902	0.0576304	Binomial	BirdID+Terr

Table 45: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	75.03187	23726567	3.2e-06	0.9999975	Binomial	BirdID
factor(fac.yr)2	-73.86935	23726567	-3.1e-06	0.9999975	Binomial	BirdID
factor(fac.yr)3	-74.32773	23726567	-3.1e-06	0.9999975	Binomial	BirdID
factor(fac.yr)4	-74.56090	23726567	-3.1e-06	0.9999975	Binomial	BirdID
factor(fac.yr)5	-73.06625	23726567	-3.1e-06	0.9999975	Binomial	BirdID
factor(fac.yr)6	-73.81131	23726567	-3.1e-06	0.9999975	Binomial	BirdID
factor(fac.yr)7	-75.21581	23726567	-3.2e-06	0.9999975	Binomial	BirdID
factor(fac.yr)8	-74.34730	23726567	-3.1e-06	0.9999975	Binomial	BirdID
factor(fac.yr)9	-75.10245	23726567	-3.2e-06	0.9999975	Binomial	BirdID
factor(fac.yr)10	-74.06278	23726567	-3.1e-06	0.9999975	Binomial	BirdID
factor(fac.yr)11	-75.83146	23726567	-3.2e-06	0.9999974	Binomial	BirdID
factor(fac.yr)12	-75.43908	23726567	-3.2e-06	0.9999975	Binomial	BirdID
factor(fac.yr)13	-74.77804	23726567	-3.2e-06	0.9999975	Binomial	BirdID
factor(fac.yr)14	-75.20111	23726567	-3.2e-06	0.9999975	Binomial	BirdID
factor(fac.yr)15	-75.38277	23726567	-3.2e-06	0.9999975	Binomial	BirdID
factor(fac.yr)16	-73.73138	23726567	-3.1e-06	0.9999975	Binomial	BirdID
factor(fac.yr)17	-75.38463	23726567	-3.2e-06	0.9999975	Binomial	BirdID
factor(fac.yr)18	-74.31589	23726567	-3.1e-06	0.9999975	Binomial	BirdID
factor(fac.yr)19	-75.11565	23726567	-3.2e-06	0.9999975	Binomial	BirdID
factor(fac.yr)20	-74.45490	23726567	-3.1e-06	0.9999975	Binomial	BirdID
(Intercept)1	56.03655	23726564	2.4e-06	0.9999981	Binomial	Terr
factor(fac.yr)21	-54.38065	23726564	-2.3e-06	0.9999982	Binomial	Terr
factor(fac.yr)31	-55.23002	23726564	-2.3e-06	0.9999981	Binomial	Terr
factor(fac.yr)41	-55.64529	23726564	-2.3e-06	0.9999981	Binomial	Terr
factor(fac.yr)51	-53.80087	23726564	-2.3e-06	0.9999982	Binomial	Terr
factor(fac.yr)61	-54.56014	23726564	-2.3e-06	0.9999982	Binomial	Terr
factor(fac.yr)71	-56.46749	23726564	-2.4e-06	0.9999981	Binomial	Terr
factor(fac.yr)81	-55.17706	23726564	-2.3e-06	0.9999981	Binomial	Terr
factor(fac.yr)91	-55.92755	23726564	-2.4e-06	0.9999981	Binomial	Terr
factor(fac.yr)101	-54.96502	23726564	-2.3e-06	0.9999982	Binomial	Terr
factor(fac.yr)111	-56.35561	23726564	-2.4e-06	0.9999981	Binomial	Terr
factor(fac.yr)121	-56.37030	23726564	-2.4e-06	0.9999981	Binomial	Terr
factor(fac.yr)131	-55.68259	23726564	-2.3e-06	0.9999981	Binomial	Terr
factor(fac.yr)141	-56.07969	23726564	-2.4e-06	0.9999981	Binomial	Terr
factor(fac.yr)151	-56.31356	23726564	-2.4e-06	0.9999981	Binomial	Terr
factor(fac.yr)161	-54.34620	23726564	-2.3e-06	0.9999982	Binomial	Terr
factor(fac.yr)171	-56.49454	23726564	-2.4e-06	0.9999981	Binomial	Terr
factor(fac.yr)181	-55.53546	23726564	-2.3e-06	0.9999981	Binomial	Terr
factor(fac.yr)191	-56.08701	23726564	-2.4e-06	0.9999981	Binomial	Terr
factor(fac.yr)201	-55.58648	23726564	-2.3e-06	0.9999981	Binomial	Terr
(Intercept)2	61.08352	23726570	2.6e-06	0.9999979	Binomial	BirdID+Terr
factor(fac.yr)22	-59.44696	23726570	-2.5e-06	0.9999980	Binomial	BirdID+Terr
factor(fac.yr)32	-60.19491	23726570	-2.5e-06	0.9999980	Binomial	BirdID+Terr
factor(fac.yr)42	-60.49426	23726570	-2.5e-06	0.9999980	Binomial	BirdID+Terr
factor(fac.yr)52	-58.73374	23726570	-2.5e-06	0.9999980	Binomial	BirdID+Terr
factor(fac.yr)62	-59.50276	23726570	-2.5e-06	0.9999980	Binomial	BirdID+Terr
factor(fac.yr)72	-61.37542	23726570	-2.6e-06	0.9999979	Binomial	BirdID+Terr
factor(fac.yr)82	-60.14259	23726570	-2.5e-06	0.9999980	Binomial	BirdID+Terr
factor(fac.yr)92	-60.99421	23726570	-2.6e-06	0.9999979	Binomial	BirdID+Terr
factor(fac.yr)102	-59.97527	23726570	-2.5e-06	0.9999980	Binomial	BirdID+Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)112	-61.79602	23726570	-2.6e-06	0.9999979	Binomial	BirdID+Terr
factor(fac.yr)122	-61.58731	23726570	-2.6e-06	0.9999979	Binomial	BirdID+Terr
factor(fac.yr)132	-60.79996	23726570	-2.6e-06	0.9999980	Binomial	BirdID+Terr
factor(fac.yr)142	-61.23346	23726570	-2.6e-06	0.9999979	Binomial	BirdID+Terr
factor(fac.yr)152	-61.58942	23726570	-2.6e-06	0.9999979	Binomial	BirdID+Terr
factor(fac.yr)162	-59.49465	23726570	-2.5e-06	0.9999980	Binomial	BirdID+Terr
factor(fac.yr)172	-61.90988	23726570	-2.6e-06	0.9999979	Binomial	BirdID+Terr
factor(fac.yr)182	-60.41071	23726570	-2.5e-06	0.9999980	Binomial	BirdID+Terr
factor(fac.yr)192	-61.02484	23726570	-2.6e-06	0.9999979	Binomial	BirdID+Terr
factor(fac.yr)202	-60.28277	23726570	-2.5e-06	0.9999980	Binomial	BirdID+Terr

Interestingly enough, the band ID only model had the lowest AIC and BIC. However, it did also have the lowest adjusted R^2 . All smoothers were significant. Note that the QQ plot for the model fitting both random effects may suggest model assumptions are not met.

Probabilites used to generate simulation 5

Simulation 1

Probability of mate fidelity: 0.373

Probability of pair dispersal: 0.655

Probability of single male dispersal: 0.667

Probability of single female dispersal: 0.031

NULL

Simulation 6

Number of fledgling as the response (Poisson)

Table 46: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
57.83014	-386.0517	892.0768	1117.181	317.1199	257.1699	315	0.2329513	143	Poisson	BirdID
70.45428	-371.1322	895.0555	1181.735	287.2809	244.5457	315	0.2970742	254	Poisson	BirdID+Terr
53.86696	-392.4486	897.1070	1107.645	329.9137	261.1330	315	0.1915766	131	Poisson	Terr

Table 47: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	37.83014	122	61.37710	0.0001462	Poisson	BirdID
s(Terrs)	33.86696	110	52.08958	0.0008209	Poisson	Terr
s(birdID)	29.51394	122	46.05121	0.0053127	Poisson	BirdID+Terr
s(Terrs)	20.94034	110	29.66448	0.0409291	Poisson	BirdID+Terr

Table 48: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.5525951	0.2359988	2.3415167	0.0192056	Poisson	BirdID
factor(fac.yr)2	-0.8723431	0.3834729	-2.2748498	0.0229149	Poisson	BirdID
factor(fac.yr)3	-0.6932599	0.3447534	-2.0108865	0.0443374	Poisson	BirdID
factor(fac.yr)4	-0.4900409	0.3197320	-1.5326618	0.1253592	Poisson	BirdID
factor(fac.yr)5	-0.1843511	0.2990273	-0.6165025	0.5375629	Poisson	BirdID
factor(fac.yr)6	-0.3660531	0.3210810	-1.1400648	0.2542593	Poisson	BirdID
factor(fac.yr)7	-1.0067617	0.3520226	-2.8599347	0.0042373	Poisson	BirdID
factor(fac.yr)8	-0.2481506	0.3177437	-0.7809772	0.4348159	Poisson	BirdID
factor(fac.yr)9	-0.2863020	0.3047866	-0.9393524	0.3475498	Poisson	BirdID
factor(fac.yr)10	-0.2024966	0.3163615	-0.6400798	0.5221207	Poisson	BirdID
factor(fac.yr)11	-0.3174189	0.3206216	-0.9900111	0.3221687	Poisson	BirdID
factor(fac.yr)12	-0.6557185	0.3409085	-1.9234444	0.0544243	Poisson	BirdID
factor(fac.yr)13	-0.4943461	0.3649710	-1.3544804	0.1755832	Poisson	BirdID
factor(fac.yr)14	-0.5163426	0.3542848	-1.4574224	0.1449998	Poisson	BirdID
factor(fac.yr)15	-1.1848084	0.4208924	-2.8149909	0.0048779	Poisson	BirdID
factor(fac.yr)16	-0.8559521	0.4267429	-2.0057795	0.0448798	Poisson	BirdID
factor(fac.yr)17	-0.9261913	0.4430427	-2.0905237	0.0365708	Poisson	BirdID
factor(fac.yr)18	-0.4199060	0.4683887	-0.8964905	0.3699908	Poisson	BirdID
factor(fac.yr)19	-2.4673797	1.0381665	-2.3766706	0.0174697	Poisson	BirdID
factor(fac.yr)20	0.0190718	0.5660818	0.0336909	0.9731236	Poisson	BirdID
(Intercept)1	0.5277017	0.2261858	2.3330446	0.0196458	Poisson	Terr
factor(fac.yr)21	-0.7312650	0.3842031	-1.9033290	0.0569976	Poisson	Terr
factor(fac.yr)31	-0.6430005	0.3396744	-1.8929905	0.0583591	Poisson	Terr
factor(fac.yr)41	-0.4343526	0.3100219	-1.4010386	0.1612025	Poisson	Terr
factor(fac.yr)51	-0.1287398	0.2887463	-0.4458578	0.6556999	Poisson	Terr
factor(fac.yr)61	-0.3873784	0.3049213	-1.2704213	0.2039346	Poisson	Terr
factor(fac.yr)71	-1.0333463	0.3405212	-3.0346014	0.0024085	Poisson	Terr
factor(fac.yr)81	-0.1833149	0.3050779	-0.6008789	0.5479206	Poisson	Terr
factor(fac.yr)91	-0.2191533	0.2850327	-0.7688705	0.4419702	Poisson	Terr
factor(fac.yr)101	-0.0079269	0.3001634	-0.0264088	0.9789313	Poisson	Terr
factor(fac.yr)111	-0.1550958	0.3037960	-0.5105260	0.6096830	Poisson	Terr
factor(fac.yr)121	-0.5965533	0.3266141	-1.8264776	0.0677784	Poisson	Terr
factor(fac.yr)131	-0.4797963	0.3563942	-1.3462519	0.1782213	Poisson	Terr
factor(fac.yr)141	-0.3356973	0.3432893	-0.9778848	0.3281313	Poisson	Terr
factor(fac.yr)151	-1.1694822	0.4061994	-2.8790841	0.0039883	Poisson	Terr
factor(fac.yr)161	-0.6380250	0.4127756	-1.5456946	0.1221783	Poisson	Terr
factor(fac.yr)171	-0.6933573	0.4343215	-1.5964149	0.1103962	Poisson	Terr
factor(fac.yr)181	-0.4250233	0.4654208	-0.9132021	0.3611362	Poisson	Terr
factor(fac.yr)191	-2.5419111	1.0289931	-2.4702899	0.0135004	Poisson	Terr
factor(fac.yr)201	-0.1088088	0.5345417	-0.2035554	0.8387010	Poisson	Terr
(Intercept)2	0.5139199	0.2431208	2.1138459	0.0345284	Poisson	BirdID+Terr
factor(fac.yr)22	-0.8052172	0.3946136	-2.0405204	0.0412985	Poisson	BirdID+Terr
factor(fac.yr)32	-0.6677077	0.3549044	-1.8813736	0.0599211	Poisson	BirdID+Terr
factor(fac.yr)42	-0.4635654	0.3276860	-1.4146634	0.1571672	Poisson	BirdID+Terr
factor(fac.yr)52	-0.1385871	0.3068207	-0.4516877	0.6514940	Poisson	BirdID+Terr
factor(fac.yr)62	-0.3496418	0.3263957	-1.0712207	0.2840702	Poisson	BirdID+Terr
factor(fac.yr)72	-0.9860501	0.3585698	-2.7499531	0.0059604	Poisson	BirdID+Terr
factor(fac.yr)82	-0.2088856	0.3262138	-0.6403333	0.5219559	Poisson	BirdID+Terr
factor(fac.yr)92	-0.2487437	0.3090989	-0.8047383	0.4209707	Poisson	BirdID+Terr
factor(fac.yr)102	-0.0981654	0.3224611	-0.3044257	0.7608036	Poisson	BirdID+Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)112	-0.2480391	0.3281638	-0.7558393	0.4497456	Poisson	BirdID+Terr
factor(fac.yr)122	-0.6255445	0.3477794	-1.7986817	0.0720690	Poisson	BirdID+Terr
factor(fac.yr)132	-0.4681849	0.3758335	-1.2457243	0.2128656	Poisson	BirdID+Terr
factor(fac.yr)142	-0.4161511	0.3648260	-1.1406837	0.2540016	Poisson	BirdID+Terr
factor(fac.yr)152	-1.1826571	0.4259638	-2.7764260	0.0054960	Poisson	BirdID+Terr
factor(fac.yr)162	-0.7874727	0.4358527	-1.8067404	0.0708028	Poisson	BirdID+Terr
factor(fac.yr)172	-0.8410322	0.4545489	-1.8502569	0.0642765	Poisson	BirdID+Terr
factor(fac.yr)182	-0.4255929	0.4850893	-0.8773496	0.3802968	Poisson	BirdID+Terr
factor(fac.yr)192	-2.4935834	1.0405308	-2.3964532	0.0165546	Poisson	BirdID+Terr
factor(fac.yr)202	0.0080111	0.5756238	0.0139173	0.9888960	Poisson	BirdID+Terr

The band ID only model has the best AIC, the territory only model has the best BIC, and the model fitting both random effects has the highest adjusted R^2 . The models are also performing somewhat similarly on the model selection criteria. All smoothers are significant.

Z-score number of fledgling (Gaussian)

Table 49: Z-score GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npars	Family	RE
77.67031	-	828.0592	1142.920	150.0130	237.3297	315	0.3272021	235	Gaussian	BandID+Terr
	330.1242									
41.29240	-	857.2860	1025.414	210.9819	273.7076	315	0.1795239	124	Gaussian	BandID
	383.8396									
37.19636	-	861.8425	1014.937	219.5706	277.8036	315	0.1587137	112	Gaussian	Terr
	390.1240									

Table 50: Z-score GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	40.29240	122	0.5631622	0.0003523	Gaussian	BandID
s(Terr)	36.19636	110	0.5385273	0.0011185	Gaussian	Terr
s(birdID)	40.73604	122	0.6921230	0.0007579	Gaussian	BandID+Terr
s(Terr)	35.93427	110	0.6856021	0.0013459	Gaussian	BandID+Terr

Table 51: Z-score GAMMs linear terms table.

	Estimate	Std. Error	t value	Pr(> t)	Family	RE
(Intercept)	-0.0181523	0.0652508	-0.2781930	0.7810742	Gaussian	BandID
(Intercept)1	0.0145099	0.0647665	0.2240337	0.8228958	Gaussian	Terr
(Intercept)2	-0.0066617	0.0739844	-0.0900414	0.9283302	Gaussian	BandID+Terr

The model fitting both random effects seems to perform the best on adjusted R^2 and AIC, but the worst on BIC, while the model only fitting territory as a random effect seems to perform well on BIC, but the worst on AIC and adjusted R^2 . All smoothers were significant.

Eggs (Binomial)

Table 52: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
69.04247	-136.76858	417.3798	687.2699	273.5372	245.9575	315	0.2934935	131	Binomial	Terr
113.57255	-85.74023	417.9065	880.2722	171.4805	201.4274	315	0.5479778	254	Binomial	BirdID+Terr
66.54343	-141.36713	421.9434	683.1395	282.7343	248.4566	315	0.2712443	143	Binomial	BirdID

Table 53: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	46.54343	121	70.38732	0.0007186	Binomial	BirdID
s(Terrs)	49.04247	109	75.76839	0.0004816	Binomial	Terr
s(birdID)	46.37830	121	80.39346	0.2395560	Binomial	BirdID+Terr
s(Terrs)	47.19425	109	82.68110	0.2636458	Binomial	BirdID+Terr

Table 54: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	1.5765414	8.580249e-01	1.8374075	0.0661498	Binomial	BirdID
factor(fac.yr)2	-1.1379280	1.055134e+00	-1.0784678	0.2808250	Binomial	BirdID
factor(fac.yr)3	-0.7741106	1.025214e+00	-0.7550724	0.4502055	Binomial	BirdID
factor(fac.yr)4	-1.2729350	9.989231e-01	-1.2743072	0.2025546	Binomial	BirdID
factor(fac.yr)5	-0.7027974	1.001895e+00	-0.7014679	0.4830110	Binomial	BirdID
factor(fac.yr)6	-0.8981968	1.010231e+00	-0.8891006	0.3739490	Binomial	BirdID
factor(fac.yr)7	-1.6405423	9.792746e-01	-1.6752628	0.0938827	Binomial	BirdID
factor(fac.yr)8	-1.1182413	1.025787e+00	-1.0901299	0.2756559	Binomial	BirdID
factor(fac.yr)9	-1.6989065	9.890327e-01	-1.7177455	0.0858430	Binomial	BirdID
factor(fac.yr)10	-1.0629962	1.040653e+00	-1.0214701	0.3070318	Binomial	BirdID
factor(fac.yr)11	-0.6012979	1.067053e+00	-0.5635127	0.5730858	Binomial	BirdID
factor(fac.yr)12	-1.8619821	1.014015e+00	-1.8362464	0.0663212	Binomial	BirdID
factor(fac.yr)13	-1.3434037	1.068832e+00	-1.2568898	0.2087935	Binomial	BirdID
factor(fac.yr)14	-0.5378318	1.099554e+00	-0.4891364	0.6247452	Binomial	BirdID
factor(fac.yr)15	-2.5044438	1.072379e+00	-2.3354087	0.0195221	Binomial	BirdID
factor(fac.yr)16	-1.4557716	1.155734e+00	-1.2596079	0.2078108	Binomial	BirdID
factor(fac.yr)17	-2.7168475	1.185279e+00	-2.2921585	0.0218965	Binomial	BirdID
factor(fac.yr)18	-2.2395012	1.312860e+00	-1.7058186	0.0880419	Binomial	BirdID
factor(fac.yr)19	-3.2025806	1.453321e+00	-2.2036288	0.0275505	Binomial	BirdID
factor(fac.yr)20	40.2297120	3.874532e+07	0.0000010	0.9999992	Binomial	BirdID
(Intercept)1	1.7511590	8.552829e-01	2.0474617	0.0406128	Binomial	Terr
factor(fac.yr)21	-0.6963390	1.069077e+00	-0.6513460	0.5148231	Binomial	Terr
factor(fac.yr)31	-0.9855328	1.037564e+00	-0.9498523	0.3421873	Binomial	Terr
factor(fac.yr)41	-1.4876754	9.859682e-01	-1.5088472	0.1313378	Binomial	Terr
factor(fac.yr)51	-0.8450511	9.819793e-01	-0.8605590	0.3894810	Binomial	Terr
factor(fac.yr)61	-1.1344474	9.906050e-01	-1.1452066	0.2521236	Binomial	Terr
factor(fac.yr)71	-1.8466920	9.694479e-01	-1.9048904	0.0567943	Binomial	Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)81	-1.0181225	1.018509e+00	-0.9996202	0.3174943	Binomial	Terr
factor(fac.yr)91	-1.6585426	9.637697e-01	-1.7208909	0.0852706	Binomial	Terr
factor(fac.yr)101	-0.5347617	1.013119e+00	-0.5278370	0.5976125	Binomial	Terr
factor(fac.yr)111	-0.1367779	1.064436e+00	-0.1284981	0.8977548	Binomial	Terr
factor(fac.yr)121	-2.0123814	9.914922e-01	-2.0296492	0.0423922	Binomial	Terr
factor(fac.yr)131	-1.6574725	1.077269e+00	-1.5385871	0.1239051	Binomial	Terr
factor(fac.yr)141	-0.2557361	1.097857e+00	-0.2329412	0.8158070	Binomial	Terr
factor(fac.yr)151	-2.7965443	1.064053e+00	-2.6282002	0.0085838	Binomial	Terr
factor(fac.yr)161	-1.0761347	1.122802e+00	-0.9584364	0.3378427	Binomial	Terr
factor(fac.yr)171	-2.4118362	1.168251e+00	-2.0644849	0.0389718	Binomial	Terr
factor(fac.yr)181	-2.4594250	1.323993e+00	-1.8575810	0.0632285	Binomial	Terr
factor(fac.yr)191	-3.9674962	1.483223e+00	-2.6749157	0.0074748	Binomial	Terr
factor(fac.yr)201	35.6210336	3.874532e+07	0.0000009	0.9999993	Binomial	Terr
(Intercept)2	1.4067915	1.032281e+00	1.3627987	0.1729460	Binomial	BirdID+Terr
factor(fac.yr)22	-0.5934093	1.282430e+00	-0.4627225	0.6435633	Binomial	BirdID+Terr
factor(fac.yr)32	-0.4337508	1.261583e+00	-0.3438147	0.7309857	Binomial	BirdID+Terr
factor(fac.yr)42	-1.2024991	1.205654e+00	-0.9973836	0.3185784	Binomial	BirdID+Terr
factor(fac.yr)52	-0.3515526	1.200427e+00	-0.2928564	0.7696319	Binomial	BirdID+Terr
factor(fac.yr)62	-0.5531812	1.229256e+00	-0.4500130	0.6527011	Binomial	BirdID+Terr
factor(fac.yr)72	-1.3913856	1.193797e+00	-1.1655123	0.2438117	Binomial	BirdID+Terr
factor(fac.yr)82	-0.7450738	1.259043e+00	-0.5917780	0.5539993	Binomial	BirdID+Terr
factor(fac.yr)92	-1.4029161	1.184643e+00	-1.1842524	0.2363132	Binomial	BirdID+Terr
factor(fac.yr)102	-0.3430272	1.236649e+00	-0.2773845	0.7814849	Binomial	BirdID+Terr
factor(fac.yr)112	0.0058345	1.299063e+00	0.0044913	0.9964165	Binomial	BirdID+Terr
factor(fac.yr)122	-1.7527611	1.235987e+00	-1.4181061	0.1561598	Binomial	BirdID+Terr
factor(fac.yr)132	-0.9590818	1.338161e+00	-0.7167162	0.4735492	Binomial	BirdID+Terr
factor(fac.yr)142	0.1636732	1.361307e+00	0.1202324	0.9042991	Binomial	BirdID+Terr
factor(fac.yr)152	-2.7581780	1.318785e+00	-2.0914530	0.0364875	Binomial	BirdID+Terr
factor(fac.yr)162	-0.9343093	1.436029e+00	-0.6506200	0.5152918	Binomial	BirdID+Terr
factor(fac.yr)172	-2.2836217	1.467169e+00	-1.5564822	0.1195935	Binomial	BirdID+Terr
factor(fac.yr)182	-2.0858312	1.623903e+00	-1.2844555	0.1989826	Binomial	BirdID+Terr
factor(fac.yr)192	-3.7394820	1.846729e+00	-2.0249217	0.0428754	Binomial	BirdID+Terr
factor(fac.yr)202	40.4784358	3.874532e+07	0.0000010	0.9999992	Binomial	BirdID+Terr

The territory only model has the best AIC, while the band ID only model has the best BIC. The QQ plot for the model fitting both random effects suggested the model assumptions were not met, so I would not consider using that model. The smoothers are significant in their respective models.

Probabilites used to generate simulation 6

Simulation 1

Probability of mate fidelity: 0.451

Probability of pair dispersal: 0.852

Probability of single male dispersal: 0.418

Probability of single female dispersal: 0.282

NULL

Once again, we had low a probability of single female dispersal and it was harder to distinguish the territory effect.

Simulation 7

Number of fledgling as the response (Poisson)

Table 55: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npars	Family	RE
51.56241	-388.9445	885.2220	1085.577	279.2227	257.4376	309	0.2598186	132	Poisson	Terr
51.56416	-388.9430	885.2343	1085.618	279.2197	257.4358	309	0.2598251	237	Poisson	BirdID+Terr
25.22733	-423.2262	906.5744	1018.802	347.7862	283.7727	309	0.0837182	125	Poisson	BirdID

Table 56: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	5.2273298	104	5.6526577	0.2956685	Poisson	BirdID
s(Terrs)	31.5624135	111	48.1782933	0.0009673	Poisson	Terr
s(birdID)	0.0019479	104	0.0016882	0.7692439	Poisson	BirdID+Terr
s(Terrs)	31.5622121	111	48.1778620	0.0009676	Poisson	BirdID+Terr

Table 57: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.1136182	0.3188786	0.3563054	0.7216119	Poisson	BirdID
factor(fac.yr)2	0.2678168	0.3930375	0.6814028	0.4956167	Poisson	BirdID
factor(fac.yr)3	-0.1128599	0.4051308	-0.2785765	0.7805698	Poisson	BirdID
factor(fac.yr)4	0.0724505	0.3689532	0.1963677	0.8443224	Poisson	BirdID
factor(fac.yr)5	0.5514390	0.3519429	1.5668420	0.1171516	Poisson	BirdID
factor(fac.yr)6	0.2306278	0.3551673	0.6493497	0.5161124	Poisson	BirdID
factor(fac.yr)7	-0.5151152	0.4054587	-1.2704504	0.2039242	Poisson	BirdID
factor(fac.yr)8	-0.1469699	0.3747997	-0.3921292	0.6949627	Poisson	BirdID
factor(fac.yr)9	0.3572730	0.3594979	0.9938111	0.3203149	Poisson	BirdID
factor(fac.yr)10	0.3192178	0.3628357	0.8797861	0.3789752	Poisson	BirdID
factor(fac.yr)11	-0.3921183	0.4234679	-0.9259692	0.3544620	Poisson	BirdID
factor(fac.yr)12	0.2891056	0.3796247	0.7615563	0.4463249	Poisson	BirdID
factor(fac.yr)13	0.2887846	0.3878128	0.7446495	0.4564837	Poisson	BirdID
factor(fac.yr)14	0.0029525	0.4509851	0.0065467	0.9947765	Poisson	BirdID
factor(fac.yr)15	-1.0963051	0.6610106	-1.6585288	0.0972108	Poisson	BirdID
factor(fac.yr)16	-0.2698772	0.5201244	-0.5188705	0.6038511	Poisson	BirdID
factor(fac.yr)17	0.2465980	0.4246539	0.5807034	0.5614403	Poisson	BirdID
factor(fac.yr)18	-0.6996491	0.5508411	-1.2701469	0.2040323	Poisson	BirdID
factor(fac.yr)19	-0.1132747	0.4967723	-0.2280214	0.8196296	Poisson	BirdID
factor(fac.yr)20	-0.0952672	0.6634994	-0.1435830	0.8858298	Poisson	BirdID
(Intercept)1	0.1223092	0.3350081	0.3650933	0.7150418	Poisson	Terr
factor(fac.yr)21	0.2015870	0.4119372	0.4893635	0.6245844	Poisson	Terr
factor(fac.yr)31	-0.0793378	0.4242849	-0.1869918	0.8516671	Poisson	Terr
factor(fac.yr)41	0.0132182	0.3837146	0.0344481	0.9725198	Poisson	Terr
factor(fac.yr)51	0.5591952	0.3724584	1.5013629	0.1332617	Poisson	Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)61	0.2224140	0.3713687	0.5989034	0.5492373	Poisson	Terr
factor(fac.yr)71	-0.4615355	0.4220856	-1.0934641	0.2741901	Poisson	Terr
factor(fac.yr)81	-0.1373731	0.3889999	-0.3531443	0.7239803	Poisson	Terr
factor(fac.yr)91	0.3705945	0.3780932	0.9801671	0.3270036	Poisson	Terr
factor(fac.yr)101	0.2871580	0.3821183	0.7514897	0.4523580	Poisson	Terr
factor(fac.yr)111	-0.3570694	0.4407266	-0.8101835	0.4178347	Poisson	Terr
factor(fac.yr)121	0.2835922	0.3984874	0.7116717	0.4766681	Poisson	Terr
factor(fac.yr)131	0.3380802	0.4019394	0.8411222	0.4002795	Poisson	Terr
factor(fac.yr)141	0.0535730	0.4702860	0.1139157	0.9093046	Poisson	Terr
factor(fac.yr)151	-1.1248017	0.6769621	-1.6615432	0.0966044	Poisson	Terr
factor(fac.yr)161	-0.2059128	0.5416719	-0.3801430	0.7038393	Poisson	Terr
factor(fac.yr)171	0.1386401	0.4477401	0.3096442	0.7568315	Poisson	Terr
factor(fac.yr)181	-0.6766560	0.5687669	-1.1896895	0.2341685	Poisson	Terr
factor(fac.yr)191	-0.0743335	0.5155152	-0.1441927	0.8853483	Poisson	Terr
factor(fac.yr)201	-0.0541135	0.6938537	-0.0779898	0.9378362	Poisson	Terr
(Intercept)2	0.1223141	0.3350092	0.3651068	0.7150317	Poisson	BirdID+Terr
factor(fac.yr)22	0.2015833	0.4119381	0.4893534	0.6245915	Poisson	BirdID+Terr
factor(fac.yr)32	-0.0793430	0.4242857	-0.1870037	0.8516577	Poisson	BirdID+Terr
factor(fac.yr)42	0.0132106	0.3837156	0.0344282	0.9725357	Poisson	BirdID+Terr
factor(fac.yr)52	0.5591882	0.3724594	1.5013400	0.1332676	Poisson	BirdID+Terr
factor(fac.yr)62	0.2224078	0.3713697	0.5988851	0.5492495	Poisson	BirdID+Terr
factor(fac.yr)72	-0.4615396	0.4220867	-1.0934711	0.2741870	Poisson	BirdID+Terr
factor(fac.yr)82	-0.1373787	0.3890010	-0.3531578	0.7239702	Poisson	BirdID+Terr
factor(fac.yr)92	0.3705864	0.3780945	0.9801422	0.3270160	Poisson	BirdID+Terr
factor(fac.yr)102	0.2871521	0.3821196	0.7514719	0.4523687	Poisson	BirdID+Terr
factor(fac.yr)112	-0.3570782	0.4407278	-0.8102011	0.4178246	Poisson	BirdID+Terr
factor(fac.yr)122	0.2835859	0.3984887	0.7116534	0.4766794	Poisson	BirdID+Terr
factor(fac.yr)132	0.3380750	0.4019410	0.8411062	0.4002884	Poisson	BirdID+Terr
factor(fac.yr)142	0.0535708	0.4702876	0.1139107	0.9093086	Poisson	BirdID+Terr
factor(fac.yr)152	-1.1248073	0.6769632	-1.6615486	0.0966033	Poisson	BirdID+Terr
factor(fac.yr)162	-0.2059198	0.5416736	-0.3801548	0.7038305	Poisson	BirdID+Terr
factor(fac.yr)172	0.1386332	0.4477418	0.3096275	0.7568442	Poisson	BirdID+Terr
factor(fac.yr)182	-0.6766606	0.5687683	-1.1896946	0.2341665	Poisson	BirdID+Terr
factor(fac.yr)192	-0.0743416	0.5155169	-0.1442078	0.8853364	Poisson	BirdID+Terr
factor(fac.yr)202	-0.0541097	0.6938558	-0.0779840	0.9378408	Poisson	BirdID+Terr

The territory only model and the model fitting both random effects are performing extremely similarly on the model selection criteria, having a better AIC and adjusted R^2 than the band ID only model. However, the band ID only model has a better BIC score. Only territory smoothers were significant.

Z-score number of fledgling (Gaussian)

Table 58: Z-score GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
54.120173	-	814.2356	1027.0853	174.4282	254.8798	309	0.2706532	113	Gaussian	Terr
	350.1046									
61.695278	-	824.6754	1093.5914	163.7101	247.3047	309	0.2945017	218	Gaussian	BandID+Terr
	340.3068									

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
8.942851	-	871.6583	931.9916	273.6268	300.0571	309	0.0281314	106	Gaussian	BandID
	419.6685									

Table 59: Z-score GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	7.942851	104	0.0857315	0.2487951	Gaussian	BandID
s(Terrs)	53.120173	111	1.0296920	0.0000013	Gaussian	Terr
s(birdID)	7.564551	104	0.0878641	0.2723184	Gaussian	BandID+Terr
s(Terrs)	53.130727	111	1.0694568	0.0000017	Gaussian	BandID+Terr

Table 60: Z-score GAMMs linear terms table.

	Estimate	Std. Error	t value	Pr(> t)	Family	RE
(Intercept)	-0.0009783	0.0576732	-0.0169635	0.9864770	Gaussian	BandID
(Intercept)1	0.0539235	0.0712730	0.7565770	0.4500021	Gaussian	Terr
(Intercept)2	0.0517676	0.0730840	0.7083302	0.4794073	Gaussian	BandID+Terr

I would say something similar to what I said for the previous batch of models.

Eggs (Binomial)

Table 61: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
56.22825	-	404.8263	627.6943	285.4329	252.7718	309	0.2049814	132	Binomial	Terr
	142.7165									
40.94778	-	412.4313	579.2224	323.0792	268.0522	309	0.1151714	125	Binomial	BirdID
	161.5396									
76.11652	-	413.2617	731.4728	242.7918	232.8835	309	0.3074378	237	Binomial	BirdID+Terr
	121.3959									

Table 62: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	20.94778	104	26.95962	0.0465855	Binomial	BirdID
s(Terrs)	36.22825	111	51.63074	0.0049726	Binomial	Terr
s(birdID)	19.91996	104	27.00995	0.1455827	Binomial	BirdID+Terr
s(Terrs)	36.19656	111	51.60757	0.0435861	Binomial	BirdID+Terr

Table 63: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	1.4057359	0.8360261	1.6814498	0.0926756	Binomial	BirdID
factor(fac.yr)2	-0.8552948	1.0169630	-0.8410285	0.4003320	Binomial	BirdID
factor(fac.yr)3	-0.5292226	0.9962827	-0.5311972	0.5952821	Binomial	BirdID
factor(fac.yr)4	-0.3095131	0.9596634	-0.3225225	0.7470569	Binomial	BirdID
factor(fac.yr)5	0.9114874	1.1152481	0.8172956	0.4137595	Binomial	BirdID
factor(fac.yr)6	-0.4545582	0.9361286	-0.4855724	0.6272704	Binomial	BirdID
factor(fac.yr)7	-1.7216291	0.9390608	-1.8333521	0.0667502	Binomial	BirdID
factor(fac.yr)8	-0.9684348	0.9298222	-1.0415268	0.2976311	Binomial	BirdID
factor(fac.yr)9	-0.8110443	0.9542070	-0.8499668	0.3953436	Binomial	BirdID
factor(fac.yr)10	-0.6407797	0.9642323	-0.6645491	0.5063389	Binomial	BirdID
factor(fac.yr)11	-0.5442598	1.0040344	-0.5420729	0.5877683	Binomial	BirdID
factor(fac.yr)12	0.1191756	1.0678717	0.1116011	0.9111397	Binomial	BirdID
factor(fac.yr)13	-0.4478421	1.0442545	-0.4288629	0.6680230	Binomial	BirdID
factor(fac.yr)14	-1.0229593	1.1042812	-0.9263575	0.3542602	Binomial	BirdID
factor(fac.yr)15	-2.5513612	1.2050286	-2.1172620	0.0342376	Binomial	BirdID
factor(fac.yr)16	-1.1109670	1.1768730	-0.9439991	0.3451701	Binomial	BirdID
factor(fac.yr)17	-1.3145570	1.1073104	-1.1871621	0.2351637	Binomial	BirdID
factor(fac.yr)18	-1.6778030	1.1060823	-1.5168880	0.1292950	Binomial	BirdID
factor(fac.yr)19	-1.1495132	1.1736408	-0.9794421	0.3273616	Binomial	BirdID
factor(fac.yr)20	-0.5205236	1.5442733	-0.3370670	0.7360664	Binomial	BirdID
(Intercept)1	1.3178496	0.8701212	1.5145586	0.1298842	Binomial	Terr
factor(fac.yr)21	-0.9424007	1.0685401	-0.8819517	0.3778029	Binomial	Terr
factor(fac.yr)31	-0.3272563	1.0480685	-0.3122470	0.7548528	Binomial	Terr
factor(fac.yr)41	-0.1934127	1.0050347	-0.1924438	0.8473946	Binomial	Terr
factor(fac.yr)51	1.2444126	1.1631813	1.0698355	0.2846934	Binomial	Terr
factor(fac.yr)61	-0.1859452	0.9742335	-0.1908630	0.8486329	Binomial	Terr
factor(fac.yr)71	-1.3797597	0.9857746	-1.3996705	0.1616120	Binomial	Terr
factor(fac.yr)81	-0.7935596	0.9665771	-0.8209998	0.4116464	Binomial	Terr
factor(fac.yr)91	-0.5287509	0.9936172	-0.5321475	0.5946239	Binomial	Terr
factor(fac.yr)101	-0.4318119	1.0078232	-0.4284600	0.6683163	Binomial	Terr
factor(fac.yr)111	-0.4407926	1.0492881	-0.4200873	0.6744217	Binomial	Terr
factor(fac.yr)121	0.2900531	1.1075782	0.2618804	0.7934136	Binomial	Terr
factor(fac.yr)131	-0.2244683	1.0617630	-0.2114109	0.8325666	Binomial	Terr
factor(fac.yr)141	-0.8134902	1.1341854	-0.7172463	0.4732222	Binomial	Terr
factor(fac.yr)151	-2.4774595	1.2606169	-1.9652754	0.0493824	Binomial	Terr
factor(fac.yr)161	-0.9086985	1.2150978	-0.7478398	0.4545568	Binomial	Terr
factor(fac.yr)171	-1.1076186	1.1503188	-0.9628797	0.3356079	Binomial	Terr
factor(fac.yr)181	-1.5602593	1.1456592	-1.3618878	0.1732333	Binomial	Terr
factor(fac.yr)191	-0.9405524	1.2116665	-0.7762469	0.4376032	Binomial	Terr
factor(fac.yr)201	-0.2348964	1.6300601	-0.1441029	0.8854192	Binomial	Terr
(Intercept)2	1.4930875	0.9185602	1.6254651	0.1040635	Binomial	BirdID+Terr
factor(fac.yr)22	-1.0568696	1.1152074	-0.9476889	0.3432879	Binomial	BirdID+Terr
factor(fac.yr)32	-0.4409312	1.0967856	-0.4020214	0.6876683	Binomial	BirdID+Terr
factor(fac.yr)42	-0.3726298	1.0501515	-0.3548344	0.7227137	Binomial	BirdID+Terr
factor(fac.yr)52	1.0754401	1.2034304	0.8936455	0.3715116	Binomial	BirdID+Terr
factor(fac.yr)62	-0.3441173	1.0239598	-0.3360652	0.7368217	Binomial	BirdID+Terr
factor(fac.yr)72	-1.5402260	1.0420273	-1.4781052	0.1393796	Binomial	BirdID+Terr
factor(fac.yr)82	-0.9260250	1.0214883	-0.9065448	0.3646476	Binomial	BirdID+Terr
factor(fac.yr)92	-0.7106363	1.0544619	-0.6739326	0.5003542	Binomial	BirdID+Terr
factor(fac.yr)102	-0.5409948	1.0691816	-0.5059896	0.6128639	Binomial	BirdID+Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)112	-0.6228437	1.1084069	-0.5619270	0.5741658	Binomial	BirdID+Terr
factor(fac.yr)122	0.1845610	1.1709109	0.1576217	0.8747549	Binomial	BirdID+Terr
factor(fac.yr)132	-0.3669462	1.1282220	-0.3252429	0.7449973	Binomial	BirdID+Terr
factor(fac.yr)142	-0.8606015	1.2074955	-0.7127161	0.4760214	Binomial	BirdID+Terr
factor(fac.yr)152	-2.7867796	1.3617410	-2.0464829	0.0407089	Binomial	BirdID+Terr
factor(fac.yr)162	-1.0969579	1.3174276	-0.8326514	0.4050414	Binomial	BirdID+Terr
factor(fac.yr)172	-1.3996823	1.2353745	-1.1330025	0.2572132	Binomial	BirdID+Terr
factor(fac.yr)182	-1.7823080	1.2367913	-1.4410741	0.1495637	Binomial	BirdID+Terr
factor(fac.yr)192	-1.1542698	1.3039564	-0.8852058	0.3760456	Binomial	BirdID+Terr
factor(fac.yr)202	-0.1212833	1.7800614	-0.0681344	0.9456787	Binomial	BirdID+Terr

Each model has a criteria where they are the best. All smoothers are significant except for the band ID smoother in the model fitting both random effects.

Probabilites used to generate simulation 7

Simulation 1

Probability of mate fidelity: 0.731

Probability of pair dispersal: 0.925

Probability of single male dispersal: 0.877

Probability of single female dispersal: 0.84

NULL

Simulation 8

Number of fledgling as the response (Poisson)

Table 64: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npars	Family	RE
28.76565	-	1011.749	1141.878	372.0337	325.2343	354	0.2252475	133	Poisson	Terr
	472.2434									
30.16422	-	1012.539	1150.239	368.9101	323.8358	354	0.2302997	147	Poisson	BirdID
	470.6816									
33.64874	-	1025.645	1202.783	361.6314	320.3513	354	0.2442538	260	Poisson	BirdID+Terr
	467.0423									

Table 65: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	10.164216	126	11.526355	0.1921266	Poisson	BirdID
s(Terrs)	8.765650	112	9.805382	0.2105483	Poisson	Terr
s(birdID)	7.566370	126	8.354794	0.2547606	Poisson	BirdID+Terr
s(Terrs)	6.082372	112	6.597239	0.2869678	Poisson	BirdID+Terr

Table 66: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.7267193	0.1975176	3.6792635	0.0002339	Poisson	BirdID
factor(fac.yr)2	-0.7271017	0.2941563	-2.4718210	0.0134427	Poisson	BirdID
factor(fac.yr)3	-0.7801833	0.2942407	-2.6515136	0.0080132	Poisson	BirdID
factor(fac.yr)4	-0.3841820	0.2618221	-1.4673398	0.1422836	Poisson	BirdID
factor(fac.yr)5	0.1781451	0.2380434	0.7483724	0.4542356	Poisson	BirdID
factor(fac.yr)6	-0.6143969	0.2762838	-2.2237894	0.0261626	Poisson	BirdID
factor(fac.yr)7	-1.1441673	0.3081862	-3.7125840	0.0002052	Poisson	BirdID
factor(fac.yr)8	-0.6988664	0.2660410	-2.6269125	0.0086163	Poisson	BirdID
factor(fac.yr)9	-0.0438001	0.2425643	-0.1805709	0.8567044	Poisson	BirdID
factor(fac.yr)10	-0.8589171	0.2917007	-2.9445149	0.0032346	Poisson	BirdID
factor(fac.yr)11	-1.5315921	0.4064782	-3.7679560	0.0001646	Poisson	BirdID
factor(fac.yr)12	-0.2571014	0.2671373	-0.9624317	0.3358328	Poisson	BirdID
factor(fac.yr)13	-0.4312546	0.2894517	-1.4899017	0.1362501	Poisson	BirdID
factor(fac.yr)14	-0.5682440	0.3636325	-1.5626877	0.1181261	Poisson	BirdID
factor(fac.yr)15	-1.6659682	0.5396564	-3.0870906	0.0020213	Poisson	BirdID
factor(fac.yr)16	-0.7362373	0.3523891	-2.0892738	0.0366831	Poisson	BirdID
factor(fac.yr)17	0.0860740	0.2847849	0.3022423	0.7624673	Poisson	BirdID
factor(fac.yr)18	-0.6125282	0.3768215	-1.6255129	0.1040533	Poisson	BirdID
factor(fac.yr)19	-1.1139062	0.5417563	-2.0561021	0.0397727	Poisson	BirdID
factor(fac.yr)20	-0.9120188	0.4935187	-1.8479921	0.0646035	Poisson	BirdID
(Intercept)1	0.7455442	0.1967347	3.7895918	0.0001509	Poisson	Terr
factor(fac.yr)21	-0.7487155	0.2948125	-2.5396330	0.0110969	Poisson	Terr
factor(fac.yr)31	-0.7880209	0.2950031	-2.6712288	0.0075574	Poisson	Terr
factor(fac.yr)41	-0.3910880	0.2616486	-1.4947069	0.1349909	Poisson	Terr
factor(fac.yr)51	0.1703835	0.2385570	0.7142258	0.4750875	Poisson	Terr
factor(fac.yr)61	-0.6289242	0.2755958	-2.2820527	0.0224862	Poisson	Terr
factor(fac.yr)71	-1.1612914	0.3075242	-3.7762598	0.0001592	Poisson	Terr
factor(fac.yr)81	-0.7100477	0.2647757	-2.6816948	0.0073250	Poisson	Terr
factor(fac.yr)91	-0.0492639	0.2415876	-0.2039174	0.8384181	Poisson	Terr
factor(fac.yr)101	-0.8783798	0.2906729	-3.0218846	0.0025121	Poisson	Terr
factor(fac.yr)111	-1.5545651	0.4054012	-3.8346339	0.0001258	Poisson	Terr
factor(fac.yr)121	-0.2807151	0.2661425	-1.0547549	0.2915375	Poisson	Terr
factor(fac.yr)131	-0.4384389	0.2871969	-1.5266144	0.1268569	Poisson	Terr
factor(fac.yr)141	-0.5382372	0.3633417	-1.4813527	0.1385126	Poisson	Terr
factor(fac.yr)151	-1.6732840	0.5390033	-3.1044042	0.0019066	Poisson	Terr
factor(fac.yr)161	-0.7293519	0.3510363	-2.0777110	0.0377360	Poisson	Terr
factor(fac.yr)171	0.0648479	0.2839614	0.2283686	0.8193597	Poisson	Terr
factor(fac.yr)181	-0.6324341	0.3751270	-1.6859198	0.0918112	Poisson	Terr
factor(fac.yr)191	-1.1639798	0.5405423	-2.1533555	0.0312907	Poisson	Terr
factor(fac.yr)201	-0.9532835	0.4916323	-1.9390171	0.0524993	Poisson	Terr
(Intercept)2	0.7376096	0.1992747	3.7014723	0.0002144	Poisson	BirdID+Terr
factor(fac.yr)22	-0.7405811	0.2960791	-2.5012953	0.0123740	Poisson	BirdID+Terr
factor(fac.yr)32	-0.7870694	0.2962980	-2.6563435	0.0078993	Poisson	BirdID+Terr
factor(fac.yr)42	-0.3889635	0.2635959	-1.4756051	0.1400499	Poisson	BirdID+Terr
factor(fac.yr)52	0.1736044	0.2404182	0.7220932	0.4702372	Poisson	BirdID+Terr
factor(fac.yr)62	-0.6244763	0.2776952	-2.2487832	0.0245263	Poisson	BirdID+Terr
factor(fac.yr)72	-1.1593280	0.3094885	-3.7459482	0.0001797	Poisson	BirdID+Terr
factor(fac.yr)82	-0.7062562	0.2672969	-2.6422164	0.0082365	Poisson	BirdID+Terr
factor(fac.yr)92	-0.0495264	0.2442862	-0.2027392	0.8393389	Poisson	BirdID+Terr
factor(fac.yr)102	-0.8725951	0.2930195	-2.9779422	0.0029019	Poisson	BirdID+Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)112	-1.5430075	0.4075178	-3.7863562	0.0001529	Poisson	BirdID+Terr
factor(fac.yr)122	-0.2723822	0.2690785	-1.0122779	0.3114052	Poisson	BirdID+Terr
factor(fac.yr)132	-0.4372328	0.2903954	-1.5056463	0.1321580	Poisson	BirdID+Terr
factor(fac.yr)142	-0.5634024	0.3661629	-1.5386661	0.1238858	Poisson	BirdID+Terr
factor(fac.yr)152	-1.6792869	0.5409764	-3.1041780	0.0019081	Poisson	BirdID+Terr
factor(fac.yr)162	-0.7343903	0.3539467	-2.0748611	0.0379994	Poisson	BirdID+Terr
factor(fac.yr)172	0.0695882	0.2874624	0.2420777	0.8087199	Poisson	BirdID+Terr
factor(fac.yr)182	-0.6198039	0.3786037	-1.6370783	0.1016141	Poisson	BirdID+Terr
factor(fac.yr)192	-1.1393261	0.5436318	-2.0957679	0.0361028	Poisson	BirdID+Terr
factor(fac.yr)202	-0.9403570	0.4952239	-1.8988523	0.0575839	Poisson	BirdID+Terr

The territory only model performs the best, barely, on AIC and the best on BIC. It does have the lowest adjusted R^2 , but all adjusted R^2 s are somewhat similar. None of the smoothers are significant.

Z-score number of fledgling (Gaussian)

Table 67: Z-score GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
33.34893	-452.5190	980.5463	1126.628	267.2073	320.6511	354	0.1192680	128	Gaussian	BandID
50.08688	-431.8929	981.1019	1208.067	237.8149	303.9131	354	0.1729765	241	Gaussian	BandID+Terr
21.49410	-468.6049	989.7562	1091.415	292.6288	332.5059	354	0.0698650	114	Gaussian	Terr

Table 68: Z-score GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	32.34893	126	0.3793898	0.0063449	Gaussian	BandID
s(Terrs)	20.49410	112	0.2367393	0.0522649	Gaussian	Terr
s(birdID)	31.32766	126	0.3810970	0.0090542	Gaussian	BandID+Terr
s(Terrs)	17.75922	112	0.2078123	0.0950966	Gaussian	BandID+Terr

Table 69: Z-score GAMMs linear terms table.

	Estimate	Std. Error	t value	Pr(> t)	Family	RE
(Intercept)	-0.0125132	0.0608184	-0.2057465	0.8371195	Gaussian	BandID
(Intercept)1	0.0053382	0.0566515	0.0942279	0.9249849	Gaussian	Terr
(Intercept)2	-0.0105577	0.0641773	-0.1645087	0.8694400	Gaussian	BandID+Terr

I would not consider the band ID only model here. When diagnostics were checked, model assumptions did not seem reasonable. The model fitting both random effects had the best AIC, the worst BIC, and the best adjusted R^2 . All smoothers were significant except the smoother on territory in the model that fit both random effects.

Eggs (Binomial)

Table 70: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
62.77260	-154.5142	441.4169	697.5420	309.0284	291.2274	354	0.2644016	147	Binomial	BirdID
76.24985	-140.4345	453.1152	786.3510	280.8690	277.7502	354	0.3225416	260	Binomial	BirdID+Terr
39.63076	-184.0211	455.6946	625.2713	368.0422	314.3692	354	0.1501219	133	Binomial	Terr

Table 71: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	42.77260	126	62.54200	0.0017791	Binomial	BirdID
s(Terrs)	19.63076	112	24.00872	0.0725215	Binomial	Terr
s(birdID)	42.42840	126	62.25403	0.0080718	Binomial	BirdID+Terr
s(Terrs)	13.82144	112	15.49444	0.4047294	Binomial	BirdID+Terr

Table 72: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	1.7566686	0.8386267	2.0946966	0.0361980	Binomial	BirdID
factor(fac.yr)2	-1.5046359	0.9605135	-1.5664912	0.1172337	Binomial	BirdID
factor(fac.yr)3	-1.4907509	0.9584851	-1.5553199	0.1198699	Binomial	BirdID
factor(fac.yr)4	-0.4162368	0.9981188	-0.4170214	0.6766628	Binomial	BirdID
factor(fac.yr)5	1.4945101	1.3524237	1.1050606	0.2691333	Binomial	BirdID
factor(fac.yr)6	-1.1329532	0.9636466	-1.1756936	0.2397173	Binomial	BirdID
factor(fac.yr)7	-2.2028241	0.9406692	-2.3417628	0.0191929	Binomial	BirdID
factor(fac.yr)8	-1.5833813	0.9312858	-1.7002099	0.0890915	Binomial	BirdID
factor(fac.yr)9	-0.0625295	1.0223972	-0.0611597	0.9512320	Binomial	BirdID
factor(fac.yr)10	-1.8599249	0.9519612	-1.9537822	0.0507270	Binomial	BirdID
factor(fac.yr)11	-2.6124036	1.0130625	-2.5787190	0.0099167	Binomial	BirdID
factor(fac.yr)12	-0.2729344	1.0479646	-0.2604424	0.7945225	Binomial	BirdID
factor(fac.yr)13	-0.5255603	1.0602295	-0.4957042	0.6201031	Binomial	BirdID
factor(fac.yr)14	-1.4293346	1.1517121	-1.2410519	0.2145866	Binomial	BirdID
factor(fac.yr)15	-2.3867769	1.1110033	-2.1483077	0.0316893	Binomial	BirdID
factor(fac.yr)16	-1.8048650	1.0655818	-1.6937837	0.0903064	Binomial	BirdID
factor(fac.yr)17	-0.3882539	1.2036574	-0.3225618	0.7470271	Binomial	BirdID
factor(fac.yr)18	-1.4839969	1.1411962	-1.3003871	0.1934683	Binomial	BirdID
factor(fac.yr)19	-1.0967904	1.2989321	-0.8443785	0.3984579	Binomial	BirdID
factor(fac.yr)20	-1.2529640	1.3006994	-0.9633002	0.3353969	Binomial	BirdID
(Intercept)1	1.7769837	0.7863650	2.2597441	0.0238371	Binomial	Terr
factor(fac.yr)21	-1.4585876	0.9075833	-1.6071114	0.1080299	Binomial	Terr
factor(fac.yr)31	-1.3897918	0.9050215	-1.5356450	0.1246255	Binomial	Terr
factor(fac.yr)41	-0.3676036	0.9361553	-0.3926737	0.6945605	Binomial	Terr
factor(fac.yr)51	1.3773556	1.2938694	1.0645244	0.2870912	Binomial	Terr
factor(fac.yr)61	-1.0464371	0.9003473	-1.1622594	0.2451301	Binomial	Terr
factor(fac.yr)71	-2.0286238	0.8808575	-2.3030101	0.0212783	Binomial	Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)81	-1.5631107	0.8680701	-1.8006734	0.0717544	Binomial	Terr
factor(fac.yr)91	-0.0599486	0.9604593	-0.0624166	0.9502311	Binomial	Terr
factor(fac.yr)101	-1.8664516	0.8864984	-2.1054203	0.0352547	Binomial	Terr
factor(fac.yr)111	-2.4490892	0.9390758	-2.6079782	0.0091079	Binomial	Terr
factor(fac.yr)121	-0.3742825	0.9744889	-0.3840808	0.7009185	Binomial	Terr
factor(fac.yr)131	-0.4989144	0.9769512	-0.5106850	0.6095716	Binomial	Terr
factor(fac.yr)141	-0.9628920	1.0819496	-0.8899602	0.3734873	Binomial	Terr
factor(fac.yr)151	-2.2421168	1.0344655	-2.1674158	0.0302032	Binomial	Terr
factor(fac.yr)161	-1.6175047	0.9868669	-1.6390303	0.1012069	Binomial	Terr
factor(fac.yr)171	-0.2344244	1.1270573	-0.2079968	0.8352314	Binomial	Terr
factor(fac.yr)181	-1.5374266	1.0529523	-1.4601104	0.1442597	Binomial	Terr
factor(fac.yr)191	-1.1423081	1.1930501	-0.9574686	0.3383308	Binomial	Terr
factor(fac.yr)201	-1.1999852	1.1916968	-1.0069552	0.3139563	Binomial	Terr
(Intercept)2	1.8107191	0.8597122	2.1061922	0.0351877	Binomial	BirdID+Terr
factor(fac.yr)22	-1.5457241	0.9870139	-1.5660611	0.1173343	Binomial	BirdID+Terr
factor(fac.yr)32	-1.5483418	0.9846993	-1.5724006	0.1158577	Binomial	BirdID+Terr
factor(fac.yr)42	-0.4326697	1.0209331	-0.4237983	0.6717129	Binomial	BirdID+Terr
factor(fac.yr)52	1.4698049	1.3752932	1.0687211	0.2851954	Binomial	BirdID+Terr
factor(fac.yr)62	-1.1745020	0.9871339	-1.1898102	0.2341210	Binomial	BirdID+Terr
factor(fac.yr)72	-2.2939802	0.9646589	-2.3780220	0.0174058	Binomial	BirdID+Terr
factor(fac.yr)82	-1.6183487	0.9535191	-1.6972379	0.0896517	Binomial	BirdID+Terr
factor(fac.yr)92	-0.0868512	1.0479181	-0.0828797	0.9339472	Binomial	BirdID+Terr
factor(fac.yr)102	-1.9388832	0.9768110	-1.9849113	0.0471543	Binomial	BirdID+Terr
factor(fac.yr)112	-2.6784443	1.0392419	-2.5773060	0.0099574	Binomial	BirdID+Terr
factor(fac.yr)122	-0.2926375	1.0781249	-0.2714319	0.7860589	Binomial	BirdID+Terr
factor(fac.yr)132	-0.5195812	1.0827464	-0.4798734	0.6313174	Binomial	BirdID+Terr
factor(fac.yr)142	-1.3685989	1.1912374	-1.1488884	0.2506020	Binomial	BirdID+Terr
factor(fac.yr)152	-2.5051814	1.1465266	-2.1850182	0.0288875	Binomial	BirdID+Terr
factor(fac.yr)162	-1.7731432	1.0954747	-1.6186071	0.1055318	Binomial	BirdID+Terr
factor(fac.yr)172	-0.4176529	1.2449176	-0.3354864	0.7372582	Binomial	BirdID+Terr
factor(fac.yr)182	-1.5076282	1.1783925	-1.2793939	0.2007584	Binomial	BirdID+Terr
factor(fac.yr)192	-1.2454011	1.3335259	-0.9339159	0.3503473	Binomial	BirdID+Terr
factor(fac.yr)202	-1.3714681	1.3403503	-1.0232161	0.3062057	Binomial	BirdID+Terr

The band ID model had the best AIC, the territory only model has the best BIC, and the model fitting both random effects had the highest adjusted R^2 . All smoothers were significant except for the territory smoother in the model that included both random effects.

Probabilities used to generate simulation 8

Simulation 1

Probability of mate fidelity: 0.808

Probability of pair dispersal: 0.801

Probability of single male dispersal: 0.091

Probability of single female dispersal: 0.763

NULL

Simulation 9

Number of fledgling as the response (Poisson)

Table 73: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
40.92434	-375.4242	838.0746	997.8278	303.8046	247.0757	288	0.1700878	126	Poisson	Terr
38.87475	-377.9086	839.3643	992.3790	308.7735	249.1253	288	0.1499008	127	Poisson	BirdID
46.50181	-369.2794	849.3665	1052.3084	291.5152	241.4982	288	0.1938037	233	Poisson	BirdID+Terr

Table 74: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	18.87475	106	24.99706	0.0314854	Poisson	BirdID
s(Terrs)	20.92434	105	28.00036	0.0202427	Poisson	Terr
s(birdID)	11.16873	106	13.54642	0.1414590	Poisson	BirdID+Terr
s(Terrs)	15.33308	105	19.39189	0.0714970	Poisson	BirdID+Terr

Table 75: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.7936424	0.2301210	3.4488045	0.0005631	Poisson	BirdID
factor(fac.yr)2	-0.6506356	0.3377016	-1.9266581	0.0540222	Poisson	BirdID
factor(fac.yr)3	-0.7878998	0.3539693	-2.2258986	0.0260210	Poisson	BirdID
factor(fac.yr)4	-0.4597330	0.3202040	-1.4357503	0.1510734	Poisson	BirdID
factor(fac.yr)5	-0.3796148	0.2932507	-1.2945063	0.1954906	Poisson	BirdID
factor(fac.yr)6	-0.3670504	0.2925137	-1.2548147	0.2095460	Poisson	BirdID
factor(fac.yr)7	-0.8770812	0.3181027	-2.7572264	0.0058294	Poisson	BirdID
factor(fac.yr)8	-0.8829190	0.3345206	-2.6393558	0.0083064	Poisson	BirdID
factor(fac.yr)9	-0.3184929	0.2896633	-1.0995282	0.2715377	Poisson	BirdID
factor(fac.yr)10	-0.6712299	0.3130024	-2.1444879	0.0319938	Poisson	BirdID
factor(fac.yr)11	-0.9407175	0.3395123	-2.7707902	0.0055920	Poisson	BirdID
factor(fac.yr)12	-0.8357506	0.3519269	-2.3747844	0.0175592	Poisson	BirdID
factor(fac.yr)13	-0.2704044	0.3123019	-0.8658430	0.3865763	Poisson	BirdID
factor(fac.yr)14	-0.9572940	0.3588008	-2.6680376	0.0076296	Poisson	BirdID
factor(fac.yr)15	-0.9735816	0.3982682	-2.4445376	0.0145038	Poisson	BirdID
factor(fac.yr)16	-0.2334350	0.4182831	-0.5580791	0.5767904	Poisson	BirdID
factor(fac.yr)17	-1.0202710	0.5630574	-1.8120195	0.0699832	Poisson	BirdID
factor(fac.yr)18	-0.8094824	0.4814350	-1.6813951	0.0926862	Poisson	BirdID
factor(fac.yr)19	-2.1794812	1.0345821	-2.1066296	0.0351497	Poisson	BirdID
factor(fac.yr)20	-2.1994807	1.0346369	-2.1258478	0.0335159	Poisson	BirdID
(Intercept)1	0.7855998	0.2296464	3.4209109	0.0006241	Poisson	Terr
factor(fac.yr)21	-0.6817364	0.3455741	-1.9727650	0.0485223	Poisson	Terr
factor(fac.yr)31	-0.8092369	0.3566175	-2.2692010	0.0232561	Poisson	Terr
factor(fac.yr)41	-0.3934681	0.3221657	-1.2213219	0.2219642	Poisson	Terr
factor(fac.yr)51	-0.3427923	0.2917501	-1.1749519	0.2400140	Poisson	Terr
factor(fac.yr)61	-0.3796293	0.2909794	-1.3046604	0.1920085	Poisson	Terr
factor(fac.yr)71	-0.8739585	0.3165370	-2.7609995	0.0057625	Poisson	Terr
factor(fac.yr)81	-0.8471533	0.3323245	-2.5491752	0.0107978	Poisson	Terr
factor(fac.yr)91	-0.3045567	0.2869540	-1.0613432	0.2885340	Poisson	Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)101	-0.6602455	0.3088364	-2.1378487	0.0325290	Poisson	Terr
factor(fac.yr)111	-0.9881478	0.3351207	-2.9486322	0.0031918	Poisson	Terr
factor(fac.yr)121	-0.8479170	0.3508738	-2.4165871	0.0156668	Poisson	Terr
factor(fac.yr)131	-0.2477391	0.3107320	-0.7972758	0.4252908	Poisson	Terr
factor(fac.yr)141	-0.9332148	0.3575015	-2.6103799	0.0090442	Poisson	Terr
factor(fac.yr)151	-1.0974267	0.3949654	-2.7785388	0.0054604	Poisson	Terr
factor(fac.yr)161	-0.4224734	0.4150435	-1.0179015	0.3087248	Poisson	Terr
factor(fac.yr)171	-0.9213107	0.5637223	-1.6343343	0.1021887	Poisson	Terr
factor(fac.yr)181	-0.7049979	0.4786854	-1.4727793	0.1408106	Poisson	Terr
factor(fac.yr)191	-2.1570188	1.0352442	-2.0835845	0.0371980	Poisson	Terr
factor(fac.yr)201	-2.1184869	1.0349437	-2.0469585	0.0406622	Poisson	Terr
(Intercept)2	0.7893684	0.2350959	3.3576436	0.0007861	Poisson	BirdID+Terr
factor(fac.yr)22	-0.6733244	0.3470084	-1.9403694	0.0523348	Poisson	BirdID+Terr
factor(fac.yr)32	-0.8039700	0.3601451	-2.2323501	0.0255918	Poisson	BirdID+Terr
factor(fac.yr)42	-0.4310121	0.3268508	-1.3186813	0.1872757	Poisson	BirdID+Terr
factor(fac.yr)52	-0.3664719	0.2981884	-1.2289943	0.2190739	Poisson	BirdID+Terr
factor(fac.yr)62	-0.3797798	0.2975464	-1.2763714	0.2018243	Poisson	BirdID+Terr
factor(fac.yr)72	-0.8757837	0.3226592	-2.7142686	0.0066422	Poisson	BirdID+Terr
factor(fac.yr)82	-0.8551552	0.3385320	-2.5260692	0.0115347	Poisson	BirdID+Terr
factor(fac.yr)92	-0.3212062	0.2941403	-1.0920171	0.2748256	Poisson	BirdID+Terr
factor(fac.yr)102	-0.6638035	0.3160731	-2.1001582	0.0357149	Poisson	BirdID+Terr
factor(fac.yr)112	-0.9695102	0.3414742	-2.8391901	0.0045228	Poisson	BirdID+Terr
factor(fac.yr)122	-0.8382400	0.3567431	-2.3497018	0.0187885	Poisson	BirdID+Terr
factor(fac.yr)132	-0.2558257	0.3175286	-0.8056776	0.4204288	Poisson	BirdID+Terr
factor(fac.yr)142	-0.9301295	0.3631754	-2.5611029	0.0104340	Poisson	BirdID+Terr
factor(fac.yr)152	-1.0369691	0.4013678	-2.5835878	0.0097779	Poisson	BirdID+Terr
factor(fac.yr)162	-0.3201074	0.4239822	-0.7550018	0.4502479	Poisson	BirdID+Terr
factor(fac.yr)172	-0.9439662	0.5702859	-1.6552508	0.0978736	Poisson	BirdID+Terr
factor(fac.yr)182	-0.7371885	0.4862349	-1.5161159	0.1294900	Poisson	BirdID+Terr
factor(fac.yr)192	-2.1561959	1.0390796	-2.0751017	0.0379771	Poisson	BirdID+Terr
factor(fac.yr)202	-2.1416261	1.0387859	-2.0616628	0.0392399	Poisson	BirdID+Terr

Each model has a criteria it performs the best in, but the territory only model does perform similarly as the best models in each of the criteria. All smoothers were significant, except for the band ID smoother in the model fitting both random effects.

Z-score number of fledgling (Gaussian)

Table 76: Z-score GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
22.46173	-372.7725	798.9221	896.6812	224.4789	265.5383	288	0.0946939	108	Gaussian	BandID
24.61153	-370.4688	799.1507	905.7668	220.9162	263.3885	288	0.1017899	107	Gaussian	Terr
38.87160	-352.6810	801.0025	976.1659	195.2454	249.1284	288	0.1607242	214	Gaussian	BandID+Terr

Table 77: Z-score GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	21.46173	106	0.2832060	0.0262224	Gaussian	BandID
s(Terrs)	23.61153	105	0.3097583	0.0272112	Gaussian	Terr
s(birdID)	18.31832	106	0.2439711	0.0508949	Gaussian	BandID+Terr
s(Terrs)	19.55328	105	0.2533957	0.0627882	Gaussian	BandID+Terr

Table 78: Z-score GAMMs linear terms table.

	Estimate	Std. Error	t value	Pr(> t)	Family	RE
(Intercept)	0.0144523	0.0642998	0.2247636	0.8223361	Gaussian	BandID
(Intercept)1	0.0071203	0.0632449	0.1125836	0.9104464	Gaussian	Terr
(Intercept)2	0.0169561	0.0678783	0.2498008	0.8029473	Gaussian	BandID+Terr

The band ID only model performs the best on AIC (barely) and BIC. However, the model fitting both random effects has a higher adjusted R^2 . All smoothers were significant.

Eggs (Binomial)

Table 79: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
68.64413	-114.45299	371.2496	631.9490	228.9060	219.3559	288	0.3254272	127	Binomial	BirdID
82.74206	-99.96884	383.9248	720.8936	199.9377	205.2579	288	0.4046252	233	Binomial	BirdID+Terr
54.62864	-136.95302	389.8008	602.0596	273.9060	233.3714	288	0.1972695	126	Binomial	Terr

Table 80: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	48.64413	106	79.83387	0.0001282	Binomial	BirdID
s(Terrs)	34.62864	105	49.05110	0.0057440	Binomial	Terr
s(birdID)	47.05881	106	77.24023	0.0051638	Binomial	BirdID+Terr
s(Terrs)	15.68325	105	19.17067	0.4281344	Binomial	BirdID+Terr

Table 81: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	1.4898512	0.9247683	1.6110534	0.1071681	Binomial	BirdID
factor(fac.yr)2	-0.3330578	1.0912049	-0.3052202	0.7601984	Binomial	BirdID
factor(fac.yr)3	-1.0481601	1.0788731	-0.9715324	0.3312832	Binomial	BirdID
factor(fac.yr)4	0.1618394	1.2088893	0.1338745	0.8935018	Binomial	BirdID
factor(fac.yr)5	-0.4267118	1.0923742	-0.3906279	0.6960723	Binomial	BirdID

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)6	-1.0550449	1.0816235	-0.9754272	0.3293484	Binomial	BirdID
factor(fac.yr)7	-1.2223136	1.0601609	-1.1529510	0.2489305	Binomial	BirdID
factor(fac.yr)8	-1.5858822	1.0853859	-1.4611229	0.1439817	Binomial	BirdID
factor(fac.yr)9	-0.9480343	1.0827271	-0.8755986	0.3812483	Binomial	BirdID
factor(fac.yr)10	-0.1280594	1.1149891	-0.1148526	0.9085619	Binomial	BirdID
factor(fac.yr)11	-0.7423736	1.0948971	-0.6780305	0.4977524	Binomial	BirdID
factor(fac.yr)12	0.1540812	1.1719364	0.1314758	0.8953990	Binomial	BirdID
factor(fac.yr)13	0.5327870	1.2233543	0.4355133	0.6631899	Binomial	BirdID
factor(fac.yr)14	-0.6988761	1.1328509	-0.6169180	0.5372888	Binomial	BirdID
factor(fac.yr)15	-1.3248421	1.1809498	-1.1218446	0.2619285	Binomial	BirdID
factor(fac.yr)16	1.4983054	1.5517485	0.9655594	0.3342647	Binomial	BirdID
factor(fac.yr)17	-2.0907028	1.4942910	-1.3991269	0.1617749	Binomial	BirdID
factor(fac.yr)18	-1.7216130	1.4005140	-1.2292723	0.2189697	Binomial	BirdID
factor(fac.yr)19	-2.6826057	1.6884890	-1.5887611	0.1121143	Binomial	BirdID
factor(fac.yr)20	-3.2765417	1.7249945	-1.8994505	0.0575053	Binomial	BirdID
(Intercept)1	1.3091637	0.8504136	1.5394436	0.1236960	Binomial	Terr
factor(fac.yr)21	-0.3752331	1.0618080	-0.3533907	0.7237955	Binomial	Terr
factor(fac.yr)31	-1.0982993	1.0278055	-1.0685867	0.2852559	Binomial	Terr
factor(fac.yr)41	0.5577627	1.1736085	0.4752545	0.6346056	Binomial	Terr
factor(fac.yr)51	-0.2912075	0.9930189	-0.2932547	0.7693275	Binomial	Terr
factor(fac.yr)61	-0.9462080	0.9749963	-0.9704734	0.3318106	Binomial	Terr
factor(fac.yr)71	-1.1521939	0.9671190	-1.1913673	0.2335094	Binomial	Terr
factor(fac.yr)81	-1.4071384	0.9810771	-1.4342791	0.1514926	Binomial	Terr
factor(fac.yr)91	-0.6696482	0.9813247	-0.6823921	0.4949911	Binomial	Terr
factor(fac.yr)101	-0.0926571	1.0102727	-0.0917150	0.9269245	Binomial	Terr
factor(fac.yr)111	-0.9226756	0.9794036	-0.9420790	0.3461522	Binomial	Terr
factor(fac.yr)121	-0.1749943	1.0577190	-0.1654450	0.8685937	Binomial	Terr
factor(fac.yr)131	0.4300668	1.0991358	0.3912772	0.6955923	Binomial	Terr
factor(fac.yr)141	-0.5869689	1.0089580	-0.5817575	0.5607300	Binomial	Terr
factor(fac.yr)151	-1.9391750	1.0541617	-1.8395423	0.0658355	Binomial	Terr
factor(fac.yr)161	0.3744583	1.4458467	0.2589889	0.7956438	Binomial	Terr
factor(fac.yr)171	-1.1597154	1.3177017	-0.8801047	0.3788026	Binomial	Terr
factor(fac.yr)181	-0.7848167	1.2402602	-0.6327839	0.5268748	Binomial	Terr
factor(fac.yr)191	-2.1901313	1.5230421	-1.4379978	0.1504347	Binomial	Terr
factor(fac.yr)201	-2.4319777	1.5142966	-1.6060114	0.1082714	Binomial	Terr
(Intercept)2	1.4609814	0.9689693	1.5077685	0.1316138	Binomial	BirdID+Terr
factor(fac.yr)22	-0.3324726	1.1396217	-0.2917395	0.7704858	Binomial	BirdID+Terr
factor(fac.yr)32	-1.0632361	1.1312798	-0.9398525	0.3472932	Binomial	BirdID+Terr
factor(fac.yr)42	0.2344854	1.2632368	0.1856227	0.8527407	Binomial	BirdID+Terr
factor(fac.yr)52	-0.3407305	1.1395400	-0.2990071	0.7649346	Binomial	BirdID+Terr
factor(fac.yr)62	-1.0236956	1.1328993	-0.9036069	0.3662039	Binomial	BirdID+Terr
factor(fac.yr)72	-1.1886109	1.1158921	-1.0651664	0.2868006	Binomial	BirdID+Terr
factor(fac.yr)82	-1.4678299	1.1390280	-1.2886688	0.1975132	Binomial	BirdID+Terr
factor(fac.yr)92	-0.8793468	1.1382448	-0.7725463	0.4397910	Binomial	BirdID+Terr
factor(fac.yr)102	-0.1129930	1.1641163	-0.0970633	0.9226761	Binomial	BirdID+Terr
factor(fac.yr)112	-0.7737386	1.1439690	-0.6763633	0.4988100	Binomial	BirdID+Terr
factor(fac.yr)122	0.2539974	1.2321234	0.2061461	0.8366768	Binomial	BirdID+Terr
factor(fac.yr)132	0.6924060	1.2780916	0.5417499	0.5879908	Binomial	BirdID+Terr
factor(fac.yr)142	-0.5173643	1.1857071	-0.4363340	0.6625944	Binomial	BirdID+Terr
factor(fac.yr)152	-1.3533684	1.2351116	-1.0957458	0.2731900	Binomial	BirdID+Terr
factor(fac.yr)162	1.5411789	1.6126711	0.9556685	0.3392398	Binomial	BirdID+Terr
factor(fac.yr)172	-1.8202147	1.5754894	-1.1553329	0.2479541	Binomial	BirdID+Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)182	-1.5007677	1.4804042	-1.0137554	0.3106995	Binomial	BirdID+Terr
factor(fac.yr)192	-2.4695722	1.7824965	-1.3854570	0.1659128	Binomial	BirdID+Terr
factor(fac.yr)202	-3.2941228	1.8177259	-1.8122219	0.0699519	Binomial	BirdID+Terr

Each model has a criteria that they are performing best in, and all smoothers are significant, except for the territory smoother in the model fitted with both random effects.

Probabilites used to generate simulation 9

Simulation 1

Probability of mate fidelity: 0.211

Probability of pair dispersal: 0.991

Probability of single male dispersal: 0.847

Probability of single female dispersal: 0.103

NULL

Simulation 10

Number of fledgling as the response (Poisson)

Table 82: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
42.38011	-	931.5994	1100.984	306.9512	285.6199	328	0.2093971	130	Poisson	Terr
	421.1427									
37.67382	-	936.1594	1090.030	319.6917	290.3262	328	0.1734980	137	Poisson	BirdID
	427.5129									
47.78662	-	941.7745	1152.568	295.2921	280.2134	328	0.2354425	247	Poisson	BirdID+Terr
	415.3131									

Table 83: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	17.673823	116	23.24267	0.0317289	Poisson	BirdID
s(Terrs)	22.380108	109	31.32496	0.0079152	Poisson	Terr
s(birdID)	9.164954	116	10.83590	0.1669198	Poisson	BirdID+Terr
s(Terrs)	18.621668	109	25.15272	0.0226676	Poisson	BirdID+Terr

Table 84: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	-0.1232260	0.2740042	-0.4497229	0.6529103	Poisson	BirdID
factor(fac.yr)2	0.1808001	0.3457420	0.5229336	0.6010205	Poisson	BirdID
factor(fac.yr)3	0.0319168	0.3442872	0.0927040	0.9261387	Poisson	BirdID
factor(fac.yr)4	0.2201802	0.3363905	0.6545375	0.5127656	Poisson	BirdID
factor(fac.yr)5	0.9273021	0.3113893	2.9779508	0.0029018	Poisson	BirdID

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)6	0.4020413	0.3341459	1.2031906	0.2289026	Poisson	BirdID
factor(fac.yr)7	0.0342426	0.3504234	0.0977178	0.9221564	Poisson	BirdID
factor(fac.yr)8	0.3239117	0.3432600	0.9436336	0.3453569	Poisson	BirdID
factor(fac.yr)9	0.7750127	0.3280991	2.3621296	0.0181703	Poisson	BirdID
factor(fac.yr)10	0.2215724	0.3454353	0.6414294	0.5212438	Poisson	BirdID
factor(fac.yr)11	-0.0800966	0.3747349	-0.2137419	0.8307483	Poisson	BirdID
factor(fac.yr)12	0.2401576	0.3663243	0.6555874	0.5120896	Poisson	BirdID
factor(fac.yr)13	0.5661384	0.3421259	1.6547663	0.0979719	Poisson	BirdID
factor(fac.yr)14	0.2533537	0.3515127	0.7207525	0.4710618	Poisson	BirdID
factor(fac.yr)15	-0.0078568	0.4240695	-0.0185273	0.9852182	Poisson	BirdID
factor(fac.yr)16	-0.2297570	0.4978679	-0.4614819	0.6444529	Poisson	BirdID
factor(fac.yr)17	0.4691065	0.3833141	1.2238176	0.2210210	Poisson	BirdID
factor(fac.yr)18	-0.0898574	0.5324477	-0.1687629	0.8659831	Poisson	BirdID
factor(fac.yr)19	-0.9973579	0.7640304	-1.3053903	0.1917600	Poisson	BirdID
factor(fac.yr)20	0.2224077	0.4762193	0.4670280	0.6404799	Poisson	BirdID
(Intercept)1	-0.0639407	0.2752636	-0.2322889	0.8163137	Poisson	Terr
factor(fac.yr)21	0.0997106	0.3508246	0.2842179	0.7762434	Poisson	Terr
factor(fac.yr)31	-0.0030339	0.3480481	-0.0087169	0.9930450	Poisson	Terr
factor(fac.yr)41	0.2083103	0.3381327	0.6160609	0.5378544	Poisson	Terr
factor(fac.yr)51	0.8528051	0.3123344	2.7304233	0.0063253	Poisson	Terr
factor(fac.yr)61	0.3350760	0.3358593	0.9976678	0.3184405	Poisson	Terr
factor(fac.yr)71	-0.0477153	0.3502425	-0.1362350	0.8916355	Poisson	Terr
factor(fac.yr)81	0.2489361	0.3433677	0.7249840	0.4684618	Poisson	Terr
factor(fac.yr)91	0.7532610	0.3290776	2.2890073	0.0220789	Poisson	Terr
factor(fac.yr)101	0.2323727	0.3448397	0.6738570	0.5004022	Poisson	Terr
factor(fac.yr)111	-0.1571754	0.3752836	-0.4188178	0.6753493	Poisson	Terr
factor(fac.yr)121	0.2277193	0.3676937	0.6193179	0.5357070	Poisson	Terr
factor(fac.yr)131	0.5502326	0.3409164	1.6139809	0.1065316	Poisson	Terr
factor(fac.yr)141	0.2051159	0.3505391	0.5851440	0.5584509	Poisson	Terr
factor(fac.yr)151	-0.0442896	0.4228101	-0.1047505	0.9165738	Poisson	Terr
factor(fac.yr)161	-0.2642290	0.4989134	-0.5296090	0.5963830	Poisson	Terr
factor(fac.yr)171	0.4371383	0.3857226	1.1332970	0.2570895	Poisson	Terr
factor(fac.yr)181	-0.1484729	0.5346957	-0.2776774	0.7812600	Poisson	Terr
factor(fac.yr)191	-1.0360074	0.7658561	-1.3527442	0.1761374	Poisson	Terr
factor(fac.yr)201	0.2408440	0.4783517	0.5034873	0.6146217	Poisson	Terr
(Intercept)2	-0.0711516	0.2773726	-0.2565201	0.7975493	Poisson	BirdID+Terr
factor(fac.yr)22	0.1146909	0.3514119	0.3263717	0.7441432	Poisson	BirdID+Terr
factor(fac.yr)32	-0.0020412	0.3494067	-0.0058418	0.9953390	Poisson	BirdID+Terr
factor(fac.yr)42	0.2015356	0.3404858	0.5919061	0.5539135	Poisson	BirdID+Terr
factor(fac.yr)52	0.8590439	0.3152728	2.7247637	0.0064348	Poisson	BirdID+Terr
factor(fac.yr)62	0.3494551	0.3386714	1.0318413	0.3021465	Poisson	BirdID+Terr
factor(fac.yr)72	-0.0359527	0.3533921	-0.1017359	0.9189663	Poisson	BirdID+Terr
factor(fac.yr)82	0.2514101	0.3468173	0.7249063	0.4685095	Poisson	BirdID+Terr
factor(fac.yr)92	0.7346914	0.3330428	2.2059969	0.0273842	Poisson	BirdID+Terr
factor(fac.yr)102	0.2178586	0.3482956	0.6254991	0.5316435	Poisson	BirdID+Terr
factor(fac.yr)112	-0.1562917	0.3783297	-0.4131099	0.6795261	Poisson	BirdID+Terr
factor(fac.yr)122	0.2050443	0.3711236	0.5524960	0.5806086	Poisson	BirdID+Terr
factor(fac.yr)132	0.5313371	0.3450199	1.5400186	0.1235558	Poisson	BirdID+Terr
factor(fac.yr)142	0.2040834	0.3540517	0.5764226	0.5643296	Poisson	BirdID+Terr
factor(fac.yr)152	-0.0450321	0.4269466	-0.1054749	0.9159990	Poisson	BirdID+Terr
factor(fac.yr)162	-0.2795266	0.5024293	-0.5563502	0.5779715	Poisson	BirdID+Terr
factor(fac.yr)172	0.4165074	0.3898791	1.0682988	0.2853857	Poisson	BirdID+Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)182	-0.1577254	0.5390397	-0.2926044	0.7698245	Poisson	BirdID+Terr
factor(fac.yr)192	-1.0351674	0.7686011	-1.3468200	0.1780382	Poisson	BirdID+Terr
factor(fac.yr)202	0.2189474	0.4831130	0.4532012	0.6504039	Poisson	BirdID+Terr

Each model had a criteria that they performed the best in. All smoothers were significant, except for the band ID smoother in the model with both random effects fitted.

Z-score number of fledgling (Gaussian)

Table 85: Z-score GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
64.63863	-369.1712	879.8614	1148.253	182.3960	263.3614	328	0.2647071	228	Gaussian	BandID+Terr
38.02142	-405.9715	894.2138	1050.241	228.2799	289.9786	328	0.1642071	111	Gaussian	Terr
33.40068	-413.6459	901.2160	1041.413	239.2162	294.5993	328	0.1379034	118	Gaussian	BandID

Table 86: Z-score GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	32.40068	116	0.4509296	0.0014483	Gaussian	BandID
s(Terrs)	37.02142	109	0.5894058	0.0002691	Gaussian	Terr
s(birdID)	29.53299	116	0.4447573	0.0042000	Gaussian	BandID+Terr
s(Terrs)	34.10564	109	0.5876172	0.0007976	Gaussian	BandID+Terr

Table 87: Z-score GAMMs linear terms table.

	Estimate	Std. Error	t value	Pr(> t)	Family	RE
(Intercept)	-0.0296429	0.0629001	-0.4712692	0.6377973	Gaussian	BandID
(Intercept)1	0.0091847	0.0634461	0.1447641	0.8849977	Gaussian	Terr
(Intercept)2	-0.0153876	0.0693242	-0.2219661	0.8245123	Gaussian	BandID+Terr

The model fitting both random effects is performing the best on AIC and adjusted R^2 , but the worst on BIC. All smoothers were significant.

Eggs (Binomial)

Table 88: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
58.67875	-145.3197	413.2758	645.8566	290.6394	269.3212	328	0.2277819	130	Binomial	Terr

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
61.43646	-	415.7062	661.6696	286.0133	266.5635	328	0.2279835	137	Binomial	BirdID
	143.0066									
88.50834	-	416.6044	780.1589	224.9075	239.4917	328	0.3865380	247	Binomial	BirdID+Terr
	112.4537									

Table 89: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	41.43646	116	60.33874	0.0023702	Binomial	BirdID
s(Terrs)	38.67875	109	58.28147	0.0011968	Binomial	Terr
s(birdID)	34.95891	116	48.84540	0.1334294	Binomial	BirdID+Terr
s(Terrs)	33.54943	109	50.41423	0.0628356	Binomial	BirdID+Terr

Table 90: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.4524505	0.5918948	0.7644103	0.4446227	Binomial	BirdID
factor(fac.yr)2	0.2290490	0.7412447	0.3090059	0.7573170	Binomial	BirdID
factor(fac.yr)3	-0.2094977	0.7252216	-0.2888740	0.7726778	Binomial	BirdID
factor(fac.yr)4	0.1712182	0.7514845	0.2278400	0.8197706	Binomial	BirdID
factor(fac.yr)5	2.5434023	1.1972652	2.1243432	0.0336415	Binomial	BirdID
factor(fac.yr)6	0.4230121	0.7834802	0.5399142	0.5892562	Binomial	BirdID
factor(fac.yr)7	-0.1630005	0.7611467	-0.2141512	0.8304292	Binomial	BirdID
factor(fac.yr)8	0.1466942	0.7982856	0.1837616	0.8542005	Binomial	BirdID
factor(fac.yr)9	0.6895497	0.8727176	0.7901178	0.4294589	Binomial	BirdID
factor(fac.yr)10	0.3509708	0.8054719	0.4357332	0.6630303	Binomial	BirdID
factor(fac.yr)11	-0.2131316	0.8015015	-0.2659154	0.7903044	Binomial	BirdID
factor(fac.yr)12	0.0070361	0.8711094	0.0080771	0.9935554	Binomial	BirdID
factor(fac.yr)13	2.2807627	1.2531608	1.8200080	0.0687578	Binomial	BirdID
factor(fac.yr)14	-0.1108944	0.8105643	-0.1368113	0.8911799	Binomial	BirdID
factor(fac.yr)15	0.0511034	0.9374225	0.0545148	0.9565250	Binomial	BirdID
factor(fac.yr)16	0.6065452	1.1221260	0.5405322	0.5888301	Binomial	BirdID
factor(fac.yr)17	1.8391490	1.3084914	1.4055493	0.1598580	Binomial	BirdID
factor(fac.yr)18	-0.6504484	1.1449942	-0.5680801	0.5699806	Binomial	BirdID
factor(fac.yr)19	-1.4174199	1.1479551	-1.2347347	0.2169293	Binomial	BirdID
factor(fac.yr)20	-0.2520641	1.1377626	-0.2215437	0.8246692	Binomial	BirdID
(Intercept)1	0.4447958	0.5807062	0.7659568	0.4437020	Binomial	Terr
factor(fac.yr)21	0.1085388	0.7678111	0.1413613	0.8875845	Binomial	Terr
factor(fac.yr)31	-0.0777890	0.7349157	-0.1058476	0.9157033	Binomial	Terr
factor(fac.yr)41	0.4200216	0.7450745	0.5637311	0.5729372	Binomial	Terr
factor(fac.yr)51	2.6466786	1.1967392	2.2115751	0.0269960	Binomial	Terr
factor(fac.yr)61	0.4265658	0.7673778	0.5558746	0.5782966	Binomial	Terr
factor(fac.yr)71	-0.1965424	0.7310193	-0.2688608	0.7880368	Binomial	Terr
factor(fac.yr)81	0.2234353	0.7773457	0.2874337	0.7737803	Binomial	Terr
factor(fac.yr)91	1.0100648	0.8565604	1.1792103	0.2383145	Binomial	Terr
factor(fac.yr)101	0.8164545	0.7737909	1.0551359	0.2913632	Binomial	Terr
factor(fac.yr)111	-0.2413369	0.7806652	-0.3091427	0.7572130	Binomial	Terr
factor(fac.yr)121	0.3408041	0.8418269	0.4048386	0.6855961	Binomial	Terr
factor(fac.yr)131	2.5194427	1.2145002	2.0744688	0.0380358	Binomial	Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)141	-0.2421196	0.7695460	-0.3146265	0.7530453	Binomial	Terr
factor(fac.yr)151	0.1995024	0.8933287	0.2233247	0.8232828	Binomial	Terr
factor(fac.yr)161	0.2597048	1.0728028	0.2420806	0.8087177	Binomial	Terr
factor(fac.yr)171	1.8032084	1.2558153	1.4358866	0.1510346	Binomial	Terr
factor(fac.yr)181	-0.3007108	1.1052135	-0.2720839	0.7855575	Binomial	Terr
factor(fac.yr)191	-0.9671613	1.1226974	-0.8614622	0.3889836	Binomial	Terr
factor(fac.yr)201	0.2489628	1.1310141	0.2201235	0.8257750	Binomial	Terr
(Intercept)2	0.5881427	0.6683127	0.8800412	0.3788370	Binomial	BirdID+Terr
factor(fac.yr)22	0.1589939	0.8393005	0.1894362	0.8497509	Binomial	BirdID+Terr
factor(fac.yr)32	-0.2189037	0.8206648	-0.2667395	0.7896697	Binomial	BirdID+Terr
factor(fac.yr)42	0.1999293	0.8280048	0.2414591	0.8091993	Binomial	BirdID+Terr
factor(fac.yr)52	2.3018679	1.2474442	1.8452672	0.0649987	Binomial	BirdID+Terr
factor(fac.yr)62	0.4444799	0.8695888	0.5111381	0.6092544	Binomial	BirdID+Terr
factor(fac.yr)72	-0.3422347	0.8493472	-0.4029385	0.6869934	Binomial	BirdID+Terr
factor(fac.yr)82	0.0937577	0.8925358	0.1050465	0.9163389	Binomial	BirdID+Terr
factor(fac.yr)92	0.8474516	0.9798039	0.8649196	0.3870829	Binomial	BirdID+Terr
factor(fac.yr)102	0.4842563	0.8830448	0.5483938	0.5834216	Binomial	BirdID+Terr
factor(fac.yr)112	-0.4076058	0.8971193	-0.4543495	0.6495773	Binomial	BirdID+Terr
factor(fac.yr)122	-0.0138361	0.9689952	-0.0142788	0.9886076	Binomial	BirdID+Terr
factor(fac.yr)132	2.4394250	1.3454945	1.8130323	0.0698268	Binomial	BirdID+Terr
factor(fac.yr)142	-0.3715804	0.8948989	-0.4152205	0.6779805	Binomial	BirdID+Terr
factor(fac.yr)152	-0.0653989	1.0187179	-0.0641973	0.9488132	Binomial	BirdID+Terr
factor(fac.yr)162	0.1092494	1.2671300	0.0862180	0.9312931	Binomial	BirdID+Terr
factor(fac.yr)172	1.7235232	1.4281120	1.2068544	0.2274882	Binomial	BirdID+Terr
factor(fac.yr)182	-0.6846466	1.3100825	-0.5225981	0.6012540	Binomial	BirdID+Terr
factor(fac.yr)192	-1.2449812	1.2679796	-0.9818622	0.3261678	Binomial	BirdID+Terr
factor(fac.yr)202	-0.1770650	1.2611550	-0.1403990	0.8883447	Binomial	BirdID+Terr

The territory only model has the best AIC and BIC scores, and the models fitting singular random effects have similar adjusted R^2 s. The model fitting both random effects had a notably higher adjusted R^2 , though the QQ plot for that model may be cause to be skeptical of that model. All smoothers are significant, except the band ID smoother in the model fitting both random effects.

Probabilites used to generate simulation 10

Simulation 1

Probability of mate fidelity: 0.526

Probability of pair dispersal: 0.024

Probability of single male dispersal: 0.312

Probability of single female dispersal: 0.664

NULL

Simulation 11

Number of fledgling as the response (Poisson)

Table 91: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npars	Family	RE
54.58774	-	982.5587	1201.467	343.7550	301.4123	356	0.2166684	130	Poisson	Terr
	434.786									
68.39232	-	983.7186	1269.070	310.6211	287.6077	356	0.2815325	252	Poisson	BirdID+Terr
	418.219									
50.65740	-	987.6858	1192.441	356.1871	305.3426	356	0.1800604	142	Poisson	BirdID
	441.002									

Table 92: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	30.65740	121	47.83179	0.0007061	Poisson	BirdID
s(Terrs)	34.58774	109	55.35885	0.0002182	Poisson	Terr
s(birdID)	20.18543	121	29.69675	0.0192942	Poisson	BirdID+Terr
s(Terrs)	28.20689	109	43.58471	0.0037943	Poisson	BirdID+Terr

Table 93: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.1087293	0.2750330	0.3953320	0.6925979	Poisson	BirdID
factor(fac.yr)2	-0.1161461	0.3463692	-0.3353247	0.7373802	Poisson	BirdID
factor(fac.yr)3	-0.2498747	0.3569480	-0.7000312	0.4839078	Poisson	BirdID
factor(fac.yr)4	0.2294465	0.3284854	0.6984981	0.4848657	Poisson	BirdID
factor(fac.yr)5	0.5186071	0.3257357	1.5921099	0.1113600	Poisson	BirdID
factor(fac.yr)6	0.1593134	0.3374189	0.4721531	0.6368175	Poisson	BirdID
factor(fac.yr)7	-0.3988060	0.3635442	-1.0969945	0.2726438	Poisson	BirdID
factor(fac.yr)8	0.1353894	0.3250522	0.4165157	0.6770327	Poisson	BirdID
factor(fac.yr)9	0.1355658	0.3293496	0.4116166	0.6806204	Poisson	BirdID
factor(fac.yr)10	-0.0665227	0.3425668	-0.1941889	0.8460280	Poisson	BirdID
factor(fac.yr)11	-0.6918045	0.3773548	-1.8332996	0.0667580	Poisson	BirdID
factor(fac.yr)12	-0.2566169	0.3460694	-0.7415185	0.4583791	Poisson	BirdID
factor(fac.yr)13	-0.5768490	0.3725001	-1.5485875	0.1214809	Poisson	BirdID
factor(fac.yr)14	0.0377062	0.3758630	0.1003190	0.9200911	Poisson	BirdID
factor(fac.yr)15	-0.3181900	0.4417677	-0.7202655	0.4713615	Poisson	BirdID
factor(fac.yr)16	0.2694470	0.4064082	0.6629960	0.5073331	Poisson	BirdID
factor(fac.yr)17	-0.1873144	0.4451695	-0.4207710	0.6739223	Poisson	BirdID
factor(fac.yr)18	-0.1551752	0.5445240	-0.2849740	0.7756641	Poisson	BirdID
factor(fac.yr)19	-1.2566970	0.7706078	-1.6307868	0.1029353	Poisson	BirdID
factor(fac.yr)20	0.1791208	0.4698638	0.3812187	0.7030410	Poisson	BirdID
(Intercept)1	0.0799139	0.2735030	0.2921864	0.7701441	Poisson	Terr
factor(fac.yr)21	0.0278059	0.3519664	0.0790015	0.9370314	Poisson	Terr
factor(fac.yr)31	-0.1497941	0.3606303	-0.4153675	0.6778729	Poisson	Terr
factor(fac.yr)41	0.2338224	0.3308471	0.7067385	0.4797290	Poisson	Terr
factor(fac.yr)51	0.5363655	0.3233126	1.6589688	0.0971221	Poisson	Terr
factor(fac.yr)61	0.2443514	0.3319276	0.7361587	0.4616341	Poisson	Terr
factor(fac.yr)71	-0.3235466	0.3589375	-0.9014008	0.3673752	Poisson	Terr
factor(fac.yr)81	0.2122017	0.3190524	0.6650997	0.5059867	Poisson	Terr
factor(fac.yr)91	0.2912436	0.3268724	0.8910008	0.3729288	Poisson	Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)101	0.1147498	0.3374267	0.3400733	0.7338013	Poisson	Terr
factor(fac.yr)111	-0.6090010	0.3711537	-1.6408325	0.1008322	Poisson	Terr
factor(fac.yr)121	-0.1501743	0.3398295	-0.4419107	0.6585538	Poisson	Terr
factor(fac.yr)131	-0.5512131	0.3681106	-1.4974120	0.1342861	Poisson	Terr
factor(fac.yr)141	0.0689457	0.3687243	0.1869843	0.8516729	Poisson	Terr
factor(fac.yr)151	-0.2670672	0.4388737	-0.6085286	0.5428369	Poisson	Terr
factor(fac.yr)161	0.3715341	0.4036935	0.9203372	0.3573966	Poisson	Terr
factor(fac.yr)171	-0.0075214	0.4447336	-0.0169122	0.9865066	Poisson	Terr
factor(fac.yr)181	-0.0706952	0.5420594	-0.1304196	0.8962345	Poisson	Terr
factor(fac.yr)191	-1.2447173	0.7659409	-1.6250828	0.1041449	Poisson	Terr
factor(fac.yr)201	0.0853617	0.4659066	0.1832164	0.8546282	Poisson	Terr
(Intercept)2	0.0651517	0.2826733	0.2304842	0.8177155	Poisson	BirdID+Terr
factor(fac.yr)22	-0.0043544	0.3560566	-0.0122295	0.9902426	Poisson	BirdID+Terr
factor(fac.yr)32	-0.1664754	0.3673223	-0.4532135	0.6503950	Poisson	BirdID+Terr
factor(fac.yr)42	0.2178135	0.3386767	0.6431309	0.5201391	Poisson	BirdID+Terr
factor(fac.yr)52	0.5485448	0.3339751	1.6424720	0.1004922	Poisson	BirdID+Terr
factor(fac.yr)62	0.2387783	0.3434276	0.6952798	0.4868800	Poisson	BirdID+Terr
factor(fac.yr)72	-0.3425781	0.3695285	-0.9270682	0.3538911	Poisson	BirdID+Terr
factor(fac.yr)82	0.1763048	0.3315079	0.5318268	0.5948460	Poisson	BirdID+Terr
factor(fac.yr)92	0.2460561	0.3385596	0.7267736	0.4673647	Poisson	BirdID+Terr
factor(fac.yr)102	0.0432366	0.3495647	0.1236871	0.9015630	Poisson	BirdID+Terr
factor(fac.yr)112	-0.6451041	0.3814587	-1.6911506	0.0908080	Poisson	BirdID+Terr
factor(fac.yr)122	-0.1838478	0.3511020	-0.5236308	0.6005354	Poisson	BirdID+Terr
factor(fac.yr)132	-0.5549970	0.3795282	-1.4623340	0.1436497	Poisson	BirdID+Terr
factor(fac.yr)142	0.0830200	0.3814658	0.2176341	0.8277142	Poisson	BirdID+Terr
factor(fac.yr)152	-0.2344367	0.4510152	-0.5197979	0.6032045	Poisson	BirdID+Terr
factor(fac.yr)162	0.3569560	0.4196200	0.8506649	0.3949555	Poisson	BirdID+Terr
factor(fac.yr)172	-0.0630611	0.4578628	-0.1377293	0.8904544	Poisson	BirdID+Terr
factor(fac.yr)182	-0.0806561	0.5592331	-0.1442262	0.8853219	Poisson	BirdID+Terr
factor(fac.yr)192	-1.2070984	0.7770474	-1.5534424	0.1203175	Poisson	BirdID+Terr
factor(fac.yr)202	0.1202413	0.4834680	0.2487057	0.8035884	Poisson	BirdID+Terr

Similar to what I said for the last batch of models. However, diagnostics look reasonable, and all smoothers are significant.

Z-score number of fledgling (Gaussian)

Table 94: Z-score GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npars	Family	RE
76.70435	-388.4935	941.2075	1259.379	184.8605	279.2957	356	0.3006912	233	Gaussian	BandID+Terr
42.74396	-437.8059	967.1172	1144.406	243.8706	313.2560	356	0.1774752	111	Gaussian	Terr
35.81586	-449.7717	977.7549	1129.287	260.8279	320.1841	356	0.1393167	123	Gaussian	BandID

Table 95: Z-score GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	34.81586	121	0.4749008	0.0007149	Gaussian	BandID
s(Terrs)	41.74396	109	0.7027332	0.0000499	Gaussian	Terr
s(birdID)	33.63469	121	0.5565014	0.0009222	Gaussian	BandID+Terr
s(Terrs)	42.06966	109	0.8348814	0.0001067	Gaussian	BandID+Terr

Table 96: Z-score GAMMs linear terms table.

	Estimate	Std. Error	t value	Pr(> t)	Family	RE
(Intercept)	-0.0323421	0.0611132	-0.5292170	0.5970217	Gaussian	BandID
(Intercept)1	0.0284036	0.0629820	0.4509794	0.6523164	Gaussian	Terr
(Intercept)2	-0.0069330	0.0703451	-0.0985568	0.9215609	Gaussian	BandID+Terr

The model fitting both random effects has the highest adjusted R^2 and the best AIC score, but the worst BIC score, with the band ID only model having the best BIC score. All smoothers are significant.

Eggs (Binomial)

Table 97: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
127.11978	-	437.7473	961.0177	167.6675	228.8802	356	0.6213343	252	Binomial	BirdID+Terr
	83.83374									
78.90210	-	446.5427	762.8593	283.2795	277.0979	356	0.3518069	142	Binomial	BirdID
	141.63977									
66.90667	-	456.2773	723.8223	318.1871	289.0933	356	0.2713755	130	Binomial	Terr
	159.09357									

Table 98: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	58.90210	121	97.13485	0.0000362	Binomial	BirdID
s(Terrs)	46.90667	109	76.97678	0.0000811	Binomial	Terr
s(birdID)	60.05898	121	117.61951	0.1439822	Binomial	BirdID+Terr
s(Terrs)	47.06080	109	96.55872	0.2770437	Binomial	BirdID+Terr

Table 99: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.3660184	0.7499290	0.4880708	0.6254997	Binomial	BirdID
factor(fac.yr)2	-0.1960150	0.8764645	-0.2236429	0.8230352	Binomial	BirdID
factor(fac.yr)3	0.5829973	0.9082902	0.6418624	0.5209626	Binomial	BirdID
factor(fac.yr)4	0.9788790	0.9509660	1.0293524	0.3033141	Binomial	BirdID
factor(fac.yr)5	2.0083275	1.1091889	1.8106271	0.0701986	Binomial	BirdID

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)6	1.0435805	0.9719309	1.0737188	0.2829488	Binomial	BirdID
factor(fac.yr)7	-0.7266132	0.9068012	-0.8012927	0.4229622	Binomial	BirdID
factor(fac.yr)8	-0.0254846	0.9042316	-0.0281837	0.9775156	Binomial	BirdID
factor(fac.yr)9	0.8678363	0.9656561	0.8987013	0.3688118	Binomial	BirdID
factor(fac.yr)10	-0.4701471	0.9173099	-0.5125281	0.6082815	Binomial	BirdID
factor(fac.yr)11	-1.1766902	0.9069683	-1.2973885	0.1944976	Binomial	BirdID
factor(fac.yr)12	-0.7150413	0.9021440	-0.7926022	0.4280096	Binomial	BirdID
factor(fac.yr)13	-1.1221123	0.9200867	-1.2195723	0.2226271	Binomial	BirdID
factor(fac.yr)14	0.3735695	1.0209253	0.3659127	0.7144302	Binomial	BirdID
factor(fac.yr)15	0.2620234	1.0414836	0.2515867	0.8013606	Binomial	BirdID
factor(fac.yr)16	1.1257579	1.1785083	0.9552397	0.3394565	Binomial	BirdID
factor(fac.yr)17	0.5804369	1.1654441	0.4980392	0.6184564	Binomial	BirdID
factor(fac.yr)18	0.1126306	1.4388690	0.0782771	0.9376076	Binomial	BirdID
factor(fac.yr)19	-1.4305699	1.3231384	-1.0811944	0.2796106	Binomial	BirdID
factor(fac.yr)20	0.6558292	1.3518662	0.4851288	0.6275850	Binomial	BirdID
(Intercept)1	0.5733082	0.6573736	0.8721192	0.3831434	Binomial	Terr
factor(fac.yr)21	0.0599755	0.8224212	0.0729255	0.9418654	Binomial	Terr
factor(fac.yr)31	0.5702421	0.8423477	0.6769677	0.4984265	Binomial	Terr
factor(fac.yr)41	0.6079730	0.8721657	0.6970843	0.4857501	Binomial	Terr
factor(fac.yr)51	1.8531982	1.0290127	1.8009478	0.0717111	Binomial	Terr
factor(fac.yr)61	1.1025911	0.8608488	1.2808185	0.2002574	Binomial	Terr
factor(fac.yr)71	-0.6616104	0.7995316	-0.8274975	0.4079551	Binomial	Terr
factor(fac.yr)81	0.3294329	0.7970484	0.4133160	0.6793751	Binomial	Terr
factor(fac.yr)91	1.0302704	0.8537290	1.2067886	0.2275136	Binomial	Terr
factor(fac.yr)101	0.0553547	0.8041978	0.0688321	0.9451232	Binomial	Terr
factor(fac.yr)111	-1.1836833	0.7904540	-1.4974727	0.1342703	Binomial	Terr
factor(fac.yr)121	-0.5808618	0.7799959	-0.7446985	0.4564540	Binomial	Terr
factor(fac.yr)131	-1.2518395	0.8039455	-1.5571199	0.1194420	Binomial	Terr
factor(fac.yr)141	0.1653894	0.8718246	0.1897048	0.8495404	Binomial	Terr
factor(fac.yr)151	-0.0546166	0.9276588	-0.0588757	0.9530511	Binomial	Terr
factor(fac.yr)161	0.4639414	1.0459234	0.4435711	0.6573527	Binomial	Terr
factor(fac.yr)171	0.4481833	1.0308930	0.4347525	0.6637421	Binomial	Terr
factor(fac.yr)181	-0.1042233	1.2347206	-0.0844105	0.9327301	Binomial	Terr
factor(fac.yr)191	-1.2812441	1.2111467	-1.0578769	0.2901116	Binomial	Terr
factor(fac.yr)201	-0.1437154	1.1984912	-0.1199136	0.9045516	Binomial	Terr
(Intercept)2	0.5185874	0.9970718	0.5201104	0.6029866	Binomial	BirdID+Terr
factor(fac.yr)22	-0.0986674	1.1378412	-0.0867146	0.9308984	Binomial	BirdID+Terr
factor(fac.yr)32	0.6811986	1.1720889	0.5811834	0.5611169	Binomial	BirdID+Terr
factor(fac.yr)42	0.6623664	1.2493927	0.5301507	0.5960075	Binomial	BirdID+Terr
factor(fac.yr)52	2.0694433	1.3653460	1.5156914	0.1295974	Binomial	BirdID+Terr
factor(fac.yr)62	1.4731539	1.2716813	1.1584301	0.2466886	Binomial	BirdID+Terr
factor(fac.yr)72	-1.2061380	1.1772328	-1.0245535	0.3055739	Binomial	BirdID+Terr
factor(fac.yr)82	0.0790837	1.1832599	0.0668354	0.9467127	Binomial	BirdID+Terr
factor(fac.yr)92	0.7609912	1.2269287	0.6202408	0.5350992	Binomial	BirdID+Terr
factor(fac.yr)102	-0.5449731	1.1894685	-0.4581652	0.6468337	Binomial	BirdID+Terr
factor(fac.yr)112	-1.4659324	1.1851341	-1.2369337	0.2161117	Binomial	BirdID+Terr
factor(fac.yr)122	-0.4822920	1.1698767	-0.4122589	0.6801497	Binomial	BirdID+Terr
factor(fac.yr)132	-1.1909152	1.2048918	-0.9884001	0.3229567	Binomial	BirdID+Terr
factor(fac.yr)142	0.8125506	1.3147383	0.6180322	0.5365541	Binomial	BirdID+Terr
factor(fac.yr)152	0.7441991	1.3688547	0.5436655	0.5866717	Binomial	BirdID+Terr
factor(fac.yr)162	1.2745146	1.5412479	0.8269368	0.4082729	Binomial	BirdID+Terr
factor(fac.yr)172	1.0549610	1.5184400	0.6947663	0.4872018	Binomial	BirdID+Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)182	0.4526040	1.9434116	0.2328915	0.8158457	Binomial	BirdID+Terr
factor(fac.yr)192	-1.3579747	1.8713069	-0.7256825	0.4680334	Binomial	BirdID+Terr
factor(fac.yr)202	0.6244587	1.8397437	0.3394270	0.7342881	Binomial	BirdID+Terr

I am only going to compare the models fitting one random effect because the QQ plot for the model fitting both random effects suggests violations of the distributional assumptions. The territory only model has a better AIC score and a higher adjusted R^2 , while the band ID only model has a lower BIC score. The smoothers are significant in their respective models.

Probabilites used to generate simulation 11

Simulation 1

Probability of mate fidelity: 0.685

Probability of pair dispersal: 0.17

Probability of single male dispersal: 0.996

Probability of single female dispersal: 0.352

NULL

Simulation 12

Number of fledgling as the response (Poisson)

Table 100: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npars	Family	RE
40.61016	-	957.6960	1121.969	348.0454	302.3898	343	0.1721815	132	Poisson	Terr
	436.0433									
35.02425	-	965.8716	1115.081	364.0710	307.9758	343	0.1279189	139	Poisson	BirdID
	444.0561									
43.59477	-	973.4496	1179.782	341.8800	299.4052	343	0.1838804	251	Poisson	BirdID+Terr
	432.9607									

Table 101: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	15.024245	118	18.310463	0.0898889	Poisson	BirdID
s(Terrs)	20.610159	111	29.158672	0.0069675	Poisson	Terr
s(birdID)	4.266876	118	4.554017	0.3438467	Poisson	BirdID+Terr
s(Terrs)	19.327890	111	27.008458	0.0107693	Poisson	BirdID+Terr

Table 102: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.5839040	2.129197e-01	2.7423673	0.0060998	Poisson	BirdID
factor(fac.yr)2	-0.5052935	2.942272e-01	-1.7173584	0.0859137	Poisson	BirdID
factor(fac.yr)3	-0.9887118	3.352126e-01	-2.9495069	0.0031828	Poisson	BirdID

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)4	-0.1627811	2.709690e-01	-0.6007370	0.5480151	Poisson	BirdID
factor(fac.yr)5	-0.0679837	2.710339e-01	-0.2508310	0.8019447	Poisson	BirdID
factor(fac.yr)6	-0.6557100	2.941012e-01	-2.2295391	0.0257781	Poisson	BirdID
factor(fac.yr)7	-1.1643404	3.439197e-01	-3.3855002	0.0007105	Poisson	BirdID
factor(fac.yr)8	-0.5410536	2.922095e-01	-1.8515948	0.0640840	Poisson	BirdID
factor(fac.yr)9	-0.1084564	2.828054e-01	-0.3835019	0.7013477	Poisson	BirdID
factor(fac.yr)10	-0.3352575	2.906641e-01	-1.1534188	0.2487385	Poisson	BirdID
factor(fac.yr)11	-1.0947777	3.618245e-01	-3.0257150	0.0024805	Poisson	BirdID
factor(fac.yr)12	-0.2350612	2.912716e-01	-0.8070173	0.4196565	Poisson	BirdID
factor(fac.yr)13	-0.2610035	2.965318e-01	-0.8801873	0.3787578	Poisson	BirdID
factor(fac.yr)14	-0.6482937	3.325444e-01	-1.9494954	0.0512363	Poisson	BirdID
factor(fac.yr)15	-0.5824461	3.545958e-01	-1.6425634	0.1004733	Poisson	BirdID
factor(fac.yr)16	-0.8396830	4.180807e-01	-2.0084234	0.0445983	Poisson	BirdID
factor(fac.yr)17	-0.3537080	3.561300e-01	-0.9931993	0.3206128	Poisson	BirdID
factor(fac.yr)18	-0.6194653	4.195179e-01	-1.4766123	0.1397796	Poisson	BirdID
factor(fac.yr)19	-0.6362634	4.195078e-01	-1.5166902	0.1293449	Poisson	BirdID
factor(fac.yr)20	-48.8401097	6.710886e+07	-0.0000007	0.9999994	Poisson	BirdID
(Intercept)1	0.5987770	2.154230e-01	2.7795402	0.0054436	Poisson	Terr
factor(fac.yr)21	-0.5104426	2.993096e-01	-1.7054000	0.0881199	Poisson	Terr
factor(fac.yr)31	-0.9883186	3.381464e-01	-2.9227538	0.0034695	Poisson	Terr
factor(fac.yr)41	-0.1823849	2.736470e-01	-0.6664970	0.5050935	Poisson	Terr
factor(fac.yr)51	-0.0884979	2.720673e-01	-0.3252794	0.7449696	Poisson	Terr
factor(fac.yr)61	-0.7230188	2.955014e-01	-2.4467527	0.0144150	Poisson	Terr
factor(fac.yr)71	-1.1776463	3.447577e-01	-3.4158667	0.0006358	Poisson	Terr
factor(fac.yr)81	-0.5411730	2.922159e-01	-1.8519629	0.0640312	Poisson	Terr
factor(fac.yr)91	-0.1479007	2.842592e-01	-0.5203022	0.6028529	Poisson	Terr
factor(fac.yr)101	-0.3988194	2.924686e-01	-1.3636313	0.1726836	Poisson	Terr
factor(fac.yr)111	-1.1150805	3.626944e-01	-3.0744355	0.0021090	Poisson	Terr
factor(fac.yr)121	-0.2321867	2.926435e-01	-0.7934116	0.4275380	Poisson	Terr
factor(fac.yr)131	-0.2865841	2.972205e-01	-0.9642137	0.3349388	Poisson	Terr
factor(fac.yr)141	-0.6704383	3.344902e-01	-2.0043583	0.0450317	Poisson	Terr
factor(fac.yr)151	-0.6783528	3.553415e-01	-1.9090166	0.0562599	Poisson	Terr
factor(fac.yr)161	-0.8347698	4.209410e-01	-1.9831042	0.0473558	Poisson	Terr
factor(fac.yr)171	-0.3624061	3.573351e-01	-1.0141912	0.3104915	Poisson	Terr
factor(fac.yr)181	-0.6283299	4.207961e-01	-1.4931932	0.1353866	Poisson	Terr
factor(fac.yr)191	-0.5574361	4.226525e-01	-1.3188992	0.1872028	Poisson	Terr
factor(fac.yr)201	-47.8290161	6.710886e+07	-0.0000007	0.9999994	Poisson	Terr
(Intercept)2	0.5925349	2.172561e-01	2.7273561	0.0063844	Poisson	BirdID+Terr
factor(fac.yr)22	-0.5071413	3.003976e-01	-1.6882334	0.0913664	Poisson	BirdID+Terr
factor(fac.yr)32	-0.9868187	3.394525e-01	-2.9070896	0.0036481	Poisson	BirdID+Terr
factor(fac.yr)42	-0.1810118	2.752610e-01	-0.6576008	0.5107947	Poisson	BirdID+Terr
factor(fac.yr)52	-0.0822512	2.741024e-01	-0.3000746	0.7641202	Poisson	BirdID+Terr
factor(fac.yr)62	-0.7128192	2.973341e-01	-2.3973683	0.0165133	Poisson	BirdID+Terr
factor(fac.yr)72	-1.1712463	3.464840e-01	-3.3803766	0.0007239	Poisson	BirdID+Terr
factor(fac.yr)82	-0.5350701	2.944170e-01	-1.8173886	0.0691576	Poisson	BirdID+Terr
factor(fac.yr)92	-0.1440474	2.867705e-01	-0.5023088	0.6154503	Poisson	BirdID+Terr
factor(fac.yr)102	-0.3906712	2.946877e-01	-1.3257127	0.1849349	Poisson	BirdID+Terr
factor(fac.yr)112	-1.1073936	3.645673e-01	-3.0375561	0.0023851	Poisson	BirdID+Terr
factor(fac.yr)122	-0.2281345	2.950360e-01	-0.7732427	0.4393788	Poisson	BirdID+Terr
factor(fac.yr)132	-0.2808813	2.996485e-01	-0.9373694	0.3485686	Poisson	BirdID+Terr
factor(fac.yr)142	-0.6648822	3.365973e-01	-1.9753045	0.0482336	Poisson	BirdID+Terr
factor(fac.yr)152	-0.6691342	3.577110e-01	-1.8705999	0.0614006	Poisson	BirdID+Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)162	-0.8395606	4.230777e-01	-1.9844123	0.0472099	Poisson	BirdID+Terr
factor(fac.yr)172	-0.3682287	3.601093e-01	-1.0225469	0.3065221	Poisson	BirdID+Terr
factor(fac.yr)182	-0.6312721	4.231836e-01	-1.4917216	0.1357721	Poisson	BirdID+Terr
factor(fac.yr)192	-0.5684537	4.250390e-01	-1.3374153	0.1810871	Poisson	BirdID+Terr
factor(fac.yr)202	-53.8162727	6.710886e+07	-0.0000008	0.9999994	Poisson	BirdID+Terr

The territory only model has the lowest AIC and performs close to the best model in each criteria. The territory smoothers are the only significant smoothers.

Z-score number of fledgling (Gaussian)

Table 103: Z-score GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
31.56563	-436.7244	943.1732	1076.863	257.4633	310.4344	342	0.1244167	113	Gaussian	Terr
48.79594	-415.4300	944.5432	1162.520	227.3178	293.2041	342	0.1815065	232	Gaussian	BandID+Terr
22.37110	-450.7351	955.8290	1060.057	279.4467	319.6289	342	0.0769935	120	Gaussian	BandID

Table 104: Z-score GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	21.37110	118	0.2410581	0.0410233	Gaussian	BandID
s(Terrs)	30.56563	111	0.4365286	0.0018812	Gaussian	Terr
s(birdID)	18.60664	118	0.2161865	0.0684084	Gaussian	BandID+Terr
s(Terrs)	29.18929	111	0.4280731	0.0037478	Gaussian	BandID+Terr

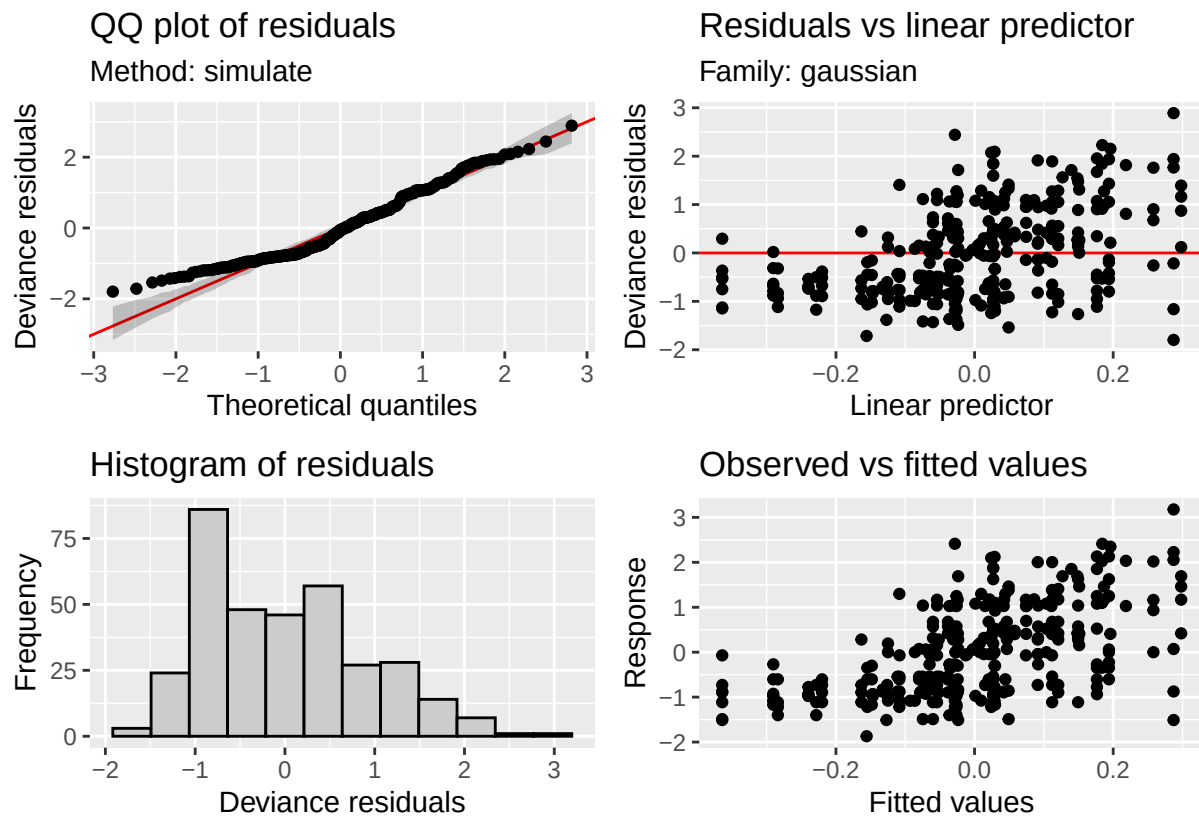
Table 105: Z-score GAMMs linear terms table.

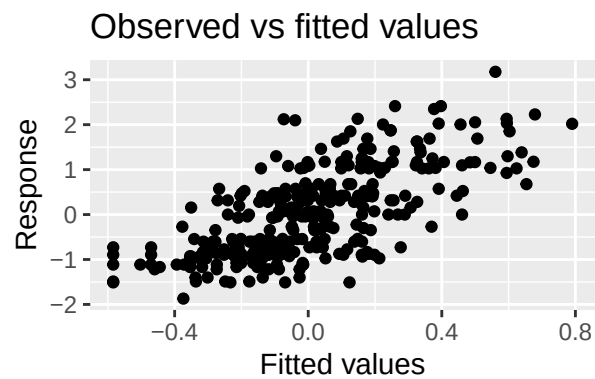
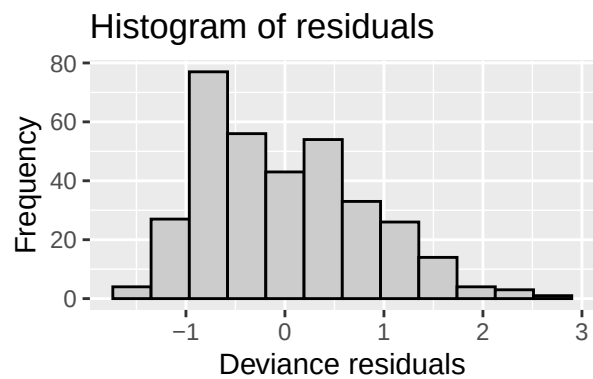
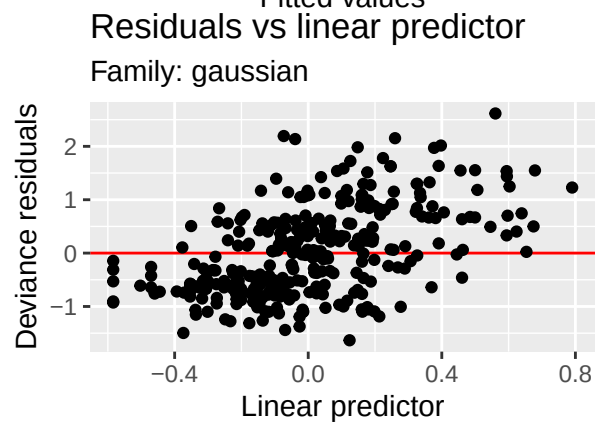
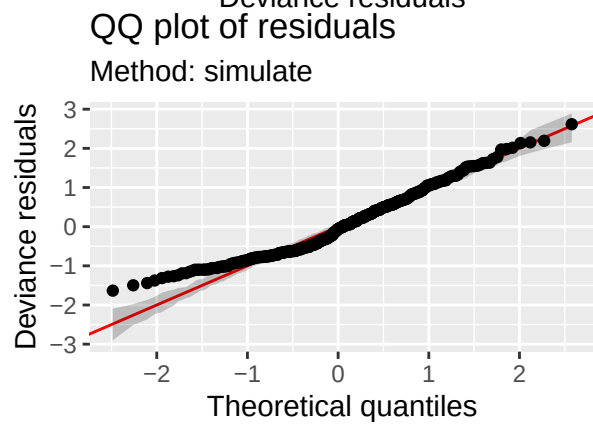
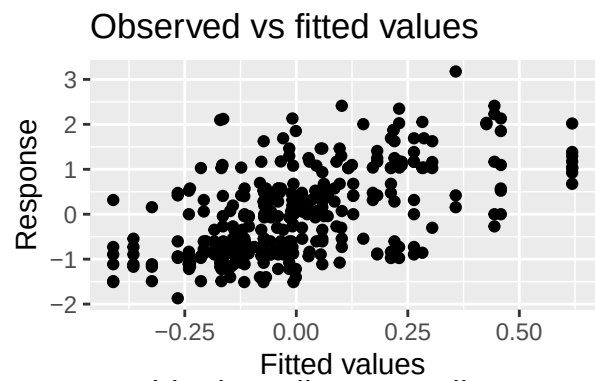
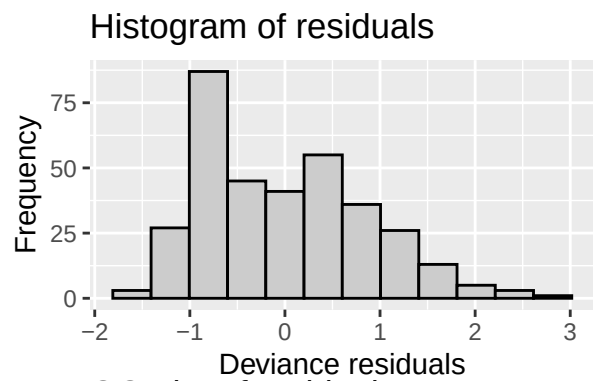
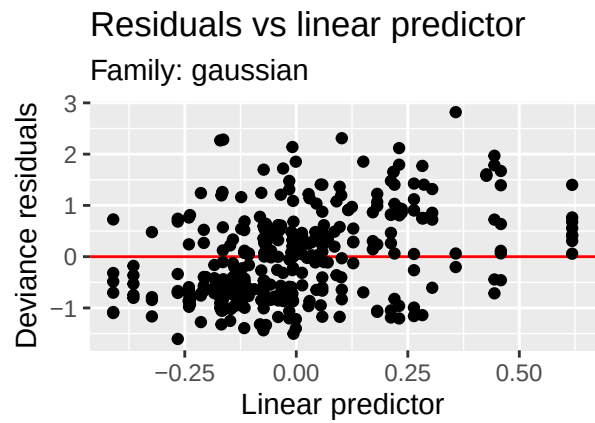
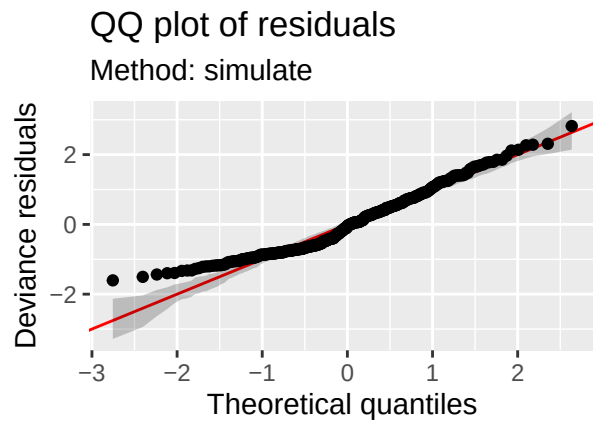
	Estimate	Std. Error	t value	Pr(> t)	Family	RE
(Intercept)	-0.0078768	0.0590567	-0.1333763	0.8939797	Gaussian	BandID
(Intercept)1	-0.0075731	0.0604872	-0.1252012	0.9004453	Gaussian	Terr
(Intercept)2	-0.0158868	0.0648716	-0.2448965	0.8067079	Gaussian	BandID+Terr

The band ID only model assumptions did not seem reasonable, so I will not include it in the comparisons. The territory only model has the best AIC and BIC scores out of the models we are considering, but a lower adjusted R^2 . All smoothers are significant.

	k'	edf	k-index	p-value
s(birdID)	119	21.3711	NA	NA
	k'	edf	k-index	p-value
s(Terrs)	112	30.56563	NA	NA
	k'	edf	k-index	p-value
s(birdID)	119	18.60664	NA	NA
s(Terrs)	112	29.18929	NA	NA

Our k-indices are still on the smaller side, but are better than those seen in the models fit with other response variables. The edf on the year term suggests that it is being shrunk out of the model, which is expected since the z-scores are calculated to remove the year effect. The QQ plots for the models including territory are better, but none of the diagnostic plots seem to suggest gross violations of assumptions.





Eggs (Binomial)

Table 106: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
52.23757	-171.7044	453.7400	665.4507	343.4088	290.7624	343	0.1774760	132	Binomial	Terr
45.79954	-180.2172	459.7098	650.2060	360.4344	297.2005	343	0.1444277	139	Binomial	BirdID
65.69729	-157.6043	466.3432	756.3500	315.2087	277.3027	343	0.2335479	251	Binomial	BirdID+Terr

Table 107: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	25.79954	118	33.97198	0.0269671	Binomial	BirdID
s(Terrs)	32.23757	111	45.19635	0.0063864	Binomial	Terr
s(birdID)	16.74292	118	20.00222	0.2232896	Binomial	BirdID+Terr
s(Terrs)	28.95437	111	39.13563	0.0384231	Binomial	BirdID+Terr

Table 108: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	2.4128490	1.063153e+00	2.2695213	0.0232366	Binomial	BirdID
factor(fac.yr)2	-2.1937035	1.149511e+00	-1.9083797	0.0563422	Binomial	BirdID
factor(fac.yr)3	-2.4338219	1.151262e+00	-2.1140467	0.0345113	Binomial	BirdID
factor(fac.yr)4	-0.9094829	1.202928e+00	-0.7560579	0.4496145	Binomial	BirdID
factor(fac.yr)5	-0.8564976	1.205686e+00	-0.7103818	0.4774674	Binomial	BirdID
factor(fac.yr)6	-2.3174383	1.137683e+00	-2.0369800	0.0416520	Binomial	BirdID
factor(fac.yr)7	-2.6016193	1.145992e+00	-2.2701899	0.0231961	Binomial	BirdID
factor(fac.yr)8	-1.7922577	1.152230e+00	-1.5554693	0.1198343	Binomial	BirdID
factor(fac.yr)9	-1.0947501	1.216967e+00	-0.8995726	0.3683477	Binomial	BirdID
factor(fac.yr)10	-1.4518045	1.180979e+00	-1.2293229	0.2189508	Binomial	BirdID
factor(fac.yr)11	-2.9918504	1.175069e+00	-2.5461067	0.0108932	Binomial	BirdID
factor(fac.yr)12	-1.3352966	1.197732e+00	-1.1148546	0.2649127	Binomial	BirdID
factor(fac.yr)13	-1.7044203	1.189341e+00	-1.4330794	0.1518352	Binomial	BirdID
factor(fac.yr)14	-2.0747902	1.187863e+00	-1.7466578	0.0806967	Binomial	BirdID
factor(fac.yr)15	-1.9213277	1.226452e+00	-1.5665742	0.1172142	Binomial	BirdID
factor(fac.yr)16	-2.5138771	1.261787e+00	-1.9923150	0.0463365	Binomial	BirdID
factor(fac.yr)17	-1.9893187	1.272482e+00	-1.5633367	0.1179734	Binomial	BirdID
factor(fac.yr)18	-1.9101595	1.320104e+00	-1.4469771	0.1479033	Binomial	BirdID
factor(fac.yr)19	-2.0068866	1.316194e+00	-1.5247650	0.1273177	Binomial	BirdID
factor(fac.yr)20	-	6.710886e+07	-0.0000007	0.9999995	Binomial	BirdID
	43.6749968					
(Intercept)1	2.6303858	1.074033e+00	2.4490737	0.0143224	Binomial	Terr
factor(fac.yr)21	-2.4585132	1.164173e+00	-2.1118117	0.0347026	Binomial	Terr
factor(fac.yr)31	-2.7166893	1.163913e+00	-2.3340997	0.0195905	Binomial	Terr
factor(fac.yr)41	-1.0601626	1.216824e+00	-0.8712542	0.3836154	Binomial	Terr
factor(fac.yr)51	-1.0481909	1.215614e+00	-0.8622726	0.3885375	Binomial	Terr
factor(fac.yr)61	-2.6468112	1.150256e+00	-2.3010631	0.0213881	Binomial	Terr
factor(fac.yr)71	-2.8346916	1.156519e+00	-2.4510550	0.0142438	Binomial	Terr
factor(fac.yr)81	-1.8807892	1.159239e+00	-1.6224342	0.1047104	Binomial	Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)91	-1.3052569	1.226382e+00	-1.0643148	0.2871861	Binomial	Terr
factor(fac.yr)101	-1.7722197	1.190540e+00	-1.4885854	0.1365966	Binomial	Terr
factor(fac.yr)111	-3.2427512	1.183325e+00	-2.7403728	0.0061370	Binomial	Terr
factor(fac.yr)121	-1.3117146	1.205959e+00	-1.0876945	0.2767300	Binomial	Terr
factor(fac.yr)131	-1.8107655	1.194193e+00	-1.5163088	0.1294413	Binomial	Terr
factor(fac.yr)141	-2.1507390	1.198392e+00	-1.7946880	0.0727034	Binomial	Terr
factor(fac.yr)151	-2.1401066	1.240961e+00	-1.7245564	0.0846074	Binomial	Terr
factor(fac.yr)161	-2.5759884	1.277555e+00	-2.0163425	0.0437642	Binomial	Terr
factor(fac.yr)171	-2.0558968	1.281706e+00	-1.6040311	0.1087072	Binomial	Terr
factor(fac.yr)181	-2.1316217	1.330754e+00	-1.6018146	0.1091966	Binomial	Terr
factor(fac.yr)191	-1.9544349	1.328822e+00	-1.4708022	0.1413446	Binomial	Terr
factor(fac.yr)201	-	6.710886e+07	-0.0000007	0.9999995	Binomial	Terr
	43.7628375					
(Intercept)2	2.5841360	1.092532e+00	2.3652729	0.0180168	Binomial	BirdID+Terr
factor(fac.yr)22	-2.3840960	1.182992e+00	-2.0153109	0.0438721	Binomial	BirdID+Terr
factor(fac.yr)32	-2.6720953	1.185305e+00	-2.2543517	0.0241740	Binomial	BirdID+Terr
factor(fac.yr)42	-1.0502218	1.236707e+00	-0.8492083	0.3957654	Binomial	BirdID+Terr
factor(fac.yr)52	-0.9986883	1.235235e+00	-0.8085009	0.4188023	Binomial	BirdID+Terr
factor(fac.yr)62	-2.5882158	1.170776e+00	-2.2106840	0.0270577	Binomial	BirdID+Terr
factor(fac.yr)72	-2.7599549	1.178478e+00	-2.3419654	0.0191825	Binomial	BirdID+Terr
factor(fac.yr)82	-1.8310932	1.183293e+00	-1.5474559	0.1217534	Binomial	BirdID+Terr
factor(fac.yr)92	-1.2759643	1.250721e+00	-1.0201829	0.3076417	Binomial	BirdID+Terr
factor(fac.yr)102	-1.6993030	1.217369e+00	-1.3958817	0.1627501	Binomial	BirdID+Terr
factor(fac.yr)112	-3.1543355	1.207329e+00	-2.6126565	0.0089842	Binomial	BirdID+Terr
factor(fac.yr)122	-1.2608485	1.231093e+00	-1.0241703	0.3057548	Binomial	BirdID+Terr
factor(fac.yr)132	-1.7537069	1.219698e+00	-1.4378212	0.1504848	Binomial	BirdID+Terr
factor(fac.yr)142	-2.1383710	1.224985e+00	-1.7456298	0.0808753	Binomial	BirdID+Terr
factor(fac.yr)152	-2.1069326	1.271492e+00	-1.6570552	0.0975083	Binomial	BirdID+Terr
factor(fac.yr)162	-2.6099581	1.310910e+00	-1.9909516	0.0464862	Binomial	BirdID+Terr
factor(fac.yr)172	-2.0520702	1.318388e+00	-1.5564997	0.1195893	Binomial	BirdID+Terr
factor(fac.yr)182	-2.0947764	1.367014e+00	-1.5323737	0.1254302	Binomial	BirdID+Terr
factor(fac.yr)192	-1.9834379	1.359555e+00	-1.4588874	0.1445961	Binomial	BirdID+Terr
factor(fac.yr)202	-	6.710886e+07	-0.0000007	0.9999995	Binomial	BirdID+Terr
	44.5854142					

Each model has a criteria that they perform the best in and the only smoother that is insignificant is the band ID smoother in the model that includes both random effects.

Probabilites used to generate simulation 12

Simulation 1

Probability of mate fidelity: 0.308

Probability of pair dispersal: 0.312

Probability of single male dispersal: 0.162

Probability of single female dispersal: 0.583

NULL

Simulation 13

Number of fledgling as the response (Poisson)

Table 109: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npars	Family	RE
48.77929	-	943.8301	1137.542	315.5576	282.2207	331	0.2051286	132	Poisson	Terr
	420.9666									
38.25221	-	952.7440	1107.991	344.7052	292.7478	331	0.1344423	127	Poisson	BirdID
	435.5404									
52.44803	-	956.2240	1188.073	307.8907	278.5520	331	0.2209067	239	Poisson	BirdID+Terr
	417.1332									

Table 110: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	18.252208	106	24.666632	0.0241600	Poisson	BirdID
s(Terrs)	28.779285	111	42.538487	0.0021572	Poisson	Terr
s(birdID)	5.924764	106	6.685256	0.2667827	Poisson	BirdID+Terr
s(Terrs)	26.523263	111	38.618712	0.0048257	Poisson	BirdID+Terr

Table 111: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.0830285	0.2436042	0.3408334	0.7332290	Poisson	BirdID
factor(fac.yr)2	0.1078164	0.3168141	0.3403144	0.7336198	Poisson	BirdID
factor(fac.yr)3	-0.6476424	0.3693296	-1.7535621	0.0795056	Poisson	BirdID
factor(fac.yr)4	0.0810199	0.3139839	0.2580384	0.7963773	Poisson	BirdID
factor(fac.yr)5	0.3546278	0.2970562	1.1938072	0.2325534	Poisson	BirdID
factor(fac.yr)6	0.5121332	0.2912976	1.7581100	0.0787288	Poisson	BirdID
factor(fac.yr)7	-0.3309749	0.3412591	-0.9698640	0.3321143	Poisson	BirdID
factor(fac.yr)8	-0.0423575	0.3067980	-0.1380630	0.8901906	Poisson	BirdID
factor(fac.yr)9	0.5087211	0.2881600	1.7654116	0.0774946	Poisson	BirdID
factor(fac.yr)10	0.2961945	0.3170472	0.9342283	0.3501861	Poisson	BirdID
factor(fac.yr)11	-0.1453982	0.3476780	-0.4181978	0.6758025	Poisson	BirdID
factor(fac.yr)12	0.1669644	0.3400397	0.4910145	0.6234162	Poisson	BirdID
factor(fac.yr)13	0.1062516	0.3286718	0.3232756	0.7464865	Poisson	BirdID
factor(fac.yr)14	0.2468872	0.3328825	0.7416648	0.4582905	Poisson	BirdID
factor(fac.yr)15	-1.0522099	0.5611681	-1.8750352	0.0607879	Poisson	BirdID
factor(fac.yr)16	0.0328881	0.3761109	0.0874426	0.9303197	Poisson	BirdID
factor(fac.yr)17	-0.3093842	0.5189354	-0.5961902	0.5510482	Poisson	BirdID
factor(fac.yr)18	-0.1051740	0.4593277	-0.2289739	0.8188892	Poisson	BirdID
factor(fac.yr)19	-0.0871015	0.5208635	-0.1672251	0.8671929	Poisson	BirdID
factor(fac.yr)20	0.6318141	0.5835110	1.0827802	0.2789060	Poisson	BirdID
(Intercept)1	0.1406906	0.2481235	0.5670183	0.5707018	Poisson	Terr
factor(fac.yr)21	0.0651140	0.3263402	0.1995280	0.8418497	Poisson	Terr
factor(fac.yr)31	-0.6734725	0.3750933	-1.7954796	0.0725773	Poisson	Terr
factor(fac.yr)41	0.0588417	0.3192098	0.1843356	0.8537502	Poisson	Terr
factor(fac.yr)51	0.3152396	0.3009392	1.0475191	0.2948602	Poisson	Terr
factor(fac.yr)61	0.4310936	0.2963495	1.4546798	0.1457579	Poisson	Terr
factor(fac.yr)71	-0.3840589	0.3458090	-1.1106101	0.2667362	Poisson	Terr
factor(fac.yr)81	-0.0888959	0.3118851	-0.2850277	0.7756229	Poisson	Terr
factor(fac.yr)91	0.4691648	0.2941673	1.5948907	0.1107367	Poisson	Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)101	0.1926895	0.3214232	0.5994885	0.5488472	Poisson	Terr
factor(fac.yr)111	-0.2612484	0.3526977	-0.7407149	0.4588663	Poisson	Terr
factor(fac.yr)121	0.0274890	0.3440014	0.0799097	0.9363091	Poisson	Terr
factor(fac.yr)131	0.0547201	0.3320552	0.1647921	0.8691076	Poisson	Terr
factor(fac.yr)141	0.2237573	0.3368190	0.6643250	0.5064823	Poisson	Terr
factor(fac.yr)151	-1.0203805	0.5626345	-1.8135760	0.0697430	Poisson	Terr
factor(fac.yr)161	0.0047399	0.3802876	0.0124640	0.9900554	Poisson	Terr
factor(fac.yr)171	-0.3007151	0.5258645	-0.5718490	0.5674243	Poisson	Terr
factor(fac.yr)181	-0.1116669	0.4658684	-0.2396962	0.8105658	Poisson	Terr
factor(fac.yr)191	-0.2067119	0.5276091	-0.3917899	0.6952135	Poisson	Terr
factor(fac.yr)201	0.4110421	0.5917102	0.6946679	0.4872635	Poisson	Terr
(Intercept)2	0.1299577	0.2497908	0.5202659	0.6028782	Poisson	BirdID+Terr
factor(fac.yr)22	0.0746373	0.3271991	0.2281099	0.8195608	Poisson	BirdID+Terr
factor(fac.yr)32	-0.6658309	0.3760972	-1.7703692	0.0766657	Poisson	BirdID+Terr
factor(fac.yr)42	0.0646200	0.3207233	0.2014822	0.8403216	Poisson	BirdID+Terr
factor(fac.yr)52	0.3194985	0.3029597	1.0545908	0.2916125	Poisson	BirdID+Terr
factor(fac.yr)62	0.4518205	0.2983897	1.5141958	0.1299762	Poisson	BirdID+Terr
factor(fac.yr)72	-0.3758742	0.3475347	-1.0815444	0.2794550	Poisson	BirdID+Terr
factor(fac.yr)82	-0.0755778	0.3135977	-0.2410025	0.8095532	Poisson	BirdID+Terr
factor(fac.yr)92	0.4830779	0.2962293	1.6307565	0.1029417	Poisson	BirdID+Terr
factor(fac.yr)102	0.2085931	0.3238250	0.6441536	0.5194758	Poisson	BirdID+Terr
factor(fac.yr)112	-0.2436402	0.3548261	-0.6866469	0.4923053	Poisson	BirdID+Terr
factor(fac.yr)122	0.0503146	0.3468109	0.1450780	0.8846493	Poisson	BirdID+Terr
factor(fac.yr)132	0.0621366	0.3347010	0.1856479	0.8527209	Poisson	BirdID+Terr
factor(fac.yr)142	0.2278648	0.3396989	0.6707847	0.5023577	Poisson	BirdID+Terr
factor(fac.yr)152	-1.0270234	0.5647044	-1.8186921	0.0689584	Poisson	BirdID+Terr
factor(fac.yr)162	0.0059647	0.3831324	0.0155682	0.9875789	Poisson	BirdID+Terr
factor(fac.yr)172	-0.3079038	0.5290615	-0.5819811	0.5605794	Poisson	BirdID+Terr
factor(fac.yr)182	-0.1115204	0.4691139	-0.2377257	0.8120938	Poisson	BirdID+Terr
factor(fac.yr)192	-0.2015719	0.5315940	-0.3791839	0.7045513	Poisson	BirdID+Terr
factor(fac.yr)202	0.4302878	0.5994232	0.7178364	0.4728581	Poisson	BirdID+Terr

Similar response as for the previous batch of models.

Z-score number of fledgling (Gaussian)

Table 112: Z-score GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npars	Family	RE
62.91465	-372.0798	882.9145	1146.275	184.9460	266.0854	329	0.2645790	219	Gaussian	BandID+Terr
43.06585	-399.1079	890.6569	1066.113	217.9724	285.9341	329	0.1934195	113	Gaussian	Terr
27.26503	-423.9630	909.3383	1025.901	253.5252	301.7350	329	0.1109879	107	Gaussian	BandID

Table 113: Z-score GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	26.26503	105	0.3899915	0.0042704	Gaussian	BandID
s(Terrs)	42.06585	111	0.7086046	0.0000380	Gaussian	Terr
s(birdID)	20.72820	105	0.3228986	0.0308701	Gaussian	BandID+Terr
s(Terrs)	41.18645	111	0.7760131	0.0000664	Gaussian	BandID+Terr

Table 114: Z-score GAMMs linear terms table.

	Estimate	Std. Error	t value	Pr(> t)	Family	RE
(Intercept)	0.0081871	0.0626934	0.1305894	0.8961871	Gaussian	BandID
(Intercept)1	0.0335095	0.0653010	0.5131542	0.6082397	Gaussian	Terr
(Intercept)2	0.0360943	0.0704879	0.5120638	0.6090311	Gaussian	BandID+Terr

The model fitting both random effects has the best AIC and best adjusted R^2 . However, it does have the worst BIC, with the band ID only model having the best BIC. All smoothers are significant.

Eggs (Binomial)

Table 115: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
98.89401	-	419.8193	819.6640	209.4919	232.1060	331	0.4493150	239	Binomial	BirdID+Terr
	104.7460									
61.45048	-	424.4458	665.7781	297.4995	269.5495	331	0.2431325	127	Binomial	BirdID
	148.7498									
67.33073	-	425.1247	690.5488	285.5057	263.6693	331	0.2598780	132	Binomial	Terr
	142.7528									

Table 116: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	41.45048	105	71.46373	0.0000406	Binomial	BirdID
s(Terrs)	47.33073	111	77.28198	0.0000890	Binomial	Terr
s(birdID)	34.66986	105	63.38406	0.0507002	Binomial	BirdID+Terr
s(Terrs)	44.22415	111	77.22951	0.0331512	Binomial	BirdID+Terr

Table 117: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	-0.1581753	5.945226e-01	-0.2660543	0.7901974	Binomial	BirdID
factor(fac.yr)2	0.7535872	7.673703e-01	0.9820385	0.3260809	Binomial	BirdID
factor(fac.yr)3	-0.0749227	7.525067e-01	-0.0995641	0.9206904	Binomial	BirdID
factor(fac.yr)4	1.1415086	7.898140e-01	1.4452878	0.1483771	Binomial	BirdID
factor(fac.yr)5	1.1796897	7.853826e-01	1.5020574	0.1330823	Binomial	BirdID

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)6	1.9004436	8.164713e-01	2.3276306	0.0199317	Binomial	BirdID
factor(fac.yr)7	0.8449492	7.803250e-01	1.0828170	0.2788896	Binomial	BirdID
factor(fac.yr)8	0.9125831	7.454351e-01	1.2242287	0.2208660	Binomial	BirdID
factor(fac.yr)9	1.6763941	7.932358e-01	2.1133617	0.0345698	Binomial	BirdID
factor(fac.yr)10	0.8244271	8.167037e-01	1.0094568	0.3127556	Binomial	BirdID
factor(fac.yr)11	1.1113245	8.279423e-01	1.3422728	0.1795075	Binomial	BirdID
factor(fac.yr)12	1.5026872	8.911583e-01	1.6862181	0.0917538	Binomial	BirdID
factor(fac.yr)13	1.2574000	8.524696e-01	1.4750086	0.1402102	Binomial	BirdID
factor(fac.yr)14	0.6754691	8.640910e-01	0.7817106	0.4343847	Binomial	BirdID
factor(fac.yr)15	-0.8636566	9.834700e-01	-0.8781729	0.3798499	Binomial	BirdID
factor(fac.yr)16	1.4658563	1.047985e+00	1.3987376	0.1618917	Binomial	BirdID
factor(fac.yr)17	0.8072040	1.208067e+00	0.6681784	0.5040197	Binomial	BirdID
factor(fac.yr)18	0.5433974	1.118562e+00	0.4858000	0.6271089	Binomial	BirdID
factor(fac.yr)19	0.9711764	1.278240e+00	0.7597762	0.4473884	Binomial	BirdID
factor(fac.yr)20	39.1013713	4.745313e+07	0.0000008	0.9999993	Binomial	BirdID
(Intercept)1	0.0494070	6.029326e-01	0.0819445	0.9346908	Binomial	Terr
factor(fac.yr)21	0.5863416	7.979205e-01	0.7348371	0.4624387	Binomial	Terr
factor(fac.yr)31	-0.2040003	7.744823e-01	-0.2634022	0.7922406	Binomial	Terr
factor(fac.yr)41	1.0206448	7.924774e-01	1.2879166	0.1977750	Binomial	Terr
factor(fac.yr)51	0.9644622	7.932580e-01	1.2158242	0.2240519	Binomial	Terr
factor(fac.yr)61	1.6579159	8.325970e-01	1.9912586	0.0464525	Binomial	Terr
factor(fac.yr)71	0.7895038	7.893628e-01	1.0001785	0.3172241	Binomial	Terr
factor(fac.yr)81	0.8748017	7.521698e-01	1.1630375	0.2448143	Binomial	Terr
factor(fac.yr)91	1.5814578	8.062203e-01	1.9615704	0.0498125	Binomial	Terr
factor(fac.yr)101	0.2471760	8.270398e-01	0.2988684	0.7650405	Binomial	Terr
factor(fac.yr)111	0.3901377	8.282346e-01	0.4710473	0.6376069	Binomial	Terr
factor(fac.yr)121	0.5572070	8.697902e-01	0.6406223	0.5217681	Binomial	Terr
factor(fac.yr)131	1.0290133	8.378428e-01	1.2281700	0.2193832	Binomial	Terr
factor(fac.yr)141	0.6945137	8.602544e-01	0.8073352	0.4194734	Binomial	Terr
factor(fac.yr)151	-0.9371265	9.605091e-01	-0.9756560	0.3292350	Binomial	Terr
factor(fac.yr)161	1.5153013	1.039083e+00	1.4583068	0.1447560	Binomial	Terr
factor(fac.yr)171	0.9988300	1.203261e+00	0.8301027	0.4064807	Binomial	Terr
factor(fac.yr)181	0.4735302	1.093735e+00	0.4329479	0.6650526	Binomial	Terr
factor(fac.yr)191	0.2865924	1.281669e+00	0.2236087	0.8230618	Binomial	Terr
factor(fac.yr)201	37.7338168	4.745313e+07	0.0000008	0.9999994	Binomial	Terr
(Intercept)2	-0.0024898	7.218924e-01	-0.0034490	0.9972481	Binomial	BirdID+Terr
factor(fac.yr)22	0.7749418	9.104628e-01	0.8511514	0.3946852	Binomial	BirdID+Terr
factor(fac.yr)32	-0.2025119	9.110338e-01	-0.2222880	0.8240897	Binomial	BirdID+Terr
factor(fac.yr)42	1.3018243	9.206863e-01	1.4139717	0.1573702	Binomial	BirdID+Terr
factor(fac.yr)52	1.1574849	9.287739e-01	1.2462505	0.2126724	Binomial	BirdID+Terr
factor(fac.yr)62	2.1687921	9.740215e-01	2.2266369	0.0259716	Binomial	BirdID+Terr
factor(fac.yr)72	0.8375309	9.255401e-01	0.9049104	0.3655128	Binomial	BirdID+Terr
factor(fac.yr)82	0.9953900	8.803746e-01	1.1306437	0.2582051	Binomial	BirdID+Terr
factor(fac.yr)92	1.8364206	9.392348e-01	1.9552305	0.0505559	Binomial	BirdID+Terr
factor(fac.yr)102	0.4674453	9.751164e-01	0.4793738	0.6316727	Binomial	BirdID+Terr
factor(fac.yr)112	0.6576673	9.737580e-01	0.6753910	0.4994274	Binomial	BirdID+Terr
factor(fac.yr)122	0.9573752	1.029561e+00	0.9298870	0.3524296	Binomial	BirdID+Terr
factor(fac.yr)132	1.2690620	1.004477e+00	1.2634058	0.2064434	Binomial	BirdID+Terr
factor(fac.yr)142	0.8534823	1.027130e+00	0.8309390	0.4060081	Binomial	BirdID+Terr
factor(fac.yr)152	-0.9407011	1.133914e+00	-0.8296051	0.4067621	Binomial	BirdID+Terr
factor(fac.yr)162	1.6554202	1.216217e+00	1.3611229	0.1734749	Binomial	BirdID+Terr
factor(fac.yr)172	1.0435990	1.483262e+00	0.7035839	0.4816920	Binomial	BirdID+Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)182	0.5419110	1.366166e+00	0.3966655	0.6916142	Binomial	BirdID+Terr
factor(fac.yr)192	0.6624405	1.620906e+00	0.4086853	0.6827706	Binomial	BirdID+Terr
factor(fac.yr)202	41.6634510	4.745313e+07	0.0000009	0.9999993	Binomial	BirdID+Terr

The model fitting both random effects has an adjusted R^2 that is nearly 20% higher and the lowest AIC score. However, the BIC score is the worst and the QQ plot may cause some concern. All smoothers are significant.

Probabilites used to generate simulation 13

Simulation 1

Probability of mate fidelity: 0.136

Probability of pair dispersal: 0.725

Probability of single male dispersal: 0.452

Probability of single female dispersal: 0.742

NULL

Simulation 14

Number of fledgling as the response (Poisson)

Table 118: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
41.44814	-352.5322	792.6852	951.2965	242.5358	234.5519	276	0.2070558	126	Poisson	Terr
41.44886	-352.5317	792.6917	951.3165	242.5348	234.5511	276	0.2070579	223	Poisson	BirdID+Terr
20.00506	-378.0814	796.1831	868.6277	293.6343	255.9949	276	0.0514822	117	Poisson	BirdID

Table 119: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	0.0050623	96	0.0048096	0.5645398	Poisson	BirdID
s(Terrs)	21.4481365	105	29.7785422	0.0109816	Poisson	Terr
s(birdID)	0.0010407	96	0.0008258	0.8632577	Poisson	BirdID+Terr
s(Terrs)	21.4478207	105	29.7780524	0.0109826	Poisson	BirdID+Terr

Table 120: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.4519919	3.015151e-01	1.4990688	0.1338558	Poisson	BirdID
factor(fac.yr)2	0.0275887	3.721956e-01	0.0741242	0.9409115	Poisson	BirdID
factor(fac.yr)3	-0.9628096	4.494693e-01	-2.1421033	0.0321852	Poisson	BirdID
factor(fac.yr)4	-0.6526544	4.494698e-01	-1.4520539	0.1464866	Poisson	BirdID
factor(fac.yr)5	0.0693116	3.495164e-01	0.1983073	0.8428047	Poisson	BirdID

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)6	-0.2696620	3.524840e-01	-0.7650335	0.4442516	Poisson	BirdID
factor(fac.yr)7	-0.4519754	3.827108e-01	-1.1809842	0.2376090	Poisson	BirdID
factor(fac.yr)8	-0.1534873	3.509554e-01	-0.4373414	0.6618638	Poisson	BirdID
factor(fac.yr)9	0.1458562	3.404580e-01	0.4284117	0.6683514	Poisson	BirdID
factor(fac.yr)10	-0.3122230	3.665924e-01	-0.8516900	0.3943862	Poisson	BirdID
factor(fac.yr)11	-0.4985047	3.721974e-01	-1.3393556	0.1804549	Poisson	BirdID
factor(fac.yr)12	-0.2776325	3.618175e-01	-0.7673276	0.4428868	Poisson	BirdID
factor(fac.yr)13	0.1076273	3.456636e-01	0.3113643	0.7555237	Poisson	BirdID
factor(fac.yr)14	-0.4519835	3.916785e-01	-1.1539654	0.2485143	Poisson	BirdID
factor(fac.yr)15	-0.5573499	4.494703e-01	-1.2400147	0.2149699	Poisson	BirdID
factor(fac.yr)16	-0.8574399	5.838778e-01	-1.4685263	0.1419613	Poisson	BirdID
factor(fac.yr)17	-0.4520071	5.838795e-01	-0.7741445	0.4388454	Poisson	BirdID
factor(fac.yr)18	-65.7545489	6.710886e+07	-0.0000010	0.9999992	Poisson	BirdID
factor(fac.yr)19	-65.7545877	6.710886e+07	-0.0000010	0.9999992	Poisson	BirdID
factor(fac.yr)20	0.2411499	5.838829e-01	0.4130107	0.6795988	Poisson	BirdID
(Intercept)1	0.4447533	3.184734e-01	1.3965165	0.1625590	Poisson	Terr
factor(fac.yr)21	-0.0007476	3.920687e-01	-0.0019069	0.9984786	Poisson	Terr
factor(fac.yr)31	-0.9973905	4.652348e-01	-2.1438432	0.0320455	Poisson	Terr
factor(fac.yr)41	-0.6413188	4.680854e-01	-1.3700893	0.1706590	Poisson	Terr
factor(fac.yr)51	0.0752927	3.685689e-01	0.2042838	0.8381317	Poisson	Terr
factor(fac.yr)61	-0.2777829	3.686724e-01	-0.7534681	0.4511687	Poisson	Terr
factor(fac.yr)71	-0.4285797	3.997229e-01	-1.0721921	0.2836337	Poisson	Terr
factor(fac.yr)81	-0.1398969	3.677882e-01	-0.3803737	0.7036681	Poisson	Terr
factor(fac.yr)91	0.1495406	3.570935e-01	0.4187717	0.6753830	Poisson	Terr
factor(fac.yr)101	-0.3129510	3.824718e-01	-0.8182329	0.4132242	Poisson	Terr
factor(fac.yr)111	-0.4982538	3.854625e-01	-1.2926130	0.1961449	Poisson	Terr
factor(fac.yr)121	-0.2578772	3.803619e-01	-0.6779786	0.4977853	Poisson	Terr
factor(fac.yr)131	0.0775967	3.641766e-01	0.2130743	0.8312690	Poisson	Terr
factor(fac.yr)141	-0.4416734	4.047553e-01	-1.0912110	0.2751801	Poisson	Terr
factor(fac.yr)151	-0.6088926	4.673012e-01	-1.3029982	0.1925754	Poisson	Terr
factor(fac.yr)161	-0.8247789	6.034252e-01	-1.3668286	0.1716790	Poisson	Terr
factor(fac.yr)171	-0.4500965	6.077782e-01	-0.7405605	0.4589600	Poisson	Terr
factor(fac.yr)181	-48.8145533	6.710886e+07	-0.0000007	0.9999994	Poisson	Terr
factor(fac.yr)191	-48.8479609	6.710886e+07	-0.0000007	0.9999994	Poisson	Terr
factor(fac.yr)201	0.3233444	6.228555e-01	0.5191323	0.6036685	Poisson	Terr
(Intercept)2	0.4447543	3.184741e-01	1.3965166	0.1625590	Poisson	BirdID+Terr
factor(fac.yr)22	-0.0007472	3.920689e-01	-0.0019058	0.9984794	Poisson	BirdID+Terr
factor(fac.yr)32	-0.9973888	4.652353e-01	-2.1438373	0.0320459	Poisson	BirdID+Terr
factor(fac.yr)42	-0.6413193	4.680860e-01	-1.3700887	0.1706592	Poisson	BirdID+Terr
factor(fac.yr)52	0.0752926	3.685695e-01	0.2042834	0.8381321	Poisson	BirdID+Terr
factor(fac.yr)62	-0.2777825	3.686731e-01	-0.7534656	0.4511702	Poisson	BirdID+Terr
factor(fac.yr)72	-0.4285778	3.997236e-01	-1.0721853	0.2836368	Poisson	BirdID+Terr
factor(fac.yr)82	-0.1398960	3.677889e-01	-0.3803705	0.7036704	Poisson	BirdID+Terr
factor(fac.yr)92	0.1495419	3.570942e-01	0.4187742	0.6753812	Poisson	BirdID+Terr
factor(fac.yr)102	-0.3129509	3.824725e-01	-0.8182311	0.4132253	Poisson	BirdID+Terr
factor(fac.yr)112	-0.4982543	3.854632e-01	-1.2926118	0.1961453	Poisson	BirdID+Terr
factor(fac.yr)122	-0.2578776	3.803626e-01	-0.6779784	0.4977854	Poisson	BirdID+Terr
factor(fac.yr)132	0.0775961	3.641773e-01	0.2130723	0.8312706	Poisson	BirdID+Terr
factor(fac.yr)142	-0.4416728	4.047560e-01	-1.0912074	0.2751816	Poisson	BirdID+Terr
factor(fac.yr)152	-0.6088921	4.673019e-01	-1.3029951	0.1925764	Poisson	BirdID+Terr
factor(fac.yr)162	-0.8247784	6.034259e-01	-1.3668263	0.1716797	Poisson	BirdID+Terr
factor(fac.yr)172	-0.4501001	6.077790e-01	-0.7405654	0.4589570	Poisson	BirdID+Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)182	-65.8145451	6.710886e+07	-0.0000010	0.9999992	Poisson	BirdID+Terr
factor(fac.yr)192	-65.8479616	6.710886e+07	-0.0000010	0.9999992	Poisson	BirdID+Terr
factor(fac.yr)202	0.3233425	6.228570e-01	0.5191281	0.6036714	Poisson	BirdID+Terr

The models with territory as a random effect are performing roughly identically on the model selection criteria. The band ID only model does have the best BIC score, but none of the band ID smoothers are significant.

Z-score number of fledgling (Gaussian)

Table 121: Z-score GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
36.279532	-330.6102	740.2592	882.7581	181.0698	235.7205	272	0.1836473	107	Gaussian	Terr
36.636754	-330.1947	748.3366	906.8965	180.5175	235.3632	272	0.1849022	203	Gaussian	BandID+Terr
3.309672	-374.8064	762.6537	786.1653	250.5990	268.6903	272	0.0088110	97	Gaussian	BandID

Table 122: Z-score GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	2.3096724	95	0.0253581	0.3936881	Gaussian	BandID
s(Terrs)	35.2795316	105	0.5806145	0.0005375	Gaussian	Terr
s(birdID)	0.3708025	95	0.0039410	0.4621587	Gaussian	BandID+Terr
s(Terrs)	35.2659514	105	0.5811819	0.0005454	Gaussian	BandID+Terr

Table 123: Z-score GAMMs linear terms table.

	Estimate	Std. Error	t value	Pr(> t)	Family	RE
(Intercept)	0.0037583	0.0599095	0.0627331	0.9500257	Gaussian	BandID
(Intercept)1	0.0057587	0.0687630	0.0837477	0.9333281	Gaussian	Terr
(Intercept)2	0.0061552	0.0689023	0.0893325	0.9288936	Gaussian	BandID+Terr

Similar for the previous batch of models. In addition to that the QQ plot for the band ID only model does perhaps suggest that model assumptions do not seem reasonable.

Eggs (Binomial)

Table 124: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
54.04165	-111.73636	337.2825	543.3011	223.4727	221.9583	276	0.2596069	126	Binomial	Terr

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
44.49681	-	342.3066	515.6886	246.5260	231.5032	276	0.2020106	117	Binomial	BirdID
	123.26298									
74.26425	-	344.4774	643.2988	179.4010	201.7357	276	0.3969489	223	Binomial	BirdID+Terr
	89.70051									

Table 125: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	24.49681	95	35.13887	0.0121555	Binomial	BirdID
s(Terrs)	34.04165	105	49.87927	0.0030944	Binomial	Terr
s(birdID)	23.10728	95	34.22884	0.1332323	Binomial	BirdID+Terr
s(Terrs)	31.15697	105	45.33801	0.0683221	Binomial	BirdID+Terr

Table 126: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	1.8818419	1.133560e+00	1.6601169	0.0968909	Binomial	BirdID
factor(fac.yr)2	-0.0282271	1.363444e+00	-0.0207028	0.9834827	Binomial	BirdID
factor(fac.yr)3	-1.9685865	1.255242e+00	-1.5682927	0.1168128	Binomial	BirdID
factor(fac.yr)4	-2.2917837	1.315068e+00	-1.7427106	0.0813842	Binomial	BirdID
factor(fac.yr)5	0.4804984	1.367446e+00	0.3513839	0.7253004	Binomial	BirdID
factor(fac.yr)6	-1.0114388	1.220992e+00	-0.8283744	0.4074585	Binomial	BirdID
factor(fac.yr)7	-0.8759286	1.254145e+00	-0.6984270	0.4849102	Binomial	BirdID
factor(fac.yr)8	0.0160014	1.268790e+00	0.0126116	0.9899377	Binomial	BirdID
factor(fac.yr)9	0.8295307	1.364245e+00	0.6080511	0.5431536	Binomial	BirdID
factor(fac.yr)10	-1.0825420	1.238550e+00	-0.8740400	0.3820965	Binomial	BirdID
factor(fac.yr)11	-1.6056005	1.227065e+00	-1.3084887	0.1907076	Binomial	BirdID
factor(fac.yr)12	-0.5821681	1.260364e+00	-0.4619046	0.6441497	Binomial	BirdID
factor(fac.yr)13	-0.9129503	1.251107e+00	-0.7297138	0.4655652	Binomial	BirdID
factor(fac.yr)14	-0.9072308	1.274844e+00	-0.7116409	0.4766872	Binomial	BirdID
factor(fac.yr)15	-1.4557063	1.328931e+00	-1.0953961	0.2733431	Binomial	BirdID
factor(fac.yr)16	-2.3543089	1.464095e+00	-1.6080298	0.1078287	Binomial	BirdID
factor(fac.yr)17	-1.8910048	1.561689e+00	-1.2108714	0.2259447	Binomial	BirdID
factor(fac.yr)18	-	6.710886e+07	-0.0000006	0.9999995	Binomial	BirdID
	43.1895315					
factor(fac.yr)19	-	6.710886e+07	-0.0000006	0.9999995	Binomial	BirdID
	43.0251689					
factor(fac.yr)20	39.9176630	4.745313e+07	0.0000008	0.9999993	Binomial	BirdID
(Intercept)1	1.9828958	1.193364e+00	1.6616011	0.0965928	Binomial	Terr
factor(fac.yr)21	-0.1566691	1.443897e+00	-0.1085043	0.9135957	Binomial	Terr
factor(fac.yr)31	-2.1339890	1.323960e+00	-1.6118223	0.1070006	Binomial	Terr
factor(fac.yr)41	-2.6324184	1.402859e+00	-1.8764668	0.0605912	Binomial	Terr
factor(fac.yr)51	0.3501105	1.420468e+00	0.2464755	0.8053142	Binomial	Terr
factor(fac.yr)61	-1.2864495	1.281517e+00	-1.0038492	0.3154513	Binomial	Terr
factor(fac.yr)71	-1.1298220	1.322905e+00	-0.8540461	0.3930794	Binomial	Terr
factor(fac.yr)81	-0.2394524	1.326080e+00	-0.1805717	0.8567038	Binomial	Terr
factor(fac.yr)91	0.5968161	1.427328e+00	0.4181353	0.6758482	Binomial	Terr
factor(fac.yr)101	-1.1867888	1.296528e+00	-0.9153593	0.3600030	Binomial	Terr
factor(fac.yr)111	-1.7057693	1.274840e+00	-1.3380266	0.1808878	Binomial	Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)121	-0.5162691	1.323528e+00	-0.3900703	0.6964846	Binomial	Terr
factor(fac.yr)131	-1.1515598	1.317361e+00	-0.8741412	0.3820413	Binomial	Terr
factor(fac.yr)141	-0.9443300	1.334075e+00	-0.7078539	0.4790360	Binomial	Terr
factor(fac.yr)151	-1.6109603	1.386490e+00	-1.1618986	0.2452767	Binomial	Terr
factor(fac.yr)161	-2.5607552	1.547154e+00	-1.6551393	0.0978963	Binomial	Terr
factor(fac.yr)171	-1.5379669	1.628979e+00	-0.9441297	0.3451034	Binomial	Terr
factor(fac.yr)181	-	6.710886e+07	-0.0000007	0.9999995	Binomial	Terr
	44.8964001					
factor(fac.yr)191	-	6.710886e+07	-0.0000007	0.9999995	Binomial	Terr
	44.4243943					
factor(fac.yr)201	40.6362275	4.745313e+07	0.0000009	0.9999993	Binomial	Terr
(Intercept)2	1.8970875	1.249431e+00	1.5183612	0.1289234	Binomial	BirdID+Terr
factor(fac.yr)22	0.0395054	1.494504e+00	0.0264338	0.9789114	Binomial	BirdID+Terr
factor(fac.yr)32	-1.9126085	1.388498e+00	-1.3774660	0.1683682	Binomial	BirdID+Terr
factor(fac.yr)42	-2.4108581	1.488752e+00	-1.6193813	0.1053652	Binomial	BirdID+Terr
factor(fac.yr)52	0.7111506	1.499266e+00	0.4743327	0.6352627	Binomial	BirdID+Terr
factor(fac.yr)62	-1.0900640	1.348939e+00	-0.8080897	0.4190390	Binomial	BirdID+Terr
factor(fac.yr)72	-0.7774518	1.394760e+00	-0.5574092	0.5772479	Binomial	BirdID+Terr
factor(fac.yr)82	0.1934106	1.391092e+00	0.1390351	0.8894224	Binomial	BirdID+Terr
factor(fac.yr)92	1.0832729	1.493459e+00	0.7253451	0.4682404	Binomial	BirdID+Terr
factor(fac.yr)102	-0.9203751	1.373668e+00	-0.6700128	0.5028496	Binomial	BirdID+Terr
factor(fac.yr)112	-1.5547864	1.350282e+00	-1.1514534	0.2495458	Binomial	BirdID+Terr
factor(fac.yr)122	-0.3217848	1.394516e+00	-0.2307501	0.8175090	Binomial	BirdID+Terr
factor(fac.yr)132	-0.9480245	1.389835e+00	-0.6821130	0.4951675	Binomial	BirdID+Terr
factor(fac.yr)142	-0.6842837	1.402962e+00	-0.4877422	0.6257325	Binomial	BirdID+Terr
factor(fac.yr)152	-1.5402996	1.473366e+00	-1.0454291	0.2958247	Binomial	BirdID+Terr
factor(fac.yr)162	-2.4501168	1.644984e+00	-1.4894475	0.1363696	Binomial	BirdID+Terr
factor(fac.yr)172	-1.4609967	1.724879e+00	-0.8470139	0.3969874	Binomial	BirdID+Terr
factor(fac.yr)182	-	6.710886e+07	-0.0000007	0.9999995	Binomial	BirdID+Terr
	44.5338974					
factor(fac.yr)192	-	6.710886e+07	-0.0000007	0.9999995	Binomial	BirdID+Terr
	43.9043870					
factor(fac.yr)202	41.0003385	4.745313e+07	0.0000009	0.9999993	Binomial	BirdID+Terr

Each model has a criteria in which they are performing the best in. All smoothers are significant, except for the band ID smoother in the model fitting both random effects.

Probabilites used to generate simulation 14

Simulation 1

Probability of mate fidelity: 0.447

Probability of pair dispersal: 0.463

Probability of single male dispersal: 0.144

Probability of single female dispersal: 0.405

NULL

Simulation 15

Number of fledgling as the response (Poisson)

Table 127: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npars	Family	RE
47.39760	-	926.4167	1112.316	320.6570	259.6024	307	0.1793084	129	Poisson	Terr
	413.3273									
42.28207	-	930.8631	1099.599	334.3137	264.7179	307	0.1432159	133	Poisson	BirdID
	420.1557									
51.74683	-	943.1905	1176.837	311.8073	255.2532	307	0.1968956	242	Poisson	BirdID+Terr
	408.9024									

Table 128: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	22.282071	112	30.39499	0.0187932	Poisson	BirdID
s(Terrs)	27.397604	108	38.97303	0.0058540	Poisson	Terr
s(birdID)	9.082917	112	10.61582	0.2138515	Poisson	BirdID+Terr
s(Terrs)	22.663916	108	30.88861	0.0228677	Poisson	BirdID+Terr

Table 129: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.7446273	0.2160385	3.4467346	0.0005674	Poisson	BirdID
factor(fac.yr)2	-0.6135366	0.3196406	-1.9194580	0.0549264	Poisson	BirdID
factor(fac.yr)3	-1.0215956	0.3434855	-2.9742033	0.0029375	Poisson	BirdID
factor(fac.yr)4	-0.3208880	0.2842402	-1.1289325	0.2589263	Poisson	BirdID
factor(fac.yr)5	-0.2025919	0.2778707	-0.7290870	0.4659484	Poisson	BirdID
factor(fac.yr)6	-0.3040451	0.2758754	-1.1021102	0.2704138	Poisson	BirdID
factor(fac.yr)7	-0.6236275	0.2945063	-2.1175351	0.0342145	Poisson	BirdID
factor(fac.yr)8	-0.5758526	0.2945344	-1.9551283	0.0505679	Poisson	BirdID
factor(fac.yr)9	-0.1433126	0.2700667	-0.5306562	0.5956570	Poisson	BirdID
factor(fac.yr)10	-0.5015324	0.2975987	-1.6852641	0.0919376	Poisson	BirdID
factor(fac.yr)11	-1.0688045	0.3491521	-3.0611435	0.0022049	Poisson	BirdID
factor(fac.yr)12	-0.7751818	0.3211068	-2.4140937	0.0157744	Poisson	BirdID
factor(fac.yr)13	-0.3273691	0.3195723	-1.0243975	0.3056476	Poisson	BirdID
factor(fac.yr)14	-0.4381716	0.3337400	-1.3129127	0.1892124	Poisson	BirdID
factor(fac.yr)15	-1.0613819	0.4222006	-2.5139279	0.0119395	Poisson	BirdID
factor(fac.yr)16	-0.9808744	0.4232732	-2.3173553	0.0204844	Poisson	BirdID
factor(fac.yr)17	-0.4338385	0.3467715	-1.2510789	0.2109057	Poisson	BirdID
factor(fac.yr)18	-1.2551238	0.5550485	-2.2612865	0.0237415	Poisson	BirdID
factor(fac.yr)19	-0.7856367	0.5093042	-1.5425687	0.1229355	Poisson	BirdID
factor(fac.yr)20	-0.2909282	0.5220142	-0.5573186	0.5773098	Poisson	BirdID
(Intercept)1	0.7416252	0.2175713	3.4086533	0.0006528	Poisson	Terr
factor(fac.yr)21	-0.6417541	0.3245776	-1.9771977	0.0480193	Poisson	Terr
factor(fac.yr)31	-1.0288647	0.3485276	-2.9520325	0.0031569	Poisson	Terr
factor(fac.yr)41	-0.3362352	0.2871675	-1.1708676	0.2416520	Poisson	Terr
factor(fac.yr)51	-0.1739764	0.2807203	-0.6197501	0.5354224	Poisson	Terr
factor(fac.yr)61	-0.2850045	0.2764599	-1.0309074	0.3025842	Poisson	Terr
factor(fac.yr)71	-0.6625556	0.2929468	-2.2616928	0.0237164	Poisson	Terr
factor(fac.yr)81	-0.6356609	0.2936072	-2.1650043	0.0303874	Poisson	Terr
factor(fac.yr)91	-0.1817585	0.2680990	-0.6779528	0.4978016	Poisson	Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)101	-0.5514360	0.2961398	-1.8620802	0.0625918	Poisson	Terr
factor(fac.yr)111	-1.0111011	0.3507971	-2.8822964	0.0039479	Poisson	Terr
factor(fac.yr)121	-0.7171369	0.3223433	-2.2247615	0.0260973	Poisson	Terr
factor(fac.yr)131	-0.2849228	0.3225184	-0.8834310	0.3770034	Poisson	Terr
factor(fac.yr)141	-0.3837240	0.3366105	-1.1399643	0.2543012	Poisson	Terr
factor(fac.yr)151	-1.0101994	0.4247231	-2.3784895	0.0173837	Poisson	Terr
factor(fac.yr)161	-0.8774486	0.4262911	-2.0583322	0.0395583	Poisson	Terr
factor(fac.yr)171	-0.2124175	0.3461498	-0.6136577	0.5394416	Poisson	Terr
factor(fac.yr)181	-1.1445683	0.5530997	-2.0693707	0.0385113	Poisson	Terr
factor(fac.yr)191	-0.7893848	0.5110821	-1.5445360	0.1224585	Poisson	Terr
factor(fac.yr)201	-0.3466710	0.5155901	-0.6723771	0.5013437	Poisson	Terr
(Intercept)2	0.7418871	0.2209204	3.3581651	0.0007846	Poisson	BirdID+Terr
factor(fac.yr)22	-0.6319961	0.3261871	-1.9375262	0.0526811	Poisson	BirdID+Terr
factor(fac.yr)32	-1.0314323	0.3500308	-2.9466904	0.0032119	Poisson	BirdID+Terr
factor(fac.yr)42	-0.3339151	0.2904457	-1.1496645	0.2502821	Poisson	BirdID+Terr
factor(fac.yr)52	-0.1807017	0.2839890	-0.6362981	0.5245821	Poisson	BirdID+Terr
factor(fac.yr)62	-0.2962642	0.2803391	-1.0568065	0.2905999	Poisson	BirdID+Terr
factor(fac.yr)72	-0.6545550	0.2972804	-2.2018102	0.0276787	Poisson	BirdID+Terr
factor(fac.yr)82	-0.6284613	0.2980615	-2.1084956	0.0349881	Poisson	BirdID+Terr
factor(fac.yr)92	-0.1752285	0.2728543	-0.6422055	0.5207398	Poisson	BirdID+Terr
factor(fac.yr)102	-0.5403895	0.3006683	-1.7972949	0.0722888	Poisson	BirdID+Terr
factor(fac.yr)112	-1.0286191	0.3542923	-2.9033063	0.0036925	Poisson	BirdID+Terr
factor(fac.yr)122	-0.7357434	0.3261547	-2.2558110	0.0240825	Poisson	BirdID+Terr
factor(fac.yr)132	-0.2923698	0.3263889	-0.8957712	0.3703750	Poisson	BirdID+Terr
factor(fac.yr)142	-0.4018280	0.3407342	-1.1793006	0.2382785	Poisson	BirdID+Terr
factor(fac.yr)152	-1.0329078	0.4281410	-2.4125414	0.0158417	Poisson	BirdID+Terr
factor(fac.yr)162	-0.9165186	0.4298986	-2.1319413	0.0330117	Poisson	BirdID+Terr
factor(fac.yr)172	-0.2726747	0.3515161	-0.7757103	0.4379201	Poisson	BirdID+Terr
factor(fac.yr)182	-1.1825219	0.5579744	-2.1193119	0.0340641	Poisson	BirdID+Terr
factor(fac.yr)192	-0.7965562	0.5158450	-1.5441775	0.1225453	Poisson	BirdID+Terr
factor(fac.yr)202	-0.3378757	0.5244685	-0.6442249	0.5194296	Poisson	BirdID+Terr

Similar to the previous batch of models.

Z-score number of fledgling (Gaussian)

Table 130: Z-score GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npars	Family	RE
30.61749	-	848.0024	977.7795	226.8612	276.3825	307	0.1248365	110	Gaussian	Terr
	389.1789									
43.32501	-	853.6810	1051.6457	205.1295	263.6750	307	0.1705337	223	Gaussian	BandID+Terr
	373.7220									
22.03031	-	854.6330	954.2648	244.3573	284.9697	307	0.0857474	114	Gaussian	BandID
	400.5829									

Table 131: Z-score GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	21.03031	112	0.2562467	0.0339734	Gaussian	BandID
s(Terrs)	29.61749	108	0.4041570	0.0087487	Gaussian	Terr
s(birdID)	16.08811	112	0.1939254	0.0844124	Gaussian	BandID+Terr
s(Terrs)	26.23690	108	0.3563240	0.0217300	Gaussian	BandID+Terr

Table 132: Z-score GAMMs linear terms table.

	Estimate	Std. Error	t value	Pr(> t)	Family	RE
(Intercept)	-0.0047004	0.0624737	-0.0752377	0.9400784	Gaussian	BandID
(Intercept)1	0.0031493	0.0632783	0.0497697	0.9603419	Gaussian	Terr
(Intercept)2	-0.0028723	0.0671135	-0.0427977	0.9658952	Gaussian	BandID+Terr

Similar to the previous batch of models, except the QQ plot suggests a strong left skew in the residuals for the band ID only model, so I would be more hesitant to trust that model.

Eggs (Binomial)

Table 133: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
67.97545	-	398.9568	663.2182	257.1418	239.0246	307	0.2574876	129	Binomial	Terr
	128.5709									
61.16745	-	402.5817	641.8438	274.1826	245.8326	307	0.2144093	133	Binomial	BirdID
	137.0913									
95.19718	-	404.4301	790.4900	197.2524	211.8028	307	0.4226484	242	Binomial	BirdID+Terr
	98.6262									

Table 134: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	41.16745	111	62.78761	0.0010003	Binomial	BirdID
s(Terrs)	47.97545	108	74.67203	0.0004142	Binomial	Terr
s(birdID)	34.32502	111	50.83223	0.1897079	Binomial	BirdID+Terr
s(Terrs)	40.87215	108	62.38752	0.1078476	Binomial	BirdID+Terr

Table 135: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	1.1034552	7.563470e-01	1.4589271	0.1445852	Binomial	BirdID
factor(fac.yr)2	-0.4407667	9.251479e-01	-0.4764284	0.6337692	Binomial	BirdID
factor(fac.yr)3	-1.2805778	8.949793e-01	-1.4308463	0.1524743	Binomial	BirdID
factor(fac.yr)4	-0.1620957	9.337137e-01	-0.1736032	0.8621773	Binomial	BirdID
factor(fac.yr)5	0.4719923	9.812742e-01	0.4809994	0.6305169	Binomial	BirdID

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)6	-0.0529113	9.289299e-01	-0.0569594	0.9545775	Binomial	BirdID
factor(fac.yr)7	-0.3263169	9.108715e-01	-0.3582469	0.7201586	Binomial	BirdID
factor(fac.yr)8	-0.5627702	9.088633e-01	-0.6192022	0.5357832	Binomial	BirdID
factor(fac.yr)9	0.8432631	9.809573e-01	0.8596328	0.3899915	Binomial	BirdID
factor(fac.yr)10	-0.1933012	9.340622e-01	-0.2069468	0.8360514	Binomial	BirdID
factor(fac.yr)11	-0.6864176	9.361045e-01	-0.7332702	0.4633936	Binomial	BirdID
factor(fac.yr)12	-1.2331399	9.312230e-01	-1.3242154	0.1854315	Binomial	BirdID
factor(fac.yr)13	0.0440840	1.039828e+00	0.0423955	0.9661834	Binomial	BirdID
factor(fac.yr)14	0.3033289	1.092815e+00	0.2775667	0.7813450	Binomial	BirdID
factor(fac.yr)15	-0.7880590	1.052294e+00	-0.7488961	0.4539199	Binomial	BirdID
factor(fac.yr)16	-1.1726155	1.074004e+00	-1.0918173	0.2749134	Binomial	BirdID
factor(fac.yr)17	-0.9790456	1.063121e+00	-0.9209168	0.3570939	Binomial	BirdID
factor(fac.yr)18	-2.3601934	1.253640e+00	-1.8826717	0.0597449	Binomial	BirdID
factor(fac.yr)19	-0.5996400	1.284831e+00	-0.4667074	0.6407093	Binomial	BirdID
factor(fac.yr)20	43.5242765	3.874532e+07	0.0000011	0.9999991	Binomial	BirdID
(Intercept)1	1.5201284	7.895144e-01	1.9253966	0.0541798	Binomial	Terr
factor(fac.yr)21	-1.0363468	9.765983e-01	-1.0611802	0.2886080	Binomial	Terr
factor(fac.yr)31	-1.6918040	9.535126e-01	-1.7742859	0.0760159	Binomial	Terr
factor(fac.yr)41	-0.5051178	9.802668e-01	-0.5152861	0.6063531	Binomial	Terr
factor(fac.yr)51	0.2205676	1.009445e+00	0.2185037	0.8270366	Binomial	Terr
factor(fac.yr)61	-0.3195335	9.613520e-01	-0.3323793	0.7396029	Binomial	Terr
factor(fac.yr)71	-0.9803556	9.248436e-01	-1.0600232	0.2891340	Binomial	Terr
factor(fac.yr)81	-1.3030481	9.288957e-01	-1.4027927	0.1606787	Binomial	Terr
factor(fac.yr)91	0.3938879	9.948800e-01	0.3959150	0.6921678	Binomial	Terr
factor(fac.yr)101	-0.6096197	9.507236e-01	-0.6412165	0.5213820	Binomial	Terr
factor(fac.yr)111	-0.9596843	9.632026e-01	-0.9963473	0.3190814	Binomial	Terr
factor(fac.yr)121	-1.3322742	9.571346e-01	-1.3919404	0.1639405	Binomial	Terr
factor(fac.yr)131	-0.4471299	1.072426e+00	-0.4169333	0.6767272	Binomial	Terr
factor(fac.yr)141	-0.3205462	1.110595e+00	-0.2886257	0.7728679	Binomial	Terr
factor(fac.yr)151	-1.1623361	1.081960e+00	-1.0742873	0.2826939	Binomial	Terr
factor(fac.yr)161	-1.2136833	1.118044e+00	-1.0855414	0.2776819	Binomial	Terr
factor(fac.yr)171	-0.4452353	1.073659e+00	-0.4146898	0.6783690	Binomial	Terr
factor(fac.yr)181	-2.1169763	1.274636e+00	-1.6608483	0.0967439	Binomial	Terr
factor(fac.yr)191	-1.0077687	1.337829e+00	-0.7532869	0.4512775	Binomial	Terr
factor(fac.yr)201	35.7671304	3.874532e+07	0.0000009	0.9999993	Binomial	Terr
(Intercept)2	1.5799497	9.027913e-01	1.7500719	0.0801059	Binomial	BirdID+Terr
factor(fac.yr)22	-0.9578165	1.073315e+00	-0.8923908	0.3721835	Binomial	BirdID+Terr
factor(fac.yr)32	-1.8144202	1.052560e+00	-1.7238167	0.0847409	Binomial	BirdID+Terr
factor(fac.yr)42	-0.4747072	1.109820e+00	-0.4277336	0.6688451	Binomial	BirdID+Terr
factor(fac.yr)52	0.3161741	1.155665e+00	0.2735864	0.7844024	Binomial	BirdID+Terr
factor(fac.yr)62	-0.3815465	1.101788e+00	-0.3462977	0.7291190	Binomial	BirdID+Terr
factor(fac.yr)72	-0.9321079	1.067150e+00	-0.8734557	0.3824147	Binomial	BirdID+Terr
factor(fac.yr)82	-1.2897540	1.075885e+00	-1.1987838	0.2306120	Binomial	BirdID+Terr
factor(fac.yr)92	0.6125104	1.151169e+00	0.5320770	0.5946727	Binomial	BirdID+Terr
factor(fac.yr)102	-0.5764390	1.095607e+00	-0.5261367	0.5987932	Binomial	BirdID+Terr
factor(fac.yr)112	-1.0754545	1.114625e+00	-0.9648576	0.3346161	Binomial	BirdID+Terr
factor(fac.yr)122	-1.5868653	1.095427e+00	-1.4486267	0.1474418	Binomial	BirdID+Terr
factor(fac.yr)132	-0.2820637	1.232879e+00	-0.2287845	0.8190365	Binomial	BirdID+Terr
factor(fac.yr)142	-0.0403853	1.282074e+00	-0.0315000	0.9748708	Binomial	BirdID+Terr
factor(fac.yr)152	-1.2882129	1.233010e+00	-1.0447705	0.2961290	Binomial	BirdID+Terr
factor(fac.yr)162	-1.4630670	1.281478e+00	-1.1417027	0.2535776	Binomial	BirdID+Terr
factor(fac.yr)172	-0.9658084	1.216196e+00	-0.7941221	0.4271243	Binomial	BirdID+Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)182	-2.7614969	1.501470e+00	-1.8391954	0.0658865	Binomial	BirdID+Terr
factor(fac.yr)192	-0.9590298	1.519442e+00	-0.6311723	0.5279279	Binomial	BirdID+Terr
factor(fac.yr)202	42.9614513	3.874532e+07	0.0000011	0.9999991	Binomial	BirdID+Terr

The QQ plot for the model fitting both random effects suggests violations of distributional assumptions. I will only compare the models with one random effect. The territory only model has the best AIC and adjusted R^2 , while the band ID only model has the best BIC. All smoothers were significant.

Probabilites used to generate simulation 15

Simulation 1

Probability of mate fidelity: 0.573

Probability of pair dispersal: 0.278

Probability of single male dispersal: 0.51

Probability of single female dispersal: 0.153

NULL

Simulation 16

Number of fledgling as the response (Poisson)

Table 136: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
20.26390	-	851.3409	927.3679	315.6650	279.7361	300	0.0948450	134	Poisson	BirdID
	405.1436									
36.85532	-	851.9781	999.2526	277.8294	263.1447	300	0.1959727	130	Poisson	Terr
	386.2258									
36.85757	-	851.9923	999.3006	277.8254	263.1424	300	0.1959823	244	Poisson	BirdID+Terr
	386.2238									

Table 137: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	0.2639021	113	0.2643380	0.4669726	Poisson	BirdID
s(Terrs)	16.8553239	109	21.5729268	0.0423342	Poisson	Terr
s(birdID)	0.0024160	113	0.0021622	0.7225970	Poisson	BirdID+Terr
s(Terrs)	16.8551578	109	21.5725727	0.0423413	Poisson	BirdID+Terr

Table 138: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.5107159	0.2237502	2.2825276	0.0224582	Poisson	BirdID
factor(fac.yr)2	-0.3285932	0.3028180	-1.0851180	0.2778694	Poisson	BirdID
factor(fac.yr)3	-0.2875722	0.3000854	-0.9583010	0.3379110	Poisson	BirdID
factor(fac.yr)4	-0.2332123	0.2835000	-0.8226184	0.4107251	Poisson	BirdID

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)5	-0.0780377	0.2776817	-0.2810328	0.7786852	Poisson	BirdID
factor(fac.yr)6	0.0721370	0.2708061	0.2663789	0.7899474	Poisson	BirdID
factor(fac.yr)7	-1.2038210	0.3652727	-3.2956772	0.0009818	Poisson	BirdID
factor(fac.yr)8	-0.1391574	0.2908356	-0.4784745	0.6323125	Poisson	BirdID
factor(fac.yr)9	0.0615511	0.2752093	0.2236518	0.8230282	Poisson	BirdID
factor(fac.yr)10	-0.8672613	0.3486132	-2.4877465	0.0128555	Poisson	BirdID
factor(fac.yr)11	-0.6108949	0.3205189	-1.9059559	0.0566559	Poisson	BirdID
factor(fac.yr)12	-0.6108153	0.3205189	-1.9057076	0.0566881	Poisson	BirdID
factor(fac.yr)13	-0.1743747	0.3486683	-0.5001163	0.6169932	Poisson	BirdID
factor(fac.yr)14	-0.7985203	0.4656468	-1.7148625	0.0863704	Poisson	BirdID
factor(fac.yr)15	-1.6091338	0.7417496	-2.1693760	0.0300542	Poisson	BirdID
factor(fac.yr)16	-0.1538889	0.3875231	-0.3971089	0.6912871	Poisson	BirdID
factor(fac.yr)17	0.0487463	0.4394472	0.1109263	0.9116748	Poisson	BirdID
factor(fac.yr)18	-0.7985226	0.6193464	-1.2892989	0.1972942	Poisson	BirdID
factor(fac.yr)19	-0.5108270	0.5479568	-0.9322394	0.3512128	Poisson	BirdID
factor(fac.yr)20	0.1823344	0.4186368	0.4355431	0.6631683	Poisson	BirdID
(Intercept)1	0.4556735	0.2338178	1.9488397	0.0513146	Poisson	Terr
factor(fac.yr)21	-0.2905131	0.3117156	-0.9319815	0.3513461	Poisson	Terr
factor(fac.yr)31	-0.2137961	0.3091078	-0.6916554	0.4891537	Poisson	Terr
factor(fac.yr)41	-0.1958919	0.2950914	-0.6638347	0.5067961	Poisson	Terr
factor(fac.yr)51	-0.0321830	0.2879812	-0.1117537	0.9110187	Poisson	Terr
factor(fac.yr)61	0.1199549	0.2825217	0.4245866	0.6711381	Poisson	Terr
factor(fac.yr)71	-1.1522863	0.3733922	-3.0859945	0.0020287	Poisson	Terr
factor(fac.yr)81	-0.0896342	0.3009180	-0.2978693	0.7658029	Poisson	Terr
factor(fac.yr)91	0.1032382	0.2868170	0.3599444	0.7188887	Poisson	Terr
factor(fac.yr)101	-0.7710903	0.3566906	-2.1617900	0.0306344	Poisson	Terr
factor(fac.yr)111	-0.5632168	0.3312614	-1.7002185	0.0890898	Poisson	Terr
factor(fac.yr)121	-0.5544517	0.3305408	-1.6774076	0.0934628	Poisson	Terr
factor(fac.yr)131	-0.1666601	0.3621028	-0.4602564	0.6453322	Poisson	Terr
factor(fac.yr)141	-0.7302818	0.4779733	-1.5278715	0.1265444	Poisson	Terr
factor(fac.yr)151	-1.6105594	0.7491292	-2.1499087	0.0315624	Poisson	Terr
factor(fac.yr)161	-0.1049344	0.4012957	-0.2614889	0.7937155	Poisson	Terr
factor(fac.yr)171	0.0879345	0.4563198	0.1927036	0.8471911	Poisson	Terr
factor(fac.yr)181	-0.7136881	0.6341601	-1.1254068	0.2604167	Poisson	Terr
factor(fac.yr)191	-0.4582984	0.5639262	-0.8126922	0.4163946	Poisson	Terr
factor(fac.yr)201	0.1700853	0.4395827	0.3869245	0.6988121	Poisson	Terr
(Intercept)2	0.4556722	0.2338192	1.9488226	0.0513166	Poisson	BirdID+Terr
factor(fac.yr)22	-0.2905138	0.3117162	-0.9319819	0.3513459	Poisson	BirdID+Terr
factor(fac.yr)32	-0.2137955	0.3091086	-0.6916516	0.4891562	Poisson	BirdID+Terr
factor(fac.yr)42	-0.1958915	0.2950927	-0.6638305	0.5067988	Poisson	BirdID+Terr
factor(fac.yr)52	-0.0321832	0.2879826	-0.1117542	0.9110183	Poisson	BirdID+Terr
factor(fac.yr)62	0.1199532	0.2825231	0.4245784	0.6711440	Poisson	BirdID+Terr
factor(fac.yr)72	-1.1522846	0.3733935	-3.0859797	0.0020288	Poisson	BirdID+Terr
factor(fac.yr)82	-0.0896330	0.3009198	-0.2978635	0.7658073	Poisson	BirdID+Terr
factor(fac.yr)92	0.1032363	0.2868188	0.3599356	0.7188953	Poisson	BirdID+Terr
factor(fac.yr)102	-0.7710890	0.3566921	-2.1617775	0.0306353	Poisson	BirdID+Terr
factor(fac.yr)112	-0.5632157	0.3312630	-1.7002071	0.0890920	Poisson	BirdID+Terr
factor(fac.yr)122	-0.5544510	0.3305424	-1.6773977	0.0934647	Poisson	BirdID+Terr
factor(fac.yr)132	-0.1666589	0.3621048	-0.4602504	0.6453365	Poisson	BirdID+Terr
factor(fac.yr)142	-0.7302822	0.4779750	-1.5278669	0.1265456	Poisson	BirdID+Terr
factor(fac.yr)152	-1.6105553	0.7491304	-2.1498997	0.0315631	Poisson	BirdID+Terr
factor(fac.yr)162	-0.1049318	0.4012978	-0.2614811	0.7937216	Poisson	BirdID+Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)172	0.0879351	0.4563227	0.1927037	0.8471910	Poisson	BirdID+Terr
factor(fac.yr)182	-0.7136892	0.6341621	-1.1254050	0.2604174	Poisson	BirdID+Terr
factor(fac.yr)192	-0.4582983	0.5639285	-0.8126887	0.4163966	Poisson	BirdID+Terr
factor(fac.yr)202	0.1700851	0.4395857	0.3869213	0.6988145	Poisson	BirdID+Terr

All models performed roughly the same on AIC, but the band ID only model performed well on BIC, but the worst on adjusted R^2 , with the other two models having roughly the same adjusted R^2 . Only territory smoothers were significant.

Z-score number of fledgling (Gaussian)

Table 139: Z-score GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
37.72665	-361.9333	805.8292	957.0648	199.9321	258.2734	296	0.1755863	111	Gaussian	Terr
55.83977	-338.7233	806.0694	1043.4016	170.9128	240.1602	296	0.2420934	225	Gaussian	BandID+Terr
21.48281	-386.7773	827.3471	926.6038	236.4751	274.5172	296	0.0826014	115	Gaussian	BandID

Table 140: Z-score GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	20.48281	113	0.2350618	0.0586696	Gaussian	BandID
s(Terrs)	36.72665	109	0.5764233	0.0004714	Gaussian	Terr
s(birdID)	18.65463	113	0.2262883	0.0900176	Gaussian	BandID+Terr
s(Terrs)	36.18514	109	0.6047501	0.0006940	Gaussian	BandID+Terr

Table 141: Z-score GAMMs linear terms table.

	Estimate	Std. Error	t value	Pr(> t)	Family	RE
(Intercept)	-0.0083017	0.0620160	-0.1338636	0.8936085	Gaussian	BandID
(Intercept)1	-0.0014618	0.0659723	-0.0221579	0.9823391	Gaussian	Terr
(Intercept)2	-0.0059835	0.0699408	-0.0855516	0.9318941	Gaussian	BandID+Terr

The band ID only model's QQ plot suggested a strong left skew in the residuals, so I will only compare the model fitting both random effects and the territory only model. The model fitting both random effects has the best AIC and adjusted R^2 , while the territory only model has the best BIC. All smoothers are significant.

Eggs (Binomial)

Table 142: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
63.65491	-116.47633	366.2320	613.0508	232.9527	236.3451	300	0.3048892	130	Binomial	Terr
51.18987	-133.05288	375.5216	578.1478	266.1058	248.8101	300	0.2227178	134	Binomial	BirdID
94.52321	-83.45498	378.0512	769.0618	166.9100	205.4768	300	0.4965020	244	Binomial	BirdID+Terr

Table 143: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	31.18987	111	42.92410	0.0105757	Binomial	BirdID
s(Terrs)	43.65491	108	65.49206	0.0012242	Binomial	Terr
s(birdID)	30.01017	111	41.19628	0.3397890	Binomial	BirdID+Terr
s(Terrs)	44.51304	108	66.41547	0.1148286	Binomial	BirdID+Terr

Table 144: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	1.1504232	7.467770e-01	1.5405176	0.1234342	Binomial	BirdID
factor(fac.yr)2	-0.7043557	8.649222e-01	-0.8143573	0.4154403	Binomial	BirdID
factor(fac.yr)3	-0.0023728	9.098865e-01	-0.0026078	0.9979193	Binomial	BirdID
factor(fac.yr)4	0.2215165	9.091060e-01	0.2436641	0.8074910	Binomial	BirdID
factor(fac.yr)5	-0.1198713	8.969157e-01	-0.1336484	0.8936806	Binomial	BirdID
factor(fac.yr)6	0.4098193	9.393953e-01	0.4362587	0.6626491	Binomial	BirdID
factor(fac.yr)7	-1.7608277	8.818479e-01	-1.9967476	0.0458526	Binomial	BirdID
factor(fac.yr)8	0.2492863	9.609277e-01	0.2594225	0.7953093	Binomial	BirdID
factor(fac.yr)9	0.2221241	9.558175e-01	0.2323917	0.8162338	Binomial	BirdID
factor(fac.yr)10	-1.0867808	9.016315e-01	-1.2053493	0.2280685	Binomial	BirdID
factor(fac.yr)11	-0.5005607	9.107310e-01	-0.5496252	0.5825765	Binomial	BirdID
factor(fac.yr)12	-1.0555036	8.960834e-01	-1.1779077	0.2388334	Binomial	BirdID
factor(fac.yr)13	0.1821062	1.133611e+00	0.1606427	0.8723748	Binomial	BirdID
factor(fac.yr)14	-1.3108844	1.087308e+00	-1.2056233	0.2279628	Binomial	BirdID
factor(fac.yr)15	-2.6574716	1.387152e+00	-1.9157750	0.0553937	Binomial	BirdID
factor(fac.yr)16	-0.1135712	1.188646e+00	-0.0955467	0.9238806	Binomial	BirdID
factor(fac.yr)17	-0.0095868	1.464396e+00	-0.0065466	0.9947766	Binomial	BirdID
factor(fac.yr)18	-1.1640152	1.360053e+00	-0.8558604	0.3920750	Binomial	BirdID
factor(fac.yr)19	42.6433720	3.355443e+07	0.0000013	0.9999990	Binomial	BirdID
factor(fac.yr)20	42.4720009	3.355443e+07	0.0000013	0.9999990	Binomial	BirdID
(Intercept)1	0.9075894	8.254337e-01	1.0995304	0.2715368	Binomial	Terr
factor(fac.yr)21	-0.4656327	9.663781e-01	-0.4818329	0.6299247	Binomial	Terr
factor(fac.yr)31	0.4146158	1.004370e+00	0.4128118	0.6797445	Binomial	Terr
factor(fac.yr)41	0.6747511	1.000285e+00	0.6745590	0.4999560	Binomial	Terr
factor(fac.yr)51	0.3055134	9.822512e-01	0.3110339	0.7557748	Binomial	Terr
factor(fac.yr)61	1.1165397	1.025230e+00	1.0890626	0.2761263	Binomial	Terr
factor(fac.yr)71	-1.4787522	9.584146e-01	-1.5429149	0.1228514	Binomial	Terr
factor(fac.yr)81	0.5957496	1.041103e+00	0.5722295	0.5671665	Binomial	Terr
factor(fac.yr)91	0.6424903	1.035258e+00	0.6206087	0.5348571	Binomial	Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)101	-0.6469322	9.642240e-01	-0.6709356	0.5022615	Binomial	Terr
factor(fac.yr)111	-0.0380603	9.882682e-01	-0.0385121	0.9692794	Binomial	Terr
factor(fac.yr)121	-0.6613979	9.717316e-01	-0.6806385	0.4961003	Binomial	Terr
factor(fac.yr)131	0.5969533	1.223744e+00	0.4878092	0.6256850	Binomial	Terr
factor(fac.yr)141	-0.9756800	1.187808e+00	-0.8214125	0.4114114	Binomial	Terr
factor(fac.yr)151	-3.1415002	1.460985e+00	-2.1502616	0.0315345	Binomial	Terr
factor(fac.yr)161	0.2280652	1.278996e+00	0.1783158	0.8584750	Binomial	Terr
factor(fac.yr)171	-0.0619366	1.537110e+00	-0.0402942	0.9678586	Binomial	Terr
factor(fac.yr)181	-0.6914693	1.462188e+00	-0.4729004	0.6362842	Binomial	Terr
factor(fac.yr)191	37.6588264	3.355443e+07	0.0000011	0.9999991	Binomial	Terr
factor(fac.yr)201	37.3094038	3.355443e+07	0.0000011	0.9999991	Binomial	Terr
(Intercept)2	0.9280151	9.841002e-01	0.9430088	0.3456764	Binomial	BirdID+Terr
factor(fac.yr)22	-0.4643594	1.111277e+00	-0.4178612	0.6760486	Binomial	BirdID+Terr
factor(fac.yr)32	0.5323713	1.156343e+00	0.4603920	0.6452348	Binomial	BirdID+Terr
factor(fac.yr)42	0.6940715	1.163672e+00	0.5964495	0.5508750	Binomial	BirdID+Terr
factor(fac.yr)52	0.2632775	1.154324e+00	0.2280793	0.8195846	Binomial	BirdID+Terr
factor(fac.yr)62	1.1538822	1.192385e+00	0.9677093	0.3331896	Binomial	BirdID+Terr
factor(fac.yr)72	-1.5905605	1.146716e+00	-1.3870567	0.1654245	Binomial	BirdID+Terr
factor(fac.yr)82	0.7361166	1.229499e+00	0.5987128	0.5493644	Binomial	BirdID+Terr
factor(fac.yr)92	0.5997067	1.226416e+00	0.4889914	0.6248478	Binomial	BirdID+Terr
factor(fac.yr)102	-0.6125559	1.150464e+00	-0.5324423	0.5944197	Binomial	BirdID+Terr
factor(fac.yr)112	-0.0276687	1.184175e+00	-0.0233654	0.9813588	Binomial	BirdID+Terr
factor(fac.yr)122	-0.6834486	1.162730e+00	-0.5877963	0.5566691	Binomial	BirdID+Terr
factor(fac.yr)132	0.7188440	1.468263e+00	0.4895881	0.6244254	Binomial	BirdID+Terr
factor(fac.yr)142	-1.1346285	1.408942e+00	-0.8053055	0.4206434	Binomial	BirdID+Terr
factor(fac.yr)152	-3.2249964	1.682749e+00	-1.9165043	0.0553009	Binomial	BirdID+Terr
factor(fac.yr)162	0.4544981	1.527313e+00	0.2975802	0.7660236	Binomial	BirdID+Terr
factor(fac.yr)172	-0.1458799	1.765164e+00	-0.0826438	0.9341348	Binomial	BirdID+Terr
factor(fac.yr)182	-0.7505313	1.756360e+00	-0.4273221	0.6691447	Binomial	BirdID+Terr
factor(fac.yr)192	42.7380725	3.355443e+07	0.0000013	0.9999990	Binomial	BirdID+Terr
factor(fac.yr)202	42.2652136	3.355443e+07	0.0000013	0.9999990	Binomial	BirdID+Terr

The QQ plot for the model fitting both random effects suggests violations of distributional assumptions. I will only compare the models with one random effect. The territory only model has the best AIC and adjusted R^2 , while the band ID only model has the best BIC. All smoothers were significant.

Probabilites used to generate simulation 16

Simulation 1
Probability of mate fidelity: 0.612
Probability of pair dispersal: 0.39
Probability of single male dispersal: 0.949
Probability of single female dispersal: 0.535
NULL

Simulation 17

Number of fledgling as the response (Poisson)

Table 145: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npars	Family	RE
20.01448	-360.0470	760.1518	831.1581	284.7972	235.9855	256	0.1236561	127	Poisson	BirdID
32.80828	-345.9467	764.5567	893.3587	256.5967	223.1917	256	0.2071068	120	Poisson	Terr
32.80944	-345.9458	764.5677	893.3927	256.5948	223.1906	256	0.2071110	227	Poisson	BirdID+Terr

Table 146: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	0.0144835	106	0.0142337	0.5069158	Poisson	BirdID
s(Terrs)	12.8082758	99	15.4601260	0.1002151	Poisson	Terr
s(birdID)	0.0018259	106	0.0016064	0.7379204	Poisson	BirdID+Terr
s(Terrs)	12.8076105	99	15.4592048	0.1002284	Poisson	BirdID+Terr

Table 147: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.7309203	0.1924593	3.7977926	0.0001460	Poisson	BirdID
factor(fac.yr)2	-0.4432186	0.2950276	-1.5022955	0.1330208	Poisson	BirdID
factor(fac.yr)3	-0.9132233	0.3220370	-2.8357715	0.0045715	Poisson	BirdID
factor(fac.yr)4	-0.4590010	0.2909667	-1.5775039	0.1146796	Poisson	BirdID
factor(fac.yr)5	-0.0618671	0.2478548	-0.2496103	0.8028887	Poisson	BirdID
factor(fac.yr)6	-0.6796213	0.2950292	-2.3035730	0.0212466	Poisson	BirdID
factor(fac.yr)7	-1.2699328	0.4241473	-2.9940842	0.0027527	Poisson	BirdID
factor(fac.yr)8	-0.4055151	0.3043001	-1.3326155	0.1826580	Poisson	BirdID
factor(fac.yr)9	-0.0377599	0.2872258	-0.1314643	0.8954080	Poisson	BirdID
factor(fac.yr)10	-0.3034995	0.2837629	-1.0695529	0.2848206	Poisson	BirdID
factor(fac.yr)11	-1.2164504	0.4025470	-3.0218840	0.0025121	Poisson	BirdID
factor(fac.yr)12	-0.3680079	0.2837634	-1.2968829	0.1946715	Poisson	BirdID
factor(fac.yr)13	0.0193782	0.2545998	0.0761126	0.9393295	Poisson	BirdID
factor(fac.yr)14	-0.7999350	0.3293512	-2.4288209	0.0151480	Poisson	BirdID
factor(fac.yr)15	-1.1363872	0.4025474	-2.8229901	0.0047578	Poisson	BirdID
factor(fac.yr)16	-0.5486047	0.3469560	-1.5811936	0.1138338	Poisson	BirdID
factor(fac.yr)17	0.0312293	0.3220462	0.0969714	0.9227491	Poisson	BirdID
factor(fac.yr)18	-0.5485886	0.4513495	-1.2154407	0.2241981	Poisson	BirdID
factor(fac.yr)19	-1.1363844	0.5357688	-2.1210351	0.0339188	Poisson	BirdID
factor(fac.yr)20	-0.7309692	0.7328462	-0.9974388	0.3185515	Poisson	BirdID
(Intercept)1	0.7043999	0.2012292	3.5004857	0.0004644	Poisson	Terr
factor(fac.yr)21	-0.4377230	0.3054608	-1.4329921	0.1518601	Poisson	Terr
factor(fac.yr)31	-0.8969586	0.3297311	-2.7202731	0.0065228	Poisson	Terr
factor(fac.yr)41	-0.4441447	0.3013760	-1.4737229	0.1405562	Poisson	Terr
factor(fac.yr)51	-0.0216164	0.2581970	-0.0837205	0.9332787	Poisson	Terr
factor(fac.yr)61	-0.6502197	0.3038644	-2.1398350	0.0323681	Poisson	Terr
factor(fac.yr)71	-1.2223239	0.4316372	-2.8318319	0.0046282	Poisson	Terr
factor(fac.yr)81	-0.3589290	0.3145231	-1.1411848	0.2537930	Poisson	Terr
factor(fac.yr)91	-0.0487640	0.2976882	-0.1638088	0.8698816	Poisson	Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)101	-0.2611600	0.2940175	-0.8882464	0.3744082	Poisson	Terr
factor(fac.yr)111	-1.2277041	0.4096875	-2.9966844	0.0027293	Poisson	Terr
factor(fac.yr)121	-0.3554429	0.2933267	-1.2117649	0.2256024	Poisson	Terr
factor(fac.yr)131	0.0132790	0.2642704	0.0502478	0.9599249	Poisson	Terr
factor(fac.yr)141	-0.7502213	0.3367379	-2.2279087	0.0258866	Poisson	Terr
factor(fac.yr)151	-1.0901861	0.4089359	-2.6659094	0.0076780	Poisson	Terr
factor(fac.yr)161	-0.5046017	0.3569836	-1.4135151	0.1575043	Poisson	Terr
factor(fac.yr)171	0.0587991	0.3350706	0.1754826	0.8607004	Poisson	Terr
factor(fac.yr)181	-0.5130518	0.4644619	-1.1046155	0.2693262	Poisson	Terr
factor(fac.yr)191	-1.0353989	0.5456087	-1.8976951	0.0577363	Poisson	Terr
factor(fac.yr)201	-0.6469373	0.7503222	-0.8622127	0.3885705	Poisson	Terr
(Intercept)2	0.7044051	0.2012300	3.5004981	0.0004644	Poisson	BirdID+Terr
factor(fac.yr)22	-0.4377243	0.3054613	-1.4329945	0.1518594	Poisson	BirdID+Terr
factor(fac.yr)32	-0.8969611	0.3297316	-2.7202767	0.0065227	Poisson	BirdID+Terr
factor(fac.yr)42	-0.4441516	0.3013768	-1.4737419	0.1405511	Poisson	BirdID+Terr
factor(fac.yr)52	-0.0216223	0.2581978	-0.0837433	0.9332605	Poisson	BirdID+Terr
factor(fac.yr)62	-0.6502244	0.3038652	-2.1398449	0.0323673	Poisson	BirdID+Terr
factor(fac.yr)72	-1.2223319	0.4316379	-2.8318456	0.0046280	Poisson	BirdID+Terr
factor(fac.yr)82	-0.3589374	0.3145239	-1.1412085	0.2537832	Poisson	BirdID+Terr
factor(fac.yr)92	-0.0487658	0.2976893	-0.1638144	0.8698772	Poisson	BirdID+Terr
factor(fac.yr)102	-0.2611690	0.2940185	-0.8882743	0.3743932	Poisson	BirdID+Terr
factor(fac.yr)112	-1.2277108	0.4096883	-2.9966947	0.0027292	Poisson	BirdID+Terr
factor(fac.yr)122	-0.3554478	0.2933277	-1.2117771	0.2255977	Poisson	BirdID+Terr
factor(fac.yr)132	0.0132741	0.2642715	0.0502291	0.9599399	Poisson	BirdID+Terr
factor(fac.yr)142	-0.7502316	0.3367389	-2.2279326	0.0258850	Poisson	BirdID+Terr
factor(fac.yr)152	-1.0901922	0.4089369	-2.6659180	0.0076778	Poisson	BirdID+Terr
factor(fac.yr)162	-0.5046088	0.3569848	-1.4135304	0.1574998	Poisson	BirdID+Terr
factor(fac.yr)172	0.0587958	0.3350720	0.1754720	0.8607087	Poisson	BirdID+Terr
factor(fac.yr)182	-0.5130574	0.4644633	-1.1046242	0.2693225	Poisson	BirdID+Terr
factor(fac.yr)192	-1.0354081	0.5456096	-1.8977086	0.0577345	Poisson	BirdID+Terr
factor(fac.yr)202	-0.6469514	0.7503238	-0.8622297	0.3885611	Poisson	BirdID+Terr

The band ID only model had the best AIC and BIC scores, while the other 2 models performed roughly the same on the model selection criteria. Only territory smoothers were significant.

Z-score number of fledgling (Gaussian)

Table 148: Z-score GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npars	Family	RE
28.78540	-	702.5732	818.5769	180.5648	227.2146	256	0.1413325	101	Gaussian	Terr
	318.5650									
36.85613	-	714.1816	885.0559	167.4218	219.1439	256	0.1745117	208	Gaussian	BandID+Terr
	308.8917									
10.92335	-	720.2174	784.0303	217.0277	245.0766	256	0.0431555	108	Gaussian	BandID
	342.1087									

Table 149: Z-score GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	9.923351	106	0.1084999	0.2029971	Gaussian	BandID
s(Terrs)	27.785397	99	0.4239573	0.0082746	Gaussian	Terr
s(birdID)	8.353369	106	0.0933312	0.2372618	Gaussian	BandID+Terr
s(Terrs)	27.502757	99	0.4285924	0.0101403	Gaussian	BandID+Terr

Table 150: Z-score GAMMs linear terms table.

	Estimate	Std. Error	t value	Pr(> t)	Family	RE
(Intercept)	-0.0010810	0.0636239	-0.0169903	0.9864582	Gaussian	BandID
(Intercept)1	0.0033624	0.0681635	0.0493277	0.9607014	Gaussian	Terr
(Intercept)2	0.0014395	0.0706384	0.0203782	0.9837602	Gaussian	BandID+Terr

Each model has a criteria that they are performing well on and only territory smoothers are significant.

Eggs (Binomial)

Warning in newton(lsp = lsp, X = G\$X, y = G\$y, Eb = G\$Eb, UrS = G\$UrS, L = G\$L,
: Fitting terminated with step failure - check results carefully

Table 151: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npars	Family	RE
66.39095	-93.81896	329.2936	580.3909	187.6379	189.6091	256	0.3174449	120	Binomial	Terr
162.51850	0.00000	331.2286	918.3606	0.0000	93.4815	256	1.0000000	227	Binomial	BirdID+Terr
35.82647	-131.96176	347.9959	497.0217	263.9235	220.1735	256	0.1025379	127	Binomial	BirdID

Table 152: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	15.82647	106	18.6790393	0.1375179	Binomial	BirdID
s(Terrs)	46.39095	99	68.0566697	0.0029523	Binomial	Terr
s(birdID)	63.38718	106	0.0000044	1.0000000	Binomial	BirdID+Terr
s(Terrs)	79.13133	99	0.0000027	1.0000000	Binomial	BirdID+Terr

Table 153: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	1.7287194	7.919825e-01	2.1827747	0.0290524	Binomial	BirdID
factor(fac.yr)2	-0.6457723	9.900144e-01	-0.6522857	0.5142169	Binomial	BirdID

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)3	-1.2745079	9.327652e-01	-1.3663758	0.1718210	Binomial	BirdID
factor(fac.yr)4	-1.2423924	9.561286e-01	-1.2993989	0.1938071	Binomial	BirdID
factor(fac.yr)5	0.0780849	1.014581e+00	0.0769627	0.9386532	Binomial	BirdID
factor(fac.yr)6	-0.6964848	9.579694e-01	-0.7270429	0.4671997	Binomial	BirdID
factor(fac.yr)7	-2.1378578	1.000276e+00	-2.1372677	0.0325762	Binomial	BirdID
factor(fac.yr)8	-0.9964024	1.006358e+00	-0.9901068	0.3221219	Binomial	BirdID
factor(fac.yr)9	-0.7538642	1.057871e+00	-0.7126237	0.4760786	Binomial	BirdID
factor(fac.yr)10	-0.3739513	1.035933e+00	-0.3609802	0.7181142	Binomial	BirdID
factor(fac.yr)11	-2.6418970	1.010748e+00	-2.6138027	0.0089541	Binomial	BirdID
factor(fac.yr)12	-0.9164147	9.752080e-01	-0.9397121	0.3473653	Binomial	BirdID
factor(fac.yr)13	-0.1809670	1.031499e+00	-0.1754408	0.8607333	Binomial	BirdID
factor(fac.yr)14	-1.3433536	9.686177e-01	-1.3868770	0.1654793	Binomial	BirdID
factor(fac.yr)15	-1.7116518	1.000321e+00	-1.7111031	0.0870621	Binomial	BirdID
factor(fac.yr)16	-0.9114232	1.073274e+00	-0.8491991	0.3957705	Binomial	BirdID
factor(fac.yr)17	0.0312293	1.366317e+00	0.0228566	0.9817647	Binomial	BirdID
factor(fac.yr)18	-1.3894254	1.248762e+00	-1.1126424	0.2658620	Binomial	BirdID
factor(fac.yr)19	-1.7531913	1.173952e+00	-1.4934095	0.1353300	Binomial	BirdID
factor(fac.yr)20	-1.8832119	1.701513e+00	-1.1067866	0.2683862	Binomial	BirdID
(Intercept)1	1.4895542	8.828874e-01	1.6871396	0.0915765	Binomial	Terr
factor(fac.yr)21	-0.6777924	1.113465e+00	-0.6087238	0.5427076	Binomial	Terr
factor(fac.yr)31	-0.9696260	1.059430e+00	-0.9152338	0.3600689	Binomial	Terr
factor(fac.yr)41	-0.9797941	1.101727e+00	-0.8893258	0.3738280	Binomial	Terr
factor(fac.yr)51	0.9142565	1.133168e+00	0.8068149	0.4197732	Binomial	Terr
factor(fac.yr)61	-0.0839686	1.103313e+00	-0.0761058	0.9393349	Binomial	Terr
factor(fac.yr)71	-1.4902492	1.114866e+00	-1.3367072	0.1813182	Binomial	Terr
factor(fac.yr)81	-0.2053463	1.120321e+00	-0.1832924	0.8545686	Binomial	Terr
factor(fac.yr)91	-0.8303365	1.179634e+00	-0.7038931	0.4814993	Binomial	Terr
factor(fac.yr)101	0.1886609	1.184195e+00	0.1593158	0.8734201	Binomial	Terr
factor(fac.yr)111	-2.9198627	1.146722e+00	-2.5462690	0.0108881	Binomial	Terr
factor(fac.yr)121	-0.4897542	1.100331e+00	-0.4450973	0.6562494	Binomial	Terr
factor(fac.yr)131	0.1862662	1.155017e+00	0.1612671	0.8718830	Binomial	Terr
factor(fac.yr)141	-0.7706855	1.088098e+00	-0.7082868	0.4787671	Binomial	Terr
factor(fac.yr)151	-0.9347432	1.095036e+00	-0.8536188	0.3933163	Binomial	Terr
factor(fac.yr)161	-0.7355660	1.200857e+00	-0.6125344	0.5401842	Binomial	Terr
factor(fac.yr)171	0.3615195	1.466149e+00	0.2465776	0.8052351	Binomial	Terr
factor(fac.yr)181	-1.0620566	1.540865e+00	-0.6892600	0.4906597	Binomial	Terr
factor(fac.yr)191	-0.3249028	1.330840e+00	-0.2441336	0.8071273	Binomial	Terr
factor(fac.yr)201	-0.4047361	1.913584e+00	-0.2115069	0.8324918	Binomial	Terr
(Intercept)2	20.2826711	1.247800e+07	0.0000016	0.9999987	Binomial	BirdID+Terr
factor(fac.yr)22	7.4027696	1.262294e+07	0.0000006	0.9999995	Binomial	BirdID+Terr
factor(fac.yr)32	7.1798682	1.262698e+07	0.0000006	0.9999995	Binomial	BirdID+Terr
factor(fac.yr)42	-	1.455659e+07	-0.0000024	0.9999981	Binomial	BirdID+Terr
	35.1183118					
factor(fac.yr)52	49.8725764	1.536942e+07	0.0000032	0.9999974	Binomial	BirdID+Terr
factor(fac.yr)62	13.2217089	1.069324e+07	0.0000012	0.9999990	Binomial	BirdID+Terr
factor(fac.yr)72	11.4489843	1.065777e+07	0.0000011	0.9999991	Binomial	BirdID+Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)82	-41.7090978	1.065655e+07	-0.0000039	0.9999969	Binomial	BirdID+Terr
factor(fac.yr)92	-33.3798634	1.608369e+07	-0.0000021	0.9999983	Binomial	BirdID+Terr
factor(fac.yr)102	-10.3665962	1.760288e+07	-0.0000006	0.9999995	Binomial	BirdID+Terr
factor(fac.yr)112	-87.9184849	1.519694e+07	-0.0000058	0.9999954	Binomial	BirdID+Terr
factor(fac.yr)122	-15.6072328	1.300417e+07	-0.0000012	0.9999990	Binomial	BirdID+Terr
factor(fac.yr)132	-15.0764785	1.300145e+07	-0.0000012	0.9999991	Binomial	BirdID+Terr
factor(fac.yr)142	-15.4141166	1.300906e+07	-0.0000012	0.9999991	Binomial	BirdID+Terr
factor(fac.yr)152	39.5778233	1.298831e+07	0.0000030	0.9999976	Binomial	BirdID+Terr
factor(fac.yr)162	37.6805508	1.312947e+07	0.0000029	0.9999977	Binomial	BirdID+Terr
factor(fac.yr)172	-7.1049120	1.824425e+07	-0.0000004	0.9999997	Binomial	BirdID+Terr
factor(fac.yr)182	16.9280819	3.562336e+07	0.0000005	0.9999996	Binomial	BirdID+Terr
factor(fac.yr)192	12.9941238	2.033641e+07	0.0000006	0.9999995	Binomial	BirdID+Terr
factor(fac.yr)202	56.9330700	3.030531e+07	0.0000019	0.9999985	Binomial	BirdID+Terr

The model fitting both random effects is clearly overfitting since there is no deviance and an adjusted R^2 of 1, so we will not consider it. The territory only model seems be performing the best on adjusted R^2 and AIC, while the band ID only model is performing the best on BIC. Only the territory smoother is significant.

Probabilites used to generate simulation 17

Simulation 1

Probability of mate fidelity: 0.099

Probability of pair dispersal: 0.648

Probability of single male dispersal: 0.286

Probability of single female dispersal: 0.246

NULL

Simulation 18

Number of fledgling as the response (Poisson)

Table 154: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
30.51644	-367.8599	803.7292	926.7164	292.4731	244.4836	275	0.1912715	126	Poisson	Terr
32.22724	-365.9792	804.8598	936.6934	288.7118	242.7728	275	0.2030162	127	Poisson	BirdID
34.55318	-363.5358	819.9702	987.9667	283.8249	240.4468	275	0.2136177	233	Poisson	BirdID+Terr

Table 155: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	12.227241	105	14.505413	0.1300998	Poisson	BirdID
s(Terrs)	10.516439	105	12.445162	0.1247276	Poisson	Terr
s(birdID)	7.613505	105	8.536981	0.2394099	Poisson	BirdID+Terr
s(Terrs)	6.939676	105	7.825338	0.2193022	Poisson	BirdID+Terr

Table 156: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.8000059	2.097306e-01	3.8144446	0.0001365	Poisson	BirdID
factor(fac.yr)2	-0.7142857	3.397643e-01	-2.1022978	0.0355272	Poisson	BirdID
factor(fac.yr)3	-1.0705926	3.490802e-01	-3.0668959	0.0021629	Poisson	BirdID
factor(fac.yr)4	-0.5184034	3.050154e-01	-1.6995976	0.0892066	Poisson	BirdID
factor(fac.yr)5	0.0329799	2.607121e-01	0.1264992	0.8993368	Poisson	BirdID
factor(fac.yr)6	-0.5654548	2.933246e-01	-1.9277442	0.0538870	Poisson	BirdID
factor(fac.yr)7	-1.2837972	3.505494e-01	-3.6622431	0.0002500	Poisson	BirdID
factor(fac.yr)8	-0.6371051	2.904440e-01	-2.1935558	0.0282674	Poisson	BirdID
factor(fac.yr)9	-0.1326277	2.609189e-01	-0.5083100	0.6112359	Poisson	BirdID
factor(fac.yr)10	-0.7526973	3.145698e-01	-2.3927829	0.0167211	Poisson	BirdID
factor(fac.yr)11	-0.9336183	3.435231e-01	-2.7177741	0.0065723	Poisson	BirdID
factor(fac.yr)12	-1.1143627	3.710475e-01	-3.0032889	0.0026708	Poisson	BirdID
factor(fac.yr)13	-0.4920596	3.034760e-01	-1.6214120	0.1049293	Poisson	BirdID
factor(fac.yr)14	-0.6377530	3.199024e-01	-1.9935860	0.0461973	Poisson	BirdID
factor(fac.yr)15	-1.1464064	4.996911e-01	-2.2942299	0.0217773	Poisson	BirdID
factor(fac.yr)16	-0.3824107	4.019260e-01	-0.9514455	0.3413783	Poisson	BirdID
factor(fac.yr)17	-0.4291299	3.872208e-01	-1.1082306	0.2677623	Poisson	BirdID
factor(fac.yr)18	-0.3575915	3.753348e-01	-0.9527267	0.3407286	Poisson	BirdID
factor(fac.yr)19	-2.1975069	1.026424e+00	-2.1409357	0.0322792	Poisson	BirdID
factor(fac.yr)20	-50.0746236	4.745313e+07	-0.0000011	0.9999992	Poisson	BirdID
(Intercept)1	0.8168661	2.078270e-01	3.9305100	0.0000848	Poisson	Terr
factor(fac.yr)21	-0.7359078	3.423126e-01	-2.1498125	0.0315701	Poisson	Terr
factor(fac.yr)31	-1.0715366	3.478284e-01	-3.0806476	0.0020655	Poisson	Terr
factor(fac.yr)41	-0.5417071	3.041803e-01	-1.7808754	0.0749328	Poisson	Terr
factor(fac.yr)51	0.0302590	2.596037e-01	0.1165586	0.9072099	Poisson	Terr
factor(fac.yr)61	-0.5733156	2.913962e-01	-1.9674780	0.0491281	Poisson	Terr
factor(fac.yr)71	-1.2866069	3.475708e-01	-3.7017120	0.0002141	Poisson	Terr
factor(fac.yr)81	-0.6411570	2.879359e-01	-2.2267354	0.0259650	Poisson	Terr
factor(fac.yr)91	-0.1359029	2.568605e-01	-0.5290922	0.5967415	Poisson	Terr
factor(fac.yr)101	-0.7510556	3.119735e-01	-2.4074342	0.0160651	Poisson	Terr
factor(fac.yr)111	-0.9429930	3.404575e-01	-2.7697820	0.0056094	Poisson	Terr
factor(fac.yr)121	-1.1203256	3.675253e-01	-3.0482952	0.0023014	Poisson	Terr
factor(fac.yr)131	-0.5016382	2.992351e-01	-1.6764014	0.0936596	Poisson	Terr
factor(fac.yr)141	-0.6358755	3.178790e-01	-2.0003700	0.0454603	Poisson	Terr
factor(fac.yr)151	-1.1546093	4.982940e-01	-2.3171246	0.0204969	Poisson	Terr
factor(fac.yr)161	-0.3983453	3.999978e-01	-0.9958686	0.3193140	Poisson	Terr
factor(fac.yr)171	-0.4674067	3.834132e-01	-1.2190681	0.2228184	Poisson	Terr
factor(fac.yr)181	-0.3685567	3.729903e-01	-0.9881134	0.3230971	Poisson	Terr
factor(fac.yr)191	-2.2214368	1.024917e+00	-2.1674315	0.0302020	Poisson	Terr
factor(fac.yr)201	-50.1272497	4.745313e+07	-0.0000011	0.9999992	Poisson	Terr
(Intercept)2	0.8058526	2.111698e-01	3.8161358	0.0001356	Poisson	BirdID+Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)22	-0.7219523	3.432342e-01	-2.1033809	0.0354325	Poisson	BirdID+Terr
factor(fac.yr)32	-1.0666124	3.502535e-01	-3.0452584	0.0023248	Poisson	BirdID+Terr
factor(fac.yr)42	-0.5255706	3.070220e-01	-1.7118341	0.0869272	Poisson	BirdID+Terr
factor(fac.yr)52	0.0343105	2.628186e-01	0.1305484	0.8961326	Poisson	BirdID+Terr
factor(fac.yr)62	-0.5620987	2.949821e-01	-1.9055347	0.0567106	Poisson	BirdID+Terr
factor(fac.yr)72	-1.2818442	3.510691e-01	-3.6512591	0.0002610	Poisson	BirdID+Terr
factor(fac.yr)82	-0.6353466	2.917181e-01	-2.1779474	0.0294100	Poisson	BirdID+Terr
factor(fac.yr)92	-0.1318380	2.615204e-01	-0.5041211	0.6141763	Poisson	BirdID+Terr
factor(fac.yr)102	-0.7486838	3.157502e-01	-2.3711271	0.0177339	Poisson	BirdID+Terr
factor(fac.yr)112	-0.9338078	3.443745e-01	-2.7116056	0.0066958	Poisson	BirdID+Terr
factor(fac.yr)122	-1.1141968	3.714451e-01	-2.9996274	0.0027031	Poisson	BirdID+Terr
factor(fac.yr)132	-0.4955340	3.038810e-01	-1.6306844	0.1029569	Poisson	BirdID+Terr
factor(fac.yr)142	-0.6361355	3.216830e-01	-1.9775229	0.0479825	Poisson	BirdID+Terr
factor(fac.yr)152	-1.1496582	5.016585e-01	-2.2917148	0.0219221	Poisson	BirdID+Terr
factor(fac.yr)162	-0.3852319	4.046361e-01	-0.9520454	0.3410739	Poisson	BirdID+Terr
factor(fac.yr)172	-0.4446873	3.885110e-01	-1.1445941	0.2523773	Poisson	BirdID+Terr
factor(fac.yr)182	-0.3604909	3.777522e-01	-0.9543053	0.3399291	Poisson	BirdID+Terr
factor(fac.yr)192	-2.2102230	1.027444e+00	-2.1511854	0.0314616	Poisson	BirdID+Terr
factor(fac.yr)202	-51.0990310	4.745313e+07	-0.0000011	0.9999991	Poisson	BirdID+Terr

The territory only model has the best AIC and BIC, but the worst adjusted R^2 . However its adjusted R^2 is only 2% lower than the best adjusted R^2 seen in this batch of models. No smoother is significant.

Z-score number of fledgling (Gaussian)

Table 157: Z-score GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
22.26956	-	761.0890	861.1004	211.9769	250.7304	273	0.0946498	107	Gaussian	BandID
	352.8365									
16.84347	-	762.1456	839.8788	222.6419	256.1565	273	0.0692423	107	Gaussian	Terr
	359.5369									
45.38857	-	803.2660	1007.6919	200.1457	227.6114	273	0.0583544	231	Gaussian	BandID+Terr
	344.9971									

Table 158: Z-score GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	21.26956	105	0.2708212	0.0471449	Gaussian	BandID
s(Terrs)	15.84347	105	0.1927144	0.0759136	Gaussian	Terr
s(birdID)	15.47647	105	0.1883004	0.1130023	Gaussian	BandID+Terr
s(Terrs)	10.91209	105	0.1277125	0.1544941	Gaussian	BandID+Terr

Table 159: Z-score GAMMs linear terms table.

	Estimate	Std. Error	t value	Pr(> t)	Family	RE
(Intercept)	0.0025855	0.0659078	0.0392289	0.9687391	Gaussian	BandID

	Estimate	Std. Error	t value	Pr(> t)	Family	RE
(Intercept)1	0.0158778	0.0629126	0.2523790	0.8009505	Gaussian	Terr
(Intercept)2	-0.0223750	0.3002972	-0.0745094	0.9406705	Gaussian	BandID+Terr
factor(fac.yr)2	0.0554696	0.4014424	0.1381758	0.8902237	Gaussian	BandID+Terr
factor(fac.yr)3	0.0416323	0.3801785	0.1095073	0.9128966	Gaussian	BandID+Terr
factor(fac.yr)4	0.0549249	0.3868282	0.1419879	0.8872152	Gaussian	BandID+Terr
factor(fac.yr)5	0.0324992	0.3746039	0.0867560	0.9309417	Gaussian	BandID+Terr
factor(fac.yr)6	0.0656533	0.3731035	0.1759654	0.8604775	Gaussian	BandID+Terr
factor(fac.yr)7	0.0331611	0.3686958	0.0899417	0.9284127	Gaussian	BandID+Terr
factor(fac.yr)8	0.0372464	0.3668339	0.1015349	0.9192153	Gaussian	BandID+Terr
factor(fac.yr)9	0.0316654	0.3633711	0.0871434	0.9306341	Gaussian	BandID+Terr
factor(fac.yr)10	0.0339929	0.3800882	0.0894343	0.9288154	Gaussian	BandID+Terr
factor(fac.yr)11	0.0410894	0.3884053	0.1057900	0.9158421	Gaussian	BandID+Terr
factor(fac.yr)12	0.0309063	0.3929071	0.0786606	0.9373717	Gaussian	BandID+Terr
factor(fac.yr)13	0.0120027	0.3878696	0.0309451	0.9753404	Gaussian	BandID+Terr
factor(fac.yr)14	-0.0007555	0.3950487	-0.0019125	0.9984757	Gaussian	BandID+Terr
factor(fac.yr)15	0.0165394	0.4804828	0.0344225	0.9725704	Gaussian	BandID+Terr
factor(fac.yr)16	0.0821282	0.5034316	0.1631368	0.8705555	Gaussian	BandID+Terr
factor(fac.yr)17	0.0613340	0.4794395	0.1279286	0.8983185	Gaussian	BandID+Terr
factor(fac.yr)18	0.0128097	0.4812148	0.0266196	0.9787865	Gaussian	BandID+Terr
factor(fac.yr)19	-0.0126661	0.5777033	-0.0219249	0.9825271	Gaussian	BandID+Terr

Each model has a criteria that they are performing the best on, but none of them are performing well. No smoothers are significant.

Eggs (Binomial)

Table 160: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
63.60657	-109.22724	350.7890	590.1010	218.4545	211.3934	275	0.3056065	126	Binomial	Terr
65.07282	-109.10811	355.4134	603.5187	218.2162	209.9272	275	0.3064898	127	Binomial	BirdID
106.92296	-60.18179	356.5038	783.5365	120.3636	168.0770	275	0.6151316	233	Binomial	BirdID+Terr

Table 161: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	45.07282	105	68.65827	0.0011669	Binomial	BirdID
s(Terrs)	43.60657	105	69.89452	0.0002869	Binomial	Terr
s(birdID)	42.86587	105	72.89251	0.5517145	Binomial	BirdID+Terr
s(Terrs)	44.05709	105	84.14339	0.3224032	Binomial	BirdID+Terr

Table 162: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	2.4488582	1.194734e+00	2.0497101	0.0403927	Binomial	BirdID
factor(fac.yr)2	-2.1501134	1.354019e+00	-1.5879498	0.1122977	Binomial	BirdID
factor(fac.yr)3	-2.4385052	1.334990e+00	-1.8266091	0.0677586	Binomial	BirdID
factor(fac.yr)4	-0.8510278	1.405624e+00	-0.6054449	0.5448834	Binomial	BirdID
factor(fac.yr)5	-0.3280935	1.425958e+00	-0.2300863	0.8180247	Binomial	BirdID
factor(fac.yr)6	-1.0336292	1.351535e+00	-0.7647815	0.4444017	Binomial	BirdID
factor(fac.yr)7	-2.6105533	1.313158e+00	-1.9879963	0.0468121	Binomial	BirdID
factor(fac.yr)8	-2.0130044	1.313620e+00	-1.5324100	0.1254213	Binomial	BirdID
factor(fac.yr)9	-1.3293708	1.332726e+00	-0.9974826	0.3185303	Binomial	BirdID
factor(fac.yr)10	-2.0060064	1.333138e+00	-1.5047253	0.1323947	Binomial	BirdID
factor(fac.yr)11	-2.4574605	1.338947e+00	-1.8353677	0.0664512	Binomial	BirdID
factor(fac.yr)12	-2.6277710	1.356772e+00	-1.9367810	0.0527721	Binomial	BirdID
factor(fac.yr)13	-1.3217588	1.382351e+00	-0.9561670	0.3389878	Binomial	BirdID
factor(fac.yr)14	-1.5743397	1.388024e+00	-1.1342308	0.2566978	Binomial	BirdID
factor(fac.yr)15	-2.8901284	1.494659e+00	-1.9336370	0.0531578	Binomial	BirdID
factor(fac.yr)16	-1.5305520	1.576510e+00	-0.9708486	0.3316237	Binomial	BirdID
factor(fac.yr)17	-1.2647883	1.569310e+00	-0.8059520	0.4202705	Binomial	BirdID
factor(fac.yr)18	-0.7771613	1.701693e+00	-0.4566989	0.6478875	Binomial	BirdID
factor(fac.yr)19	-3.7809020	1.820533e+00	-2.0768103	0.0378191	Binomial	BirdID
factor(fac.yr)20	-	4.745313e+07	-0.0000009	0.9999993	Binomial	BirdID
	40.6318367					
(Intercept)1	2.5045919	1.149039e+00	2.1797282	0.0292776	Binomial	Terr
factor(fac.yr)21	-2.1170581	1.331246e+00	-1.5902830	0.1117710	Binomial	Terr
factor(fac.yr)31	-2.1439007	1.276357e+00	-1.6797030	0.0930151	Binomial	Terr
factor(fac.yr)41	-1.0812794	1.336334e+00	-0.8091387	0.4184354	Binomial	Terr
factor(fac.yr)51	-0.3637838	1.352261e+00	-0.2690188	0.7879152	Binomial	Terr
factor(fac.yr)61	-0.9592107	1.298109e+00	-0.7389290	0.4599501	Binomial	Terr
factor(fac.yr)71	-2.3720445	1.247728e+00	-1.9010904	0.0572902	Binomial	Terr
factor(fac.yr)81	-2.0294002	1.256606e+00	-1.6149854	0.1063139	Binomial	Terr
factor(fac.yr)91	-1.0622879	1.266375e+00	-0.8388416	0.4015582	Binomial	Terr
factor(fac.yr)101	-1.7133503	1.291932e+00	-1.3261922	0.1847761	Binomial	Terr
factor(fac.yr)111	-2.3915515	1.293206e+00	-1.8493191	0.0644118	Binomial	Terr
factor(fac.yr)121	-2.5034435	1.289541e+00	-1.9413440	0.0522166	Binomial	Terr
factor(fac.yr)131	-1.2093434	1.336808e+00	-0.9046503	0.3656507	Binomial	Terr
factor(fac.yr)141	-1.5079352	1.332338e+00	-1.1317962	0.2577201	Binomial	Terr
factor(fac.yr)151	-2.5075139	1.464657e+00	-1.7120145	0.0868940	Binomial	Terr
factor(fac.yr)161	-1.4040393	1.562309e+00	-0.8986952	0.3688150	Binomial	Terr
factor(fac.yr)171	-1.4184398	1.502269e+00	-0.9441984	0.3450683	Binomial	Terr
factor(fac.yr)181	-0.6736656	1.661866e+00	-0.4053671	0.6852077	Binomial	Terr
factor(fac.yr)191	-3.9916895	1.769186e+00	-2.2562300	0.0240562	Binomial	Terr
factor(fac.yr)201	-	4.745313e+07	-0.0000009	0.9999993	Binomial	Terr
	40.5416850					
(Intercept)2	3.2695149	1.653307e+00	1.9775603	0.0479783	Binomial	BirdID+Terr
factor(fac.yr)22	-3.0557165	1.873816e+00	-1.6307454	0.1029440	Binomial	BirdID+Terr
factor(fac.yr)32	-3.0709346	1.821666e+00	-1.6857833	0.0918375	Binomial	BirdID+Terr
factor(fac.yr)42	-1.1998953	1.935030e+00	-0.6200914	0.5351976	Binomial	BirdID+Terr
factor(fac.yr)52	-0.4448296	1.952148e+00	-0.2278667	0.8197498	Binomial	BirdID+Terr
factor(fac.yr)62	-1.0026642	1.845960e+00	-0.5431668	0.5870150	Binomial	BirdID+Terr
factor(fac.yr)72	-3.2213263	1.788283e+00	-1.8013512	0.0716475	Binomial	BirdID+Terr
factor(fac.yr)82	-2.6604514	1.799196e+00	-1.4786892	0.1392234	Binomial	BirdID+Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)92	-1.8506492	1.839449e+00	-1.0060887	0.3143729	Binomial	BirdID+Terr
factor(fac.yr)102	-2.4771146	1.846556e+00	-1.3414780	0.1797653	Binomial	BirdID+Terr
factor(fac.yr)112	-3.1011093	1.834864e+00	-1.6901030	0.0910083	Binomial	BirdID+Terr
factor(fac.yr)122	-3.3441955	1.846898e+00	-1.8107090	0.0701859	Binomial	BirdID+Terr
factor(fac.yr)132	-1.8474607	1.894682e+00	-0.9750770	0.3295221	Binomial	BirdID+Terr
factor(fac.yr)142	-2.4194761	1.908968e+00	-1.2674265	0.2050028	Binomial	BirdID+Terr
factor(fac.yr)152	-3.3730378	2.060885e+00	-1.6366940	0.1016944	Binomial	BirdID+Terr
factor(fac.yr)162	-1.5331631	2.246456e+00	-0.6824809	0.4949349	Binomial	BirdID+Terr
factor(fac.yr)172	-1.6511665	2.197716e+00	-0.7513100	0.4524661	Binomial	BirdID+Terr
factor(fac.yr)182	-1.5506559	2.236633e+00	-0.6932991	0.4881219	Binomial	BirdID+Terr
factor(fac.yr)192	-5.1626046	2.549181e+00	-2.0252016	0.0428467	Binomial	BirdID+Terr
factor(fac.yr)202	-	4.745313e+07	-0.0000011	0.9999992	Binomial	BirdID+Terr
	49.9907939					

Honestly, model selection here is a bit of a tossup.

The QQ plot for the model fitting both random effects suggests violations of distributional assumptions. I will only compare the models with one random effect. The territory only model has the best AIC and adjusted R^2 , while the band ID only model has the best BIC. Only the smoothers in models with a single random effect are significant.

Probabilites used to generate simulation 18

Simulation 1

Probability of mate fidelity: 0.994

Probability of pair dispersal: 0.1

Probability of single male dispersal: 0.033

Probability of single female dispersal: 0.912

NULL

Simulation 19

Number of fledgling as the response (Poisson)

Table 163: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
36.45994	-	902.0441	1049.249	305.9485	266.5401	303	0.1053203	133	Poisson	BirdID
	411.3841									
36.57491	-	903.6747	1052.854	306.5156	266.4251	303	0.1009046	132	Poisson	Terr
	411.6676									
41.36207	-	916.8412	1110.504	295.7261	261.6379	303	0.1296400	245	Poisson	BirdID+Terr
	406.2728									

Table 164: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	16.459940	112	20.82878	0.0557006	Poisson	BirdID

term	edf	ref.df	statistic	p.value	Family	RE
s(Terrs)	16.574915	111	20.14414	0.0766889	Poisson	Terr
s(birdID)	12.010392	112	14.45140	0.1249209	Poisson	BirdID+Terr
s(Terrs)	9.351683	111	10.55027	0.2110596	Poisson	BirdID+Terr

Table 165: No. fledged GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.3447143	0.3283690	1.0497774	0.2938205	Poisson	BirdID
factor(fac.yr)2	0.1446588	0.3851363	0.3756041	0.7072112	Poisson	BirdID
factor(fac.yr)3	-0.3564132	0.3947272	-0.9029354	0.3665602	Poisson	BirdID
factor(fac.yr)4	-0.3104603	0.3806170	-0.8156765	0.4146852	Poisson	BirdID
factor(fac.yr)5	0.2049117	0.3674339	0.5576832	0.5770607	Poisson	BirdID
factor(fac.yr)6	0.0803867	0.3645818	0.2204902	0.8254894	Poisson	BirdID
factor(fac.yr)7	-0.4090413	0.3891624	-1.0510812	0.2932213	Poisson	BirdID
factor(fac.yr)8	-0.0681519	0.3857510	-0.1766733	0.8597650	Poisson	BirdID
factor(fac.yr)9	0.2681048	0.3687035	0.7271557	0.4671306	Poisson	BirdID
factor(fac.yr)10	-0.0157611	0.3881492	-0.0406058	0.9676102	Poisson	BirdID
factor(fac.yr)11	-0.1878049	0.3923526	-0.4786635	0.6321780	Poisson	BirdID
factor(fac.yr)12	-0.1460242	0.3924522	-0.3720814	0.7098322	Poisson	BirdID
factor(fac.yr)13	-0.2280342	0.4221993	-0.5401103	0.5891210	Poisson	BirdID
factor(fac.yr)14	-0.0930876	0.4291258	-0.2169238	0.8282677	Poisson	BirdID
factor(fac.yr)15	-1.0318589	0.6035867	-1.7095454	0.0873500	Poisson	BirdID
factor(fac.yr)16	-0.0896917	0.4757504	-0.1885267	0.8504638	Poisson	BirdID
factor(fac.yr)17	0.0955491	0.4770111	0.2003079	0.8412398	Poisson	BirdID
factor(fac.yr)18	-0.0216251	0.4913527	-0.0440113	0.9648954	Poisson	BirdID
factor(fac.yr)19	-0.7704166	0.7897505	-0.9755191	0.3293028	Poisson	BirdID
factor(fac.yr)20	-0.3393169	0.6773712	-0.5009321	0.6164189	Poisson	BirdID
(Intercept)1	0.3744184	0.3270299	1.1449058	0.2522482	Poisson	Terr
factor(fac.yr)21	0.1048556	0.3860301	0.2716256	0.7859099	Poisson	Terr
factor(fac.yr)31	-0.3932272	0.3949865	-0.9955460	0.3194708	Poisson	Terr
factor(fac.yr)41	-0.3566252	0.3778331	-0.9438695	0.3452363	Poisson	Terr
factor(fac.yr)51	0.1869423	0.3652553	0.5118128	0.6087820	Poisson	Terr
factor(fac.yr)61	0.0591849	0.3625192	0.1632601	0.8703137	Poisson	Terr
factor(fac.yr)71	-0.4654300	0.3872038	-1.2020284	0.2293525	Poisson	Terr
factor(fac.yr)81	-0.1183044	0.3844601	-0.3077156	0.7582988	Poisson	Terr
factor(fac.yr)91	0.2231352	0.3662200	0.6092929	0.5423303	Poisson	Terr
factor(fac.yr)101	-0.0595360	0.3857306	-0.1543461	0.8773369	Poisson	Terr
factor(fac.yr)111	-0.2463518	0.3901204	-0.6314763	0.5277292	Poisson	Terr
factor(fac.yr)121	-0.1589310	0.3883936	-0.4092008	0.6823923	Poisson	Terr
factor(fac.yr)131	-0.2590855	0.4196022	-0.6174551	0.5369346	Poisson	Terr
factor(fac.yr)141	-0.1170916	0.4274631	-0.2739222	0.7841444	Poisson	Terr
factor(fac.yr)151	-1.0056904	0.6010249	-1.6732926	0.0942697	Poisson	Terr
factor(fac.yr)161	-0.1298988	0.4744243	-0.2738031	0.7842360	Poisson	Terr
factor(fac.yr)171	0.0571155	0.4755507	0.1201040	0.9044008	Poisson	Terr
factor(fac.yr)181	-0.1369552	0.4900736	-0.2794585	0.7798930	Poisson	Terr
factor(fac.yr)191	-0.7468085	0.7889913	-0.9465358	0.3438754	Poisson	Terr
factor(fac.yr)201	-0.2931872	0.6757326	-0.4338805	0.6643752	Poisson	Terr
(Intercept)2	0.3589075	0.3311865	1.0837022	0.2784969	Poisson	BirdID+Terr
factor(fac.yr)22	0.1191517	0.3891295	0.3062006	0.7594519	Poisson	BirdID+Terr
factor(fac.yr)32	-0.3762999	0.3980463	-0.9453671	0.3444715	Poisson	BirdID+Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)42	-0.3347368	0.3827126	-0.8746427	0.3817683	Poisson	BirdID+Terr
factor(fac.yr)52	0.1966276	0.3703353	0.5309449	0.5954570	Poisson	BirdID+Terr
factor(fac.yr)62	0.0715288	0.3674337	0.1946715	0.8456502	Poisson	BirdID+Terr
factor(fac.yr)72	-0.4317536	0.3918841	-1.1017382	0.2705755	Poisson	BirdID+Terr
factor(fac.yr)82	-0.0899059	0.3891242	-0.2310469	0.8172784	Poisson	BirdID+Terr
factor(fac.yr)92	0.2442256	0.3715166	0.6573748	0.5109400	Poisson	BirdID+Terr
factor(fac.yr)102	-0.0368619	0.3910514	-0.0942636	0.9248998	Poisson	BirdID+Terr
factor(fac.yr)112	-0.2136389	0.3951776	-0.5406150	0.5887730	Poisson	BirdID+Terr
factor(fac.yr)122	-0.1480120	0.3942978	-0.3753813	0.7073769	Poisson	BirdID+Terr
factor(fac.yr)132	-0.2313680	0.4248236	-0.5446213	0.5860140	Poisson	BirdID+Terr
factor(fac.yr)142	-0.1022757	0.4326182	-0.2364110	0.8131138	Poisson	BirdID+Terr
factor(fac.yr)152	-1.0150234	0.6057263	-1.6757130	0.0937944	Poisson	BirdID+Terr
factor(fac.yr)162	-0.1059502	0.4798021	-0.2208207	0.8252320	Poisson	BirdID+Terr
factor(fac.yr)172	0.0832250	0.4813894	0.1728850	0.8627418	Poisson	BirdID+Terr
factor(fac.yr)182	-0.0765053	0.4956933	-0.1543401	0.8773416	Poisson	BirdID+Terr
factor(fac.yr)192	-0.7599112	0.7940207	-0.9570420	0.3385460	Poisson	BirdID+Terr
factor(fac.yr)202	-0.3095065	0.6819767	-0.4538374	0.6499459	Poisson	BirdID+Terr

The models with only one random effect are performing similarly, with the band ID only model just performing slightly better. The model with both random effects performs the best on adjusted R^2 , but the worst on the other two criteria. No smoothers were significant.

Z-score number of fledgling (Gaussian)

Table 166: Z-score GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
52.60531	-	829.6542	1054.0909	184.0270	250.3947	303	0.2157094	226	Gaussian	BandID+Terr
	354.3929									
32.05271	-	833.0476	966.4361	218.7890	270.9473	303	0.1382899	114	Gaussian	BandID
	380.6062									
31.34137	-	836.0583	967.4228	221.7701	271.6586	303	0.1288357	113	Gaussian	Terr
	382.6565									

Table 167: Z-score GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	31.05271	112	0.4327326	0.0035775	Gaussian	BandID
s(Terrs)	30.34137	111	0.4023657	0.0077617	Gaussian	Terr
s(birdID)	27.06908	112	0.3848338	0.0122107	Gaussian	BandID+Terr
s(Terrs)	24.53622	111	0.3276655	0.0256437	Gaussian	BandID+Terr

Table 168: Z-score GAMMs linear terms table.

	Estimate	Std. Error	t value	Pr(> t)	Family	RE
(Intercept)	0.0169332	0.0652441	0.2595355	0.7954192	Gaussian	BandID
(Intercept)1	0.0062062	0.0626008	0.0991391	0.9211010	Gaussian	Terr

	Estimate	Std. Error	t value	Pr(> t)	Family	RE
(Intercept)2	0.0182237	0.0684236	0.2663371	0.7901988	Gaussian	BandID+Terr

The model fitting both random effects has the best AIC and adjusted R^2 , while the band ID only model has the best BIC score. All smoothers are significant.

Eggs (Binomial)

Table 169: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
61.42865	-130.7013	390.5614	630.3923	261.4025	241.5714	303	0.2016100	132	Binomial	Terr
52.18318	-141.1302	392.4715	597.1186	282.2605	250.8168	303	0.1593877	133	Binomial	BirdID
80.58043	-110.2214	397.7572	727.0063	220.4427	222.4196	303	0.3148996	245	Binomial	BirdID+Terr

Table 170: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	32.18318	112	46.65664	0.0040748	Binomial	BirdID
s(Terrs)	41.42865	111	60.10021	0.0022771	Binomial	Terr
s(birdID)	25.23242	112	34.83990	0.1270679	Binomial	BirdID+Terr
s(Terrs)	35.34801	111	49.22549	0.0691086	Binomial	BirdID+Terr

Table 171: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	1.6766309	1.139776	1.4710184	0.1412861	Binomial	BirdID
factor(fac.yr)2	-0.4830977	1.312557	-0.3680585	0.7128296	Binomial	BirdID
factor(fac.yr)3	-0.7929616	1.244287	-0.6372821	0.5239411	Binomial	BirdID
factor(fac.yr)4	-1.3768803	1.216174	-1.1321408	0.2575752	Binomial	BirdID
factor(fac.yr)5	-0.3061360	1.269809	-0.2410882	0.8094868	Binomial	BirdID
factor(fac.yr)6	-0.6539959	1.232473	-0.5306369	0.5956704	Binomial	BirdID
factor(fac.yr)7	-1.1577753	1.227587	-0.9431307	0.3456141	Binomial	BirdID
factor(fac.yr)8	-1.0342676	1.251892	-0.8261638	0.4087112	Binomial	BirdID
factor(fac.yr)9	0.1866359	1.322499	0.1411237	0.8877723	Binomial	BirdID
factor(fac.yr)10	-0.3355858	1.301856	-0.2577749	0.7965806	Binomial	BirdID
factor(fac.yr)11	-0.6608744	1.267680	-0.5213259	0.6021398	Binomial	BirdID
factor(fac.yr)12	-0.7957929	1.269569	-0.6268214	0.5307763	Binomial	BirdID
factor(fac.yr)13	-1.1779542	1.290747	-0.9126141	0.3614455	Binomial	BirdID
factor(fac.yr)14	-1.1294033	1.335898	-0.8454264	0.3978728	Binomial	BirdID
factor(fac.yr)15	-1.6489997	1.385219	-1.1904253	0.2338793	Binomial	BirdID
factor(fac.yr)16	-1.3045875	1.422257	-0.9172658	0.3590034	Binomial	BirdID
factor(fac.yr)17	0.1253300	1.636207	0.0765979	0.9389434	Binomial	BirdID
factor(fac.yr)18	0.1105109	1.636986	0.0675088	0.9461767	Binomial	BirdID
factor(fac.yr)19	-2.7759747	1.749588	-1.5866449	0.1125931	Binomial	BirdID

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)20	-1.3321635	1.755527	-0.7588397	0.4479485	Binomial	BirdID
(Intercept)1	1.9371672	1.157744	1.6732265	0.0942827	Binomial	Terr
factor(fac.yr)21	-0.6321709	1.348870	-0.4686669	0.6393078	Binomial	Terr
factor(fac.yr)31	-0.9919420	1.275386	-0.7777584	0.4367115	Binomial	Terr
factor(fac.yr)41	-1.7463230	1.225733	-1.4247176	0.1542389	Binomial	Terr
factor(fac.yr)51	-0.3763053	1.277431	-0.2945798	0.7683149	Binomial	Terr
factor(fac.yr)61	-0.9970769	1.248248	-0.7987808	0.4244175	Binomial	Terr
factor(fac.yr)71	-1.3908396	1.245082	-1.1170665	0.2639659	Binomial	Terr
factor(fac.yr)81	-1.2631112	1.280488	-0.9864292	0.3239225	Binomial	Terr
factor(fac.yr)91	-0.0790348	1.333731	-0.0592584	0.9527463	Binomial	Terr
factor(fac.yr)101	-0.5548094	1.320543	-0.4201372	0.6743852	Binomial	Terr
factor(fac.yr)111	-1.0371069	1.286046	-0.8064307	0.4199945	Binomial	Terr
factor(fac.yr)121	-0.8438068	1.265993	-0.6665178	0.5050802	Binomial	Terr
factor(fac.yr)131	-1.2116099	1.312886	-0.9228597	0.3560803	Binomial	Terr
factor(fac.yr)141	-1.1962637	1.375967	-0.8693987	0.3846291	Binomial	Terr
factor(fac.yr)151	-1.5280137	1.418192	-1.0774381	0.2812846	Binomial	Terr
factor(fac.yr)161	-1.6843105	1.457774	-1.1553991	0.2479270	Binomial	Terr
factor(fac.yr)171	-0.2538147	1.694331	-0.1498023	0.8809206	Binomial	Terr
factor(fac.yr)181	-0.6016790	1.688254	-0.3563913	0.7215476	Binomial	Terr
factor(fac.yr)191	-2.7447247	1.833725	-1.4968027	0.1344446	Binomial	Terr
factor(fac.yr)201	-0.5092441	1.777562	-0.2864846	0.7745070	Binomial	Terr
(Intercept)2	1.8925129	1.217693	1.5541783	0.1201419	Binomial	BirdID+Terr
factor(fac.yr)22	-0.7521881	1.399317	-0.5375393	0.5908951	Binomial	BirdID+Terr
factor(fac.yr)32	-0.9640650	1.335804	-0.7217114	0.4704719	Binomial	BirdID+Terr
factor(fac.yr)42	-1.6511223	1.294461	-1.2755287	0.2021222	Binomial	BirdID+Terr
factor(fac.yr)52	-0.3091883	1.352825	-0.2285501	0.8192186	Binomial	BirdID+Terr
factor(fac.yr)62	-0.8730575	1.318820	-0.6619992	0.5079717	Binomial	BirdID+Terr
factor(fac.yr)72	-1.2924473	1.318904	-0.9799407	0.3271154	Binomial	BirdID+Terr
factor(fac.yr)82	-1.2037606	1.352622	-0.8899460	0.3734949	Binomial	BirdID+Terr
factor(fac.yr)92	0.0084227	1.418153	0.0059392	0.9952613	Binomial	BirdID+Terr
factor(fac.yr)102	-0.4352793	1.403604	-0.3101154	0.7564732	Binomial	BirdID+Terr
factor(fac.yr)112	-0.8804358	1.370545	-0.6423985	0.5206144	Binomial	BirdID+Terr
factor(fac.yr)122	-0.8027916	1.338220	-0.5998951	0.5485761	Binomial	BirdID+Terr
factor(fac.yr)132	-1.0510674	1.395299	-0.7532920	0.4512745	Binomial	BirdID+Terr
factor(fac.yr)142	-1.1521466	1.453644	-0.7925918	0.4280157	Binomial	BirdID+Terr
factor(fac.yr)152	-1.5026220	1.508342	-0.9962075	0.3191493	Binomial	BirdID+Terr
factor(fac.yr)162	-1.5551172	1.555028	-1.0000573	0.3172828	Binomial	BirdID+Terr
factor(fac.yr)172	-0.0238561	1.801712	-0.0132408	0.9894357	Binomial	BirdID+Terr
factor(fac.yr)182	-0.4356396	1.755793	-0.2481157	0.8040449	Binomial	BirdID+Terr
factor(fac.yr)192	-2.9740546	1.961358	-1.5163246	0.1294373	Binomial	BirdID+Terr
factor(fac.yr)202	-0.8605529	1.890561	-0.4551839	0.6489770	Binomial	BirdID+Terr

Each model has a criteria that they are performing the best in. Only models with one random effect has significant smoothers.

Probabilites used to generate simulation 19

[1] "This dataset was generated with a probability of mate fidelity of 0.871, a probability of pair disj"

Simulation 20

Number of fledgling as the response (Poisson)

Table 172: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npars	Family	RE
49.50829	-	949.2837	1144.487	300.3748	279.4917	329	0.2490112	126	Poisson	Terr
	423.2192									
49.52063	-	949.3929	1144.847	300.3517	279.4794	329	0.2490638	241	Poisson	BirdID+Terr
	423.2076									
36.57512	-	963.0600	1113.279	337.8515	292.4249	329	0.1448679	135	Poisson	BirdID
	441.9575									

Table 173: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	16.5751242	114	21.0711113	0.0480970	Poisson	BirdID
s(Terrs)	29.5082915	105	46.1653227	0.0005898	Poisson	Terr
s(birdID)	0.0138878	114	0.0135597	0.5324135	Poisson	BirdID+Terr
s(Terrs)	29.5067422	105	46.1609181	0.0005912	Poisson	BirdID+Terr

Table 174: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.3563138	0.2436667	1.4622996	0.1436591	Poisson	BirdID
factor(fac.yr)2	0.0364894	0.3024142	0.1206603	0.9039601	Poisson	BirdID
factor(fac.yr)3	-0.6474668	0.3564173	-1.8165973	0.0692788	Poisson	BirdID
factor(fac.yr)4	0.0887807	0.3153719	0.2815113	0.7783183	Poisson	BirdID
factor(fac.yr)5	0.3667640	0.2860855	1.2820084	0.1998397	Poisson	BirdID
factor(fac.yr)6	-0.0830891	0.2973438	-0.2794377	0.7799089	Poisson	BirdID
factor(fac.yr)7	-0.1909050	0.3071026	-0.6216325	0.5341835	Poisson	BirdID
factor(fac.yr)8	-0.1602426	0.3036765	-0.5276753	0.5977247	Poisson	BirdID
factor(fac.yr)9	0.1244498	0.2938813	0.4234696	0.6719527	Poisson	BirdID
factor(fac.yr)10	-0.0737662	0.3142958	-0.2347032	0.8144391	Poisson	BirdID
factor(fac.yr)11	-0.5454673	0.3525937	-1.5470138	0.1218599	Poisson	BirdID
factor(fac.yr)12	-0.2817489	0.3275357	-0.8602082	0.3896743	Poisson	BirdID
factor(fac.yr)13	0.1298769	0.3176828	0.4088256	0.6826676	Poisson	BirdID
factor(fac.yr)14	-0.5172686	0.3823281	-1.3529445	0.1760734	Poisson	BirdID
factor(fac.yr)15	-1.5244703	0.5123231	-2.9756030	0.0029241	Poisson	BirdID
factor(fac.yr)16	-0.2412843	0.4200438	-0.5744265	0.5656792	Poisson	BirdID
factor(fac.yr)17	-0.5025604	0.4825064	-1.0415622	0.2976147	Poisson	BirdID
factor(fac.yr)18	0.1944039	0.4626151	0.4202282	0.6743188	Poisson	BirdID
factor(fac.yr)19	0.0384064	0.4882938	0.0786543	0.9373076	Poisson	BirdID
factor(fac.yr)20	-0.2152593	0.4591345	-0.4688370	0.6391861	Poisson	BirdID
(Intercept)1	0.3275452	0.2510977	1.3044529	0.1920792	Poisson	Terr
factor(fac.yr)21	0.0481637	0.3146072	0.1530916	0.8783260	Poisson	Terr
factor(fac.yr)31	-0.6079770	0.3646737	-1.6671807	0.0954785	Poisson	Terr
factor(fac.yr)41	0.1138670	0.3249819	0.3503795	0.7260539	Poisson	Terr
factor(fac.yr)51	0.3586043	0.2940511	1.2195303	0.2226430	Poisson	Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)61	-0.0930576	0.3048110	-0.3052960	0.7601407	Poisson	Terr
factor(fac.yr)71	-0.1749822	0.3150181	-0.5554672	0.5785752	Poisson	Terr
factor(fac.yr)81	-0.1250958	0.3093078	-0.4044378	0.6858908	Poisson	Terr
factor(fac.yr)91	0.2068204	0.3011368	0.6867986	0.4922096	Poisson	Terr
factor(fac.yr)101	-0.0612296	0.3215024	-0.1904484	0.8489578	Poisson	Terr
factor(fac.yr)111	-0.5109115	0.3583716	-1.4256472	0.1539702	Poisson	Terr
factor(fac.yr)121	-0.2879855	0.3358562	-0.8574668	0.3911870	Poisson	Terr
factor(fac.yr)131	0.1986968	0.3258424	0.6097942	0.5419981	Poisson	Terr
factor(fac.yr)141	-0.5235193	0.3865174	-1.3544520	0.1755922	Poisson	Terr
factor(fac.yr)151	-1.5625601	0.5158949	-3.0288344	0.0024550	Poisson	Terr
factor(fac.yr)161	-0.2157311	0.4324175	-0.4988955	0.6178530	Poisson	Terr
factor(fac.yr)171	-0.6032845	0.4910273	-1.2286170	0.2192154	Poisson	Terr
factor(fac.yr)181	0.3430121	0.4758080	0.7209045	0.4709683	Poisson	Terr
factor(fac.yr)191	0.0056710	0.4998501	0.0113454	0.9909478	Poisson	Terr
factor(fac.yr)201	-0.0994377	0.4699707	-0.2115828	0.8324326	Poisson	Terr
(Intercept)2	0.3275633	0.2511041	1.3044918	0.1920659	Poisson	BirdID+Terr
factor(fac.yr)22	0.0481376	0.3146114	0.1530065	0.8783932	Poisson	BirdID+Terr
factor(fac.yr)32	-0.6080016	0.3646791	-1.6672235	0.0954700	Poisson	BirdID+Terr
factor(fac.yr)42	0.1138538	0.3249896	0.3503305	0.7260907	Poisson	BirdID+Terr
factor(fac.yr)52	0.3585949	0.2940586	1.2194675	0.2226668	Poisson	BirdID+Terr
factor(fac.yr)62	-0.0930878	0.3048180	-0.3053881	0.7600706	Poisson	BirdID+Terr
factor(fac.yr)72	-0.1750123	0.3150253	-0.5555499	0.5785186	Poisson	BirdID+Terr
factor(fac.yr)82	-0.1251297	0.3093153	-0.4045376	0.6858174	Poisson	BirdID+Terr
factor(fac.yr)92	0.2067790	0.3011446	0.6866434	0.4923075	Poisson	BirdID+Terr
factor(fac.yr)102	-0.0612672	0.3215104	-0.1905606	0.8488699	Poisson	BirdID+Terr
factor(fac.yr)112	-0.5109508	0.3583793	-1.4257265	0.1539473	Poisson	BirdID+Terr
factor(fac.yr)122	-0.2880122	0.3358641	-0.8575261	0.3911542	Poisson	BirdID+Terr
factor(fac.yr)132	0.1986722	0.3258514	0.6097017	0.5420594	Poisson	BirdID+Terr
factor(fac.yr)142	-0.5235558	0.3865255	-1.3545183	0.1755711	Poisson	BirdID+Terr
factor(fac.yr)152	-1.5625920	0.5159006	-3.0288624	0.0024548	Poisson	BirdID+Terr
factor(fac.yr)162	-0.2157669	0.4324271	-0.4989672	0.6178025	Poisson	BirdID+Terr
factor(fac.yr)172	-0.6033062	0.4910371	-1.2286367	0.2192080	Poisson	BirdID+Terr
factor(fac.yr)182	0.3429819	0.4758213	0.7208209	0.4710197	Poisson	BirdID+Terr
factor(fac.yr)192	0.0056761	0.4998628	0.0113554	0.9909399	Poisson	BirdID+Terr
factor(fac.yr)202	-0.0994700	0.4699813	-0.2116468	0.8323826	Poisson	BirdID+Terr

The territory only and both random effects models performed nearly identically on the selection criteria. Having the best AIC and adjusted R^2 s. However, the band ID only model had the best BIC. All smoothers are significant, except for the band ID smoother in the model fitting both random effects.

Z-score number of fledgling (Gaussian)

Table 175: Z-score GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npars	Family	RE
44.56624	-395.4137	885.6796	1065.712	213.1319	284.4338	329	0.2046053	107	Gaussian	Terr
58.33237	-378.1611	887.0516	1135.180	191.9111	270.6676	329	0.2473741	222	Gaussian	BandID+Terr

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
28.39236	-	909.7056	1031.288	251.7760	300.6076	329	0.1109427	116	Gaussian	BandID
	422.8241									

Table 176: Z-score GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	27.39236	114	0.3590369	0.0065403	Gaussian	BandID
s(Terrs)	43.56624	105	0.8035611	0.0000158	Gaussian	Terr
s(birdID)	16.82574	114	0.2091233	0.0621583	Gaussian	BandID+Terr
s(Terrs)	40.50663	105	0.7661651	0.0000691	Gaussian	BandID+Terr

Table 177: Z-score GAMMs linear terms table.

	Estimate	Std. Error	t value	Pr(> t)	Family	RE
(Intercept)	-0.0070428	0.0609344	-0.1155794	0.9080631	Gaussian	BandID
(Intercept)1	-0.0022272	0.0657773	-0.0338604	0.9730122	Gaussian	Terr
(Intercept)2	-0.0100044	0.0680656	-0.1469824	0.8832553	Gaussian	BandID+Terr

Each model has a criteria that they are performing well in. The band ID smoother in the model fitting both random effects is not significant.

Eggs (Binomial)

Table 178: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
56.45577	-	419.1138	644.3420	300.4496	272.5442	329	0.2012127	126	Binomial	Terr
	150.2248									
47.28368	-	425.5088	616.8887	324.6780	281.7163	329	0.1470914	135	Binomial	BirdID
	162.3390									
65.27394	-	432.4274	716.1034	282.9692	263.7261	329	0.2398019	241	Binomial	BirdID+Terr
	141.4846									

Table 179: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	27.28368	113	37.19476	0.0133392	Binomial	BirdID
s(Terrs)	36.45577	105	53.11019	0.0027744	Binomial	Terr
s(birdID)	13.47223	113	15.72956	0.2604706	Binomial	BirdID+Terr
s(Terrs)	31.80172	105	44.54387	0.0201438	Binomial	BirdID+Terr

Table 180: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.9298813	6.467456e-01	1.4377854	0.1504950	Binomial	BirdID
factor(fac.yr)2	0.2088286	8.217065e-01	0.2541402	0.7993873	Binomial	BirdID
factor(fac.yr)3	-1.1178154	8.031261e-01	-1.3918306	0.1639737	Binomial	BirdID
factor(fac.yr)4	0.1385703	8.562017e-01	0.1618430	0.8714295	Binomial	BirdID
factor(fac.yr)5	0.7532400	8.627559e-01	0.8730627	0.3826289	Binomial	BirdID
factor(fac.yr)6	-0.0903323	7.861369e-01	-0.1149066	0.9085192	Binomial	BirdID
factor(fac.yr)7	-0.0160536	8.031760e-01	-0.0199877	0.9840532	Binomial	BirdID
factor(fac.yr)8	-0.1053024	7.901012e-01	-0.1332771	0.8939742	Binomial	BirdID
factor(fac.yr)9	0.3482742	8.386060e-01	0.4153014	0.6779213	Binomial	BirdID
factor(fac.yr)10	-0.3083313	8.233880e-01	-0.3744666	0.7080572	Binomial	BirdID
factor(fac.yr)11	-0.3968816	8.276158e-01	-0.4795481	0.6315488	Binomial	BirdID
factor(fac.yr)12	0.2212668	8.510285e-01	0.2599993	0.7948643	Binomial	BirdID
factor(fac.yr)13	-0.1042004	8.702034e-01	-0.1197425	0.9046871	Binomial	BirdID
factor(fac.yr)14	-0.3434785	8.837740e-01	-0.3886497	0.6975353	Binomial	BirdID
factor(fac.yr)15	-2.0269168	8.923589e-01	-2.2714143	0.0231219	Binomial	BirdID
factor(fac.yr)16	-0.8918202	9.997635e-01	-0.8920312	0.3723762	Binomial	BirdID
factor(fac.yr)17	-1.1605972	1.046480e+00	-1.1090485	0.2674092	Binomial	BirdID
factor(fac.yr)18	41.5724083	3.355443e+07	0.0000012	0.9999990	Binomial	BirdID
factor(fac.yr)19	0.3050551	1.390113e+00	0.2194463	0.8263024	Binomial	BirdID
factor(fac.yr)20	0.7494765	1.320053e+00	0.5677623	0.5701964	Binomial	BirdID
(Intercept)1	0.8142953	6.656062e-01	1.2233890	0.2211828	Binomial	Terr
factor(fac.yr)21	0.3651883	8.608373e-01	0.4242246	0.6714020	Binomial	Terr
factor(fac.yr)31	-0.9031072	8.355700e-01	-1.0808277	0.2797738	Binomial	Terr
factor(fac.yr)41	0.2095797	8.849720e-01	0.2368207	0.8127959	Binomial	Terr
factor(fac.yr)51	0.8511797	8.867049e-01	0.9599358	0.3370875	Binomial	Terr
factor(fac.yr)61	0.0481438	8.078172e-01	0.0595973	0.9524763	Binomial	Terr
factor(fac.yr)71	0.2390978	8.235412e-01	0.2903288	0.7715647	Binomial	Terr
factor(fac.yr)81	0.0931880	8.070702e-01	0.1154645	0.9080769	Binomial	Terr
factor(fac.yr)91	0.8030514	8.538296e-01	0.9405288	0.3469464	Binomial	Terr
factor(fac.yr)101	-0.0692390	8.469827e-01	-0.0817478	0.9348472	Binomial	Terr
factor(fac.yr)111	-0.1196810	8.444313e-01	-0.1417297	0.8872935	Binomial	Terr
factor(fac.yr)121	0.1883777	8.771524e-01	0.2147606	0.8299540	Binomial	Terr
factor(fac.yr)131	0.1937804	8.872719e-01	0.2184003	0.8271172	Binomial	Terr
factor(fac.yr)141	-0.1612955	9.014112e-01	-0.1789367	0.8579874	Binomial	Terr
factor(fac.yr)151	-2.1570483	9.150215e-01	-2.3573744	0.0184047	Binomial	Terr
factor(fac.yr)161	-0.9301386	1.043278e+00	-0.8915543	0.3726319	Binomial	Terr
factor(fac.yr)171	-1.3297065	1.083635e+00	-1.2270803	0.2197924	Binomial	Terr
factor(fac.yr)181	44.0779505	3.355443e+07	0.0000013	0.9999990	Binomial	Terr
factor(fac.yr)191	0.2514252	1.453777e+00	0.1729461	0.8626938	Binomial	Terr
factor(fac.yr)201	1.1690130	1.336826e+00	0.8744688	0.3818630	Binomial	Terr
(Intercept)2	0.8555724	6.792171e-01	1.2596450	0.2077974	Binomial	BirdID+Terr
factor(fac.yr)22	0.3330874	8.668463e-01	0.3842520	0.7007917	Binomial	BirdID+Terr
factor(fac.yr)32	-0.9670511	8.520715e-01	-1.1349412	0.2564000	Binomial	BirdID+Terr
factor(fac.yr)42	0.2394327	9.042653e-01	0.2647814	0.7911778	Binomial	BirdID+Terr
factor(fac.yr)52	0.8426771	9.006733e-01	0.9356080	0.3494750	Binomial	BirdID+Terr
factor(fac.yr)62	-0.0089984	8.254082e-01	-0.0109017	0.9913019	Binomial	BirdID+Terr
factor(fac.yr)72	0.1592371	8.421998e-01	0.1890728	0.8500357	Binomial	BirdID+Terr
factor(fac.yr)82	0.0331529	8.255814e-01	0.0401571	0.9679679	Binomial	BirdID+Terr
factor(fac.yr)92	0.6753137	8.718485e-01	0.7745769	0.4385897	Binomial	BirdID+Terr
factor(fac.yr)102	-0.1284670	8.667851e-01	-0.1482109	0.8821763	Binomial	BirdID+Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)112	-0.2072861	8.649775e-01	-0.2396434	0.8106067	Binomial	BirdID+Terr
factor(fac.yr)122	0.1838654	8.954229e-01	0.2053392	0.8373071	Binomial	BirdID+Terr
factor(fac.yr)132	0.1558648	9.105812e-01	0.1711707	0.8640895	Binomial	BirdID+Terr
factor(fac.yr)142	-0.2307074	9.222642e-01	-0.2501532	0.8024688	Binomial	BirdID+Terr
factor(fac.yr)152	-2.1728148	9.364070e-01	-2.3203743	0.0203206	Binomial	BirdID+Terr
factor(fac.yr)162	-0.9597307	1.063469e+00	-0.9024529	0.3668163	Binomial	BirdID+Terr
factor(fac.yr)172	-1.3498963	1.108733e+00	-1.2175128	0.2234092	Binomial	BirdID+Terr
factor(fac.yr)182	45.8672339	3.355443e+07	0.0000014	0.9999989	Binomial	BirdID+Terr
factor(fac.yr)192	0.3408163	1.479919e+00	0.2302939	0.8178634	Binomial	BirdID+Terr
factor(fac.yr)202	1.1115158	1.364321e+00	0.8147029	0.4152424	Binomial	BirdID+Terr

Similar to the batch of models before.

Probabilites used to generate simulation 20

Simulation 1

Probability of mate fidelity: 0.777

Probability of pair dispersal: 0.201

Probability of single male dispersal: 0.608

Probability of single female dispersal: 0.496

NULL

Simulation 21

Number of fledgling as the response (Poisson)

Table 181: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npars	Family	RE
42.78233	-	993.0198	1166.801	333.6958	297.2177	340	0.1983159	130	Poisson	Terr
	451.1237									
42.80663	-	993.3170	1167.757	333.6490	297.1934	340	0.1984095	241	Poisson	BirdID+Terr
	451.1003									
32.96150	-	1001.1731	1142.458	358.8231	307.0385	340	0.1468089	131	Poisson	BirdID
	463.6873									

Table 182: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	12.9614982	110	15.4683723	0.1161489	Poisson	BirdID
s(Terr)	22.7823268	109	30.8527619	0.0148203	Poisson	Terr
s(birdID)	0.0293508	110	0.0290747	0.4939072	Poisson	BirdID+Terr
s(Terr)	22.7772750	109	30.8425753	0.0148694	Poisson	BirdID+Terr

Table 183: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.9832499	0.1940727	5.0664009	0.0000004	Poisson	BirdID
factor(fac.yr)2	-1.2340717	0.3032463	-4.0695358	0.0000471	Poisson	BirdID
factor(fac.yr)3	-1.2102001	0.2881842	-4.1993978	0.0000268	Poisson	BirdID
factor(fac.yr)4	-0.6522559	0.2550224	-2.5576413	0.0105385	Poisson	BirdID
factor(fac.yr)5	-0.6030287	0.2497291	-2.4147313	0.0157468	Poisson	BirdID
factor(fac.yr)6	-0.7718174	0.2512918	-3.0713989	0.0021306	Poisson	BirdID
factor(fac.yr)7	-1.0037973	0.2832788	-3.5434965	0.0003949	Poisson	BirdID
factor(fac.yr)8	-0.6199205	0.2648908	-2.3402866	0.0192689	Poisson	BirdID
factor(fac.yr)9	-0.4421284	0.2503234	-1.7662284	0.0773575	Poisson	BirdID
factor(fac.yr)10	-0.8634759	0.2840425	-3.0399533	0.0023661	Poisson	BirdID
factor(fac.yr)11	-1.0210261	0.3511739	-2.9074660	0.0036437	Poisson	BirdID
factor(fac.yr)12	-1.0929784	0.3337593	-3.2747502	0.0010576	Poisson	BirdID
factor(fac.yr)13	-0.5390630	0.2771106	-1.9452995	0.0517389	Poisson	BirdID
factor(fac.yr)14	-0.7824981	0.3147616	-2.4860021	0.0129187	Poisson	BirdID
factor(fac.yr)15	-0.6291576	0.3153838	-1.9948949	0.0460543	Poisson	BirdID
factor(fac.yr)16	-0.3588975	0.3299181	-1.0878379	0.2766667	Poisson	BirdID
factor(fac.yr)17	-0.1779722	0.3245126	-0.5484291	0.5833973	Poisson	BirdID
factor(fac.yr)18	-1.6915775	0.7391876	-2.2884278	0.0221126	Poisson	BirdID
factor(fac.yr)19	-0.9907031	0.4947468	-2.0024449	0.0452369	Poisson	BirdID
factor(fac.yr)20	-1.8313292	0.6133489	-2.9857871	0.0028285	Poisson	BirdID
(Intercept)1	0.9332715	0.2000402	4.6654207	0.0000031	Poisson	Terr
factor(fac.yr)21	-1.2007469	0.3126612	-3.8404085	0.0001228	Poisson	Terr
factor(fac.yr)31	-1.1732190	0.2918483	-4.0199613	0.0000582	Poisson	Terr
factor(fac.yr)41	-0.6168071	0.2619018	-2.3551086	0.0185173	Poisson	Terr
factor(fac.yr)51	-0.5618135	0.2537595	-2.2139605	0.0268315	Poisson	Terr
factor(fac.yr)61	-0.7234443	0.2555220	-2.8312414	0.0046368	Poisson	Terr
factor(fac.yr)71	-0.9159447	0.2894099	-3.1648698	0.0015515	Poisson	Terr
factor(fac.yr)81	-0.5590882	0.2701328	-2.0696788	0.0384824	Poisson	Terr
factor(fac.yr)91	-0.3933464	0.2559607	-1.5367452	0.1243557	Poisson	Terr
factor(fac.yr)101	-0.8374463	0.2883345	-2.9044262	0.0036793	Poisson	Terr
factor(fac.yr)111	-0.9281657	0.3557745	-2.6088598	0.0090844	Poisson	Terr
factor(fac.yr)121	-1.0313080	0.3377529	-3.0534388	0.0022623	Poisson	Terr
factor(fac.yr)131	-0.4511938	0.2827390	-1.5957959	0.1105344	Poisson	Terr
factor(fac.yr)141	-0.7432294	0.3185489	-2.3331718	0.0196391	Poisson	Terr
factor(fac.yr)151	-0.5716807	0.3210381	-1.7807255	0.0749573	Poisson	Terr
factor(fac.yr)161	-0.3359787	0.3352009	-1.0023206	0.3161888	Poisson	Terr
factor(fac.yr)171	-0.0806956	0.3284683	-0.2456725	0.8059357	Poisson	Terr
factor(fac.yr)181	-1.6254153	0.7462250	-2.1781840	0.0293923	Poisson	Terr
factor(fac.yr)191	-0.8539252	0.5023343	-1.6999141	0.0891471	Poisson	Terr
factor(fac.yr)201	-1.7923066	0.6163853	-2.9077700	0.0036402	Poisson	Terr
(Intercept)2	0.9333102	0.2000565	4.6652341	0.0000031	Poisson	BirdID+Terr
factor(fac.yr)22	-1.2007896	0.3126675	-3.8404683	0.0001228	Poisson	BirdID+Terr
factor(fac.yr)32	-1.1732192	0.2918588	-4.0198175	0.0000582	Poisson	BirdID+Terr
factor(fac.yr)42	-0.6168346	0.2619124	-2.3551179	0.0185168	Poisson	BirdID+Terr
factor(fac.yr)52	-0.5618584	0.2537741	-2.2140097	0.0268281	Poisson	BirdID+Terr
factor(fac.yr)62	-0.7235098	0.2555379	-2.8313205	0.0046356	Poisson	BirdID+Terr
factor(fac.yr)72	-0.9160363	0.2894243	-3.1650287	0.0015507	Poisson	BirdID+Terr
factor(fac.yr)82	-0.5591501	0.2701496	-2.0697791	0.0384730	Poisson	BirdID+Terr
factor(fac.yr)92	-0.3934160	0.2559784	-1.5369109	0.1243151	Poisson	BirdID+Terr
factor(fac.yr)102	-0.8375092	0.2883507	-2.9044813	0.0036786	Poisson	BirdID+Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)112	-0.9282969	0.3557908	-2.6091091	0.0090778	Poisson	BirdID+Terr
factor(fac.yr)122	-1.0314204	0.3377695	-3.0536224	0.0022610	Poisson	BirdID+Terr
factor(fac.yr)132	-0.4512836	0.2827583	-1.5960049	0.1104877	Poisson	BirdID+Terr
factor(fac.yr)142	-0.7432534	0.3185685	-2.3331036	0.0196427	Poisson	BirdID+Terr
factor(fac.yr)152	-0.5717136	0.3210583	-1.7807158	0.0749589	Poisson	BirdID+Terr
factor(fac.yr)162	-0.3360113	0.3352246	-1.0023467	0.3161762	Poisson	BirdID+Terr
factor(fac.yr)172	-0.0807870	0.3284931	-0.2459322	0.8057347	Poisson	BirdID+Terr
factor(fac.yr)182	-1.6254932	0.7462408	-2.1782423	0.0293880	Poisson	BirdID+Terr
factor(fac.yr)192	-0.8540092	0.5023542	-1.7000140	0.0891283	Poisson	BirdID+Terr
factor(fac.yr)202	-1.7923499	0.6163995	-2.9077732	0.0036401	Poisson	BirdID+Terr

The territory only model performs very similarly to the model fitting both random effects on the selection criteria. Having the best AIC and roughly the best adjusted R^2 , but the band ID only model has the best BIC. However, none of the band ID smoothers are significant, and the edf is close to 0 for the band ID smoother in the model with both random effects, suggesting that it is being shrunk out of the model.

Z-score number of fledgling (Gaussian)

Table 184: Z-score GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
41.13190	-418.6908	925.9065	1095.385	233.6807	298.8681	340	0.1716897	111	Gaussian	Terr
51.71398	-405.9156	933.4852	1166.388	216.7636	288.2860	340	0.2034508	222	Gaussian	BandID+Terr
23.25512	-446.1691	947.9059	1054.289	274.6761	316.7449	340	0.0813270	112	Gaussian	BandID

Table 185: Z-score GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	22.25512	110	0.2728242	0.0337024	Gaussian	BandID
s(Terrs)	40.13190	109	0.6446506	0.0001801	Gaussian	Terr
s(birdID)	12.23877	110	0.1373917	0.1845038	Gaussian	BandID+Terr
s(Terrs)	38.47521	109	0.6352747	0.0003197	Gaussian	BandID+Terr

Table 186: Z-score GAMMs linear terms table.

	Estimate	Std. Error	t value	Pr(> t)	Family	RE
(Intercept)	-0.0101223	0.0589651	-0.1716667	0.8638092	Gaussian	BandID
(Intercept)1	0.0062653	0.0632463	0.0990611	0.9211561	Gaussian	Terr
(Intercept)2	-0.0007332	0.0653685	-0.0112157	0.9910591	Gaussian	BandID+Terr

Each model has a criteria that they are performing well on, but the band ID smoother in the model with both random effects was insignificant.

Eggs (Binomial)

Table 187: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
72.49128	-131.6134	412.6118	698.6053	263.2269	267.5087	340	0.3138148	130	Binomial	Terr
88.55047	-114.8591	425.5858	800.5690	229.7182	251.4495	340	0.3931273	241	Binomial	BirdID+Terr
49.57334	-166.2387	438.2872	640.8571	332.4775	290.4267	340	0.1467857	131	Binomial	BirdID

Table 188: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	29.57334	110	40.81882	0.0104839	Binomial	BirdID
s(Terrs)	52.49128	109	86.97018	0.0000544	Binomial	Terr
s(birdID)	16.91081	110	20.27830	0.4737808	Binomial	BirdID+Terr
s(Terrs)	51.63966	109	87.23240	0.0015650	Binomial	BirdID+Terr

Table 189: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	2.3005848	1.078129e+00	2.1338686	0.0328535	Binomial	BirdID
factor(fac.yr)2	-2.6028567	1.149208e+00	-2.2649136	0.0235180	Binomial	BirdID
factor(fac.yr)3	-1.7308722	1.145267e+00	-1.5113267	0.1307052	Binomial	BirdID
factor(fac.yr)4	-1.6165210	1.155288e+00	-1.3992370	0.1617419	Binomial	BirdID
factor(fac.yr)5	-1.4893458	1.158503e+00	-1.2855773	0.1985906	Binomial	BirdID
factor(fac.yr)6	-1.0167672	1.168092e+00	-0.8704514	0.3840538	Binomial	BirdID
factor(fac.yr)7	-1.8585084	1.169480e+00	-1.5891756	0.1120208	Binomial	BirdID
factor(fac.yr)8	-1.5352693	1.186069e+00	-1.2944182	0.1955210	Binomial	BirdID
factor(fac.yr)9	-0.9819751	1.202315e+00	-0.8167367	0.4140789	Binomial	BirdID
factor(fac.yr)10	-1.8019826	1.181994e+00	-1.5245275	0.1273770	Binomial	BirdID
factor(fac.yr)11	-1.2307782	1.291303e+00	-0.9531289	0.3405248	Binomial	BirdID
factor(fac.yr)12	-2.0620473	1.221235e+00	-1.6884933	0.0913166	Binomial	BirdID
factor(fac.yr)13	-1.7032494	1.211506e+00	-1.4058943	0.1597555	Binomial	BirdID
factor(fac.yr)14	-1.2256194	1.255033e+00	-0.9765635	0.3287853	Binomial	BirdID
factor(fac.yr)15	-1.0820027	1.295935e+00	-0.8349205	0.4037625	Binomial	BirdID
factor(fac.yr)16	-0.4359570	1.554041e+00	-0.2805311	0.7790700	Binomial	BirdID
factor(fac.yr)17	40.9982566	2.536477e+07	0.0000016	0.9999987	Binomial	BirdID
factor(fac.yr)18	-2.5157802	1.528759e+00	-1.6456360	0.0998387	Binomial	BirdID
factor(fac.yr)19	-1.0161697	1.598007e+00	-0.6358984	0.5248427	Binomial	BirdID
factor(fac.yr)20	-3.2567115	1.402988e+00	-2.3212683	0.0202724	Binomial	BirdID
(Intercept)1	2.0830905	1.122394e+00	1.8559359	0.0634627	Binomial	Terr
factor(fac.yr)21	-2.5123384	1.227037e+00	-2.0474834	0.0406106	Binomial	Terr
factor(fac.yr)31	-1.4152611	1.191477e+00	-1.1878204	0.2349042	Binomial	Terr
factor(fac.yr)41	-1.1726146	1.219315e+00	-0.9616994	0.3362006	Binomial	Terr
factor(fac.yr)51	-1.0117784	1.207692e+00	-0.8377785	0.4021551	Binomial	Terr
factor(fac.yr)61	-0.5573121	1.228177e+00	-0.4537717	0.6499931	Binomial	Terr
factor(fac.yr)71	-1.2448377	1.231650e+00	-1.0107075	0.3121564	Binomial	Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)81	-1.0045387	1.238397e+00	-0.8111606	0.4172734	Binomial	Terr
factor(fac.yr)91	-0.5054940	1.254668e+00	-0.4028908	0.6870286	Binomial	Terr
factor(fac.yr)101	-1.7175050	1.237024e+00	-1.3884170	0.1650101	Binomial	Terr
factor(fac.yr)111	-0.8538463	1.336368e+00	-0.6389303	0.5228683	Binomial	Terr
factor(fac.yr)121	-1.6825752	1.276762e+00	-1.3178457	0.1875553	Binomial	Terr
factor(fac.yr)131	-1.0052206	1.262821e+00	-0.7960122	0.4260249	Binomial	Terr
factor(fac.yr)141	-0.6827910	1.311146e+00	-0.5207587	0.6025349	Binomial	Terr
factor(fac.yr)151	-0.2612429	1.357781e+00	-0.1924043	0.8474255	Binomial	Terr
factor(fac.yr)161	-0.0005476	1.630255e+00	-0.0003359	0.9997320	Binomial	Terr
factor(fac.yr)171	40.7015705	2.536477e+07	0.0000016	0.9999987	Binomial	Terr
factor(fac.yr)181	-2.2056510	1.655938e+00	-1.3319649	0.1828717	Binomial	Terr
factor(fac.yr)191	0.4048061	1.690182e+00	0.2395045	0.8107144	Binomial	Terr
factor(fac.yr)201	-3.4637324	1.492052e+00	-2.3214563	0.0202622	Binomial	Terr
(Intercept)2	2.0042349	1.153442e+00	1.7376120	0.0822792	Binomial	BirdID+Terr
factor(fac.yr)22	-2.4229314	1.254815e+00	-1.9309066	0.0534946	Binomial	BirdID+Terr
factor(fac.yr)32	-1.2730434	1.220002e+00	-1.0434767	0.2967276	Binomial	BirdID+Terr
factor(fac.yr)42	-1.0859256	1.252050e+00	-0.8673180	0.3857678	Binomial	BirdID+Terr
factor(fac.yr)52	-0.8986889	1.241902e+00	-0.7236390	0.4692874	Binomial	BirdID+Terr
factor(fac.yr)62	-0.4434664	1.263343e+00	-0.3510260	0.7255688	Binomial	BirdID+Terr
factor(fac.yr)72	-1.2598806	1.273454e+00	-0.9893415	0.3224961	Binomial	BirdID+Terr
factor(fac.yr)82	-0.9474689	1.282912e+00	-0.7385298	0.4601926	Binomial	BirdID+Terr
factor(fac.yr)92	-0.4209965	1.296004e+00	-0.3248420	0.7453007	Binomial	BirdID+Terr
factor(fac.yr)102	-1.6152602	1.281112e+00	-1.2608266	0.2073713	Binomial	BirdID+Terr
factor(fac.yr)112	-0.8188812	1.391995e+00	-0.5882786	0.5563453	Binomial	BirdID+Terr
factor(fac.yr)122	-1.6864321	1.322240e+00	-1.2754356	0.2021551	Binomial	BirdID+Terr
factor(fac.yr)132	-0.9354104	1.307453e+00	-0.7154445	0.4743344	Binomial	BirdID+Terr
factor(fac.yr)142	-0.4084811	1.366529e+00	-0.2989186	0.7650021	Binomial	BirdID+Terr
factor(fac.yr)152	-0.0384246	1.413476e+00	-0.0271845	0.9783126	Binomial	BirdID+Terr
factor(fac.yr)162	0.0097588	1.680210e+00	0.0058081	0.9953658	Binomial	BirdID+Terr
factor(fac.yr)172	36.7195122	2.536477e+07	0.0000014	0.9999988	Binomial	BirdID+Terr
factor(fac.yr)182	-2.3319560	1.737561e+00	-1.3420859	0.1795681	Binomial	BirdID+Terr
factor(fac.yr)192	0.3291775	1.747986e+00	0.1883181	0.8506273	Binomial	BirdID+Terr
factor(fac.yr)202	-3.4237132	1.556748e+00	-2.1992722	0.0278586	Binomial	BirdID+Terr

Similar to the previous batch of models.

Probabilites used to generate simulation 21

Simulation 1

Probability of mate fidelity: 0.1

Probability of pair dispersal: 0.9

Probability of single male dispersal: 0.95

Probability of single female dispersal: 0.95

NULL

This run was designed to allow for the territory effect to be present.

Simulation 22

Number of fledgling as the response (Poisson)

Table 190: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npars	Family	RE
44.42590	-413.0141	920.1570	1098.529	322.4912	282.5741	327	0.1768416	140	Poisson	BirdID
36.91965	-422.8420	925.3643	1076.357	342.1471	290.0804	327	0.1415633	125	Poisson	Terr
50.49239	-406.4508	928.4161	1147.314	309.3648	276.5076	327	0.2081473	245	Poisson	BirdID+Terr

Table 191: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	24.42590	119	33.99100	0.0100152	Poisson	BirdID
s(Terrs)	16.91965	104	21.63449	0.0448693	Poisson	Terr
s(birdID)	20.81381	119	28.16636	0.0267926	Poisson	BirdID+Terr
s(Terrs)	9.67858	104	11.40329	0.1731121	Poisson	BirdID+Terr

Table 192: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	0.3736662	0.2514096	1.4862849	0.1372038	Poisson	BirdID
factor(fac.yr)2	-0.4941800	0.3750187	-1.3177476	0.1875882	Poisson	BirdID
factor(fac.yr)3	-0.6396946	0.3508251	-1.8234006	0.0682427	Poisson	BirdID
factor(fac.yr)4	-0.4597744	0.3317231	-1.3860185	0.1657412	Poisson	BirdID
factor(fac.yr)5	-0.2564152	0.3178467	-0.8067259	0.4198244	Poisson	BirdID
factor(fac.yr)6	-0.1571032	0.3020502	-0.5201229	0.6029779	Poisson	BirdID
factor(fac.yr)7	-0.4925531	0.3259158	-1.5112892	0.1307148	Poisson	BirdID
factor(fac.yr)8	-0.1162365	0.3128329	-0.3715611	0.7102196	Poisson	BirdID
factor(fac.yr)9	0.1885058	0.3056759	0.6166852	0.5374424	Poisson	BirdID
factor(fac.yr)10	0.3161385	0.3080567	1.0262346	0.3047810	Poisson	BirdID
factor(fac.yr)11	-0.6706877	0.3500679	-1.9158791	0.0553805	Poisson	BirdID
factor(fac.yr)12	-0.4998787	0.3558527	-1.4047349	0.1601001	Poisson	BirdID
factor(fac.yr)13	-0.2282721	0.3583067	-0.6370857	0.5240690	Poisson	BirdID
factor(fac.yr)14	-0.3469124	0.4153195	-0.8352904	0.4035542	Poisson	BirdID
factor(fac.yr)15	-0.4936485	0.4021855	-1.2274151	0.2196666	Poisson	BirdID
factor(fac.yr)16	-1.0272639	0.5216182	-1.9693789	0.0489096	Poisson	BirdID
factor(fac.yr)17	0.1306327	0.3818206	0.3421310	0.7322523	Poisson	BirdID
factor(fac.yr)18	-0.0041878	0.3970940	-0.0105462	0.9915855	Poisson	BirdID
factor(fac.yr)19	-1.6670560	1.0415782	-1.6005097	0.1094856	Poisson	BirdID
factor(fac.yr)20	-1.3779435	1.0443063	-1.3194822	0.1870080	Poisson	BirdID
(Intercept)1	0.4154070	0.2446979	1.6976317	0.0895773	Poisson	Terr
factor(fac.yr)21	-0.4873623	0.3737905	-1.3038384	0.1922887	Poisson	Terr
factor(fac.yr)31	-0.6546564	0.3464992	-1.8893444	0.0588457	Poisson	Terr
factor(fac.yr)41	-0.4759504	0.3264435	-1.4579871	0.1448441	Poisson	Terr
factor(fac.yr)51	-0.3167421	0.3107428	-1.0193061	0.3080577	Poisson	Terr
factor(fac.yr)61	-0.2097858	0.2942326	-0.7129929	0.4758501	Poisson	Terr
factor(fac.yr)71	-0.5853420	0.3167913	-1.8477212	0.0646427	Poisson	Terr
factor(fac.yr)81	-0.2268548	0.3025692	-0.7497618	0.4533982	Poisson	Terr
factor(fac.yr)91	0.1543467	0.2949367	0.5233217	0.6007504	Poisson	Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)101	0.2946814	0.2967425	0.9930541	0.3206836	Poisson	Terr
factor(fac.yr)111	-0.6468620	0.3408832	-1.8976060	0.0577480	Poisson	Terr
factor(fac.yr)121	-0.4744874	0.3471300	-1.3668870	0.1716607	Poisson	Terr
factor(fac.yr)131	-0.1120401	0.3492896	-0.3207658	0.7483879	Poisson	Terr
factor(fac.yr)141	-0.2644659	0.4067090	-0.6502584	0.5155253	Poisson	Terr
factor(fac.yr)151	-0.4107065	0.3921817	-1.0472353	0.2949911	Poisson	Terr
factor(fac.yr)161	-0.9521206	0.5142865	-1.8513429	0.0641202	Poisson	Terr
factor(fac.yr)171	0.0832513	0.3710519	0.2243656	0.8224728	Poisson	Terr
factor(fac.yr)181	0.0054046	0.3870477	0.0139638	0.9888589	Poisson	Terr
factor(fac.yr)191	-1.7327868	1.0348213	-1.6744792	0.0940365	Poisson	Terr
factor(fac.yr)201	-1.4854911	1.0377272	-1.4314852	0.1522912	Poisson	Terr
(Intercept)2	0.3843193	0.2545727	1.5096644	0.1311291	Poisson	BirdID+Terr
factor(fac.yr)22	-0.4975234	0.3795305	-1.3108917	0.1898944	Poisson	BirdID+Terr
factor(fac.yr)32	-0.6385221	0.3541062	-1.8031936	0.0713578	Poisson	BirdID+Terr
factor(fac.yr)42	-0.4553087	0.3351492	-1.3585254	0.1742970	Poisson	BirdID+Terr
factor(fac.yr)52	-0.2782002	0.3211986	-0.8661313	0.3864182	Poisson	BirdID+Terr
factor(fac.yr)62	-0.1689517	0.3046768	-0.5545275	0.5792179	Poisson	BirdID+Terr
factor(fac.yr)72	-0.5346846	0.3284274	-1.6280146	0.1035218	Poisson	BirdID+Terr
factor(fac.yr)82	-0.1612872	0.3151850	-0.5117224	0.6088453	Poisson	BirdID+Terr
factor(fac.yr)92	0.1869220	0.3082455	0.6064064	0.5442449	Poisson	BirdID+Terr
factor(fac.yr)102	0.3066204	0.3108023	0.9865448	0.3238658	Poisson	BirdID+Terr
factor(fac.yr)112	-0.6655795	0.3520307	-1.8906862	0.0586663	Poisson	BirdID+Terr
factor(fac.yr)122	-0.5007980	0.3587771	-1.3958470	0.1627606	Poisson	BirdID+Terr
factor(fac.yr)132	-0.1973237	0.3622874	-0.5446608	0.5859868	Poisson	BirdID+Terr
factor(fac.yr)142	-0.3200138	0.4200108	-0.7619180	0.4461090	Poisson	BirdID+Terr
factor(fac.yr)152	-0.4887642	0.4050763	-1.2065979	0.2275871	Poisson	BirdID+Terr
factor(fac.yr)162	-1.0031838	0.5249441	-1.9110297	0.0560008	Poisson	BirdID+Terr
factor(fac.yr)172	0.0951862	0.3870398	0.2459339	0.8057334	Poisson	BirdID+Terr
factor(fac.yr)182	-0.0034484	0.4023771	-0.0085702	0.9931621	Poisson	BirdID+Terr
factor(fac.yr)192	-1.6535805	1.0437505	-1.5842680	0.1131327	Poisson	BirdID+Terr
factor(fac.yr)202	-1.3880880	1.0484517	-1.3239408	0.1855227	Poisson	BirdID+Terr

Overall, I wouldn't say there's a clear superior model when considering multiple selection criteria. All smoothers are significant except for the territory smoother in the model that included both random effects.

Z-score number of fledgling (Gaussian)

Table 193: Z-score GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npars	Family	RE
57.04409	-381.7065	890.2035	1130.469	197.6859	269.9559	327	0.2223897	226	Gaussian	BandID+Terr
31.86525	-415.3851	902.1046	1037.282	242.9036	295.1348	327	0.1260375	121	Gaussian	BandID
25.84249	-423.6874	906.7269	1019.198	255.5564	301.1575	327	0.0989015	106	Gaussian	Terr

Table 194: Z-score GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	30.86525	119	0.3950732	0.0043259	Gaussian	BandID
s(Terrs)	24.84249	104	0.3440475	0.0132057	Gaussian	Terr
s(birdID)	30.93729	119	0.4382607	0.0038132	Gaussian	BandID+Terr
s(Terrs)	25.10680	104	0.3837429	0.0146479	Gaussian	BandID+Terr

Table 195: Z-score GAMMs linear terms table.

	Estimate	Std. Error	t value	Pr(> t)	Family	RE
(Intercept)	-0.0143835	0.0619550	-0.2321612	0.8165737	Gaussian	BandID
(Intercept)1	0.0096378	0.0606257	0.1589728	0.8737969	Gaussian	Terr
(Intercept)2	-0.0069147	0.0680813	-0.1015647	0.9191776	Gaussian	BandID+Terr

The model fitting both random effects had the best AIC and adjusted R^2 , where the territory only model had the best BIC. No smoother was significant.

Eggs (Binomial)

Table 196: No. fledge GAMMs summary table.

df	logLik	AIC	BIC	deviance	df.residual	nobs	adj.r.squared	npar	Family	RE
56.28816	-161.8413	443.3432	670.0978	323.6826	270.7118	327	0.1677193	140	Binomial	BirdID
87.23582	-127.2471	445.5103	807.4822	254.4941	239.7642	327	0.3231870	245	Binomial	BirdID+Terr
46.76539	-173.6686	447.8302	638.2624	347.3372	280.2346	327	0.1158686	125	Binomial	Terr

Table 197: No. fledge GAMMs smooth terms table.

term	edf	ref.df	statistic	p.value	Family	RE
s(birdID)	36.28816	119	50.73013	0.0062714	Binomial	BirdID
s(Terrs)	26.76539	104	35.85767	0.0193835	Binomial	Terr
s(birdID)	39.12776	119	58.30004	0.0388085	Binomial	BirdID+Terr
s(Terrs)	28.10806	104	41.62176	0.0819853	Binomial	BirdID+Terr

Table 198: No. fledge GAMMs linear terms table.

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
(Intercept)	1.5209748	0.8336560	1.8244632	0.0680821	Binomial	BirdID
factor(fac.yr)2	-1.3261466	1.0087980	-1.3145809	0.1886508	Binomial	BirdID
factor(fac.yr)3	-0.8422455	0.9590941	-0.8781677	0.3798527	Binomial	BirdID
factor(fac.yr)4	-0.8872915	0.9531258	-0.9309280	0.3518908	Binomial	BirdID
factor(fac.yr)5	-1.3839246	0.9434744	-1.4668385	0.1424200	Binomial	BirdID

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)6	-0.7032916	0.9341641	-0.7528566	0.4515361	Binomial	BirdID
factor(fac.yr)7	-1.3210589	0.9353932	-1.4123033	0.1578607	Binomial	BirdID
factor(fac.yr)8	-0.6598738	0.9577617	-0.6889750	0.4908390	Binomial	BirdID
factor(fac.yr)9	-0.2513616	0.9984596	-0.2517494	0.8012348	Binomial	BirdID
factor(fac.yr)10	0.4927139	1.1447932	0.4303955	0.6669079	Binomial	BirdID
factor(fac.yr)11	-1.5684755	0.9574524	-1.6381760	0.1013850	Binomial	BirdID
factor(fac.yr)12	-1.5073959	0.9892297	-1.5238078	0.1275567	Binomial	BirdID
factor(fac.yr)13	-1.0357515	1.0715061	-0.9666314	0.3337283	Binomial	BirdID
factor(fac.yr)14	-1.0713386	1.1530178	-0.9291605	0.3528059	Binomial	BirdID
factor(fac.yr)15	-1.6970299	1.0778151	-1.5745093	0.1153698	Binomial	BirdID
factor(fac.yr)16	-1.9990248	1.1234474	-1.7793666	0.0751797	Binomial	BirdID
factor(fac.yr)17	-0.4935110	1.2264847	-0.4023785	0.6874055	Binomial	BirdID
factor(fac.yr)18	-0.3221295	1.2344525	-0.2609493	0.7941316	Binomial	BirdID
factor(fac.yr)19	-2.1986011	1.5048072	-1.4610517	0.1440013	Binomial	BirdID
factor(fac.yr)20	-2.0019199	1.5864558	-1.2618819	0.2069913	Binomial	BirdID
(Intercept)1	1.7409251	0.8151704	2.1356579	0.0327073	Binomial	Terr
factor(fac.yr)21	-1.3753401	0.9976954	-1.3785171	0.1680437	Binomial	Terr
factor(fac.yr)31	-1.0827185	0.9376378	-1.1547300	0.2482010	Binomial	Terr
factor(fac.yr)41	-1.1086898	0.9270099	-1.1959849	0.2317024	Binomial	Terr
factor(fac.yr)51	-1.5631677	0.9156309	-1.7072029	0.0877843	Binomial	Terr
factor(fac.yr)61	-0.9314979	0.9037661	-1.0306847	0.3026887	Binomial	Terr
factor(fac.yr)71	-1.7041010	0.9053922	-1.8821690	0.0598131	Binomial	Terr
factor(fac.yr)81	-1.1077458	0.9281025	-1.1935598	0.2326502	Binomial	Terr
factor(fac.yr)91	-0.3181120	0.9689494	-0.3283061	0.7426802	Binomial	Terr
factor(fac.yr)101	0.3863404	1.1198299	0.3449992	0.7300950	Binomial	Terr
factor(fac.yr)111	-1.6291278	0.9235265	-1.7640293	0.0777271	Binomial	Terr
factor(fac.yr)121	-1.4078392	0.9538978	-1.4758806	0.1399759	Binomial	Terr
factor(fac.yr)131	-0.7494547	1.0351295	-0.7240203	0.4690533	Binomial	Terr
factor(fac.yr)141	-0.9342594	1.1070525	-0.8439161	0.3987163	Binomial	Terr
factor(fac.yr)151	-1.5478284	1.0357148	-1.4944543	0.1350569	Binomial	Terr
factor(fac.yr)161	-1.5672799	1.0857995	-1.4434340	0.1488982	Binomial	Terr
factor(fac.yr)171	-0.5840209	1.1845427	-0.4930349	0.6219879	Binomial	Terr
factor(fac.yr)181	-0.5778418	1.1811663	-0.4892129	0.6246910	Binomial	Terr
factor(fac.yr)191	-2.7606068	1.4494756	-1.9045556	0.0568379	Binomial	Terr
factor(fac.yr)201	-2.5204488	1.5240352	-1.6537996	0.0981682	Binomial	Terr
(Intercept)2	1.7362092	0.9437890	1.8396158	0.0658247	Binomial	BirdID+Terr
factor(fac.yr)22	-1.3989640	1.1406496	-1.2264625	0.2200247	Binomial	BirdID+Terr
factor(fac.yr)32	-0.8805315	1.0762846	-0.8181214	0.4132879	Binomial	BirdID+Terr
factor(fac.yr)42	-0.9534482	1.0706784	-0.8905085	0.3731929	Binomial	BirdID+Terr
factor(fac.yr)52	-1.5118891	1.0584431	-1.4284084	0.1531743	Binomial	BirdID+Terr
factor(fac.yr)62	-0.8326155	1.0444606	-0.7971727	0.4253507	Binomial	BirdID+Terr
factor(fac.yr)72	-1.6142265	1.0521527	-1.5342132	0.1249772	Binomial	BirdID+Terr
factor(fac.yr)82	-0.9801058	1.0739339	-0.9126314	0.3614364	Binomial	BirdID+Terr
factor(fac.yr)92	-0.2264044	1.1132464	-0.2033731	0.8388434	Binomial	BirdID+Terr
factor(fac.yr)102	0.4003979	1.2630045	0.3170201	0.7512283	Binomial	BirdID+Terr
factor(fac.yr)112	-1.7890808	1.0742978	-1.6653490	0.0958432	Binomial	BirdID+Terr
factor(fac.yr)122	-1.6389200	1.1109898	-1.4751891	0.1401617	Binomial	BirdID+Terr
factor(fac.yr)132	-0.9933762	1.1977032	-0.8294010	0.4068776	Binomial	BirdID+Terr
factor(fac.yr)142	-1.1900191	1.2936319	-0.9199055	0.3576222	Binomial	BirdID+Terr
factor(fac.yr)152	-1.9751836	1.2149304	-1.6257587	0.1040010	Binomial	BirdID+Terr
factor(fac.yr)162	-1.7820258	1.2666814	-1.4068461	0.1594730	Binomial	BirdID+Terr
factor(fac.yr)172	-0.5746425	1.3703957	-0.4193260	0.6749779	Binomial	BirdID+Terr

	Estimate	Std. Error	z value	Pr(> z)	Family	RE
factor(fac.yr)182	-0.4446712	1.3847670	-0.3211163	0.7481223	Binomial	BirdID+Terr
factor(fac.yr)192	-2.3431442	1.6554849	-1.4153824	0.1569564	Binomial	BirdID+Terr
factor(fac.yr)202	-2.3928155	1.7491805	-1.3679638	0.1713234	Binomial	BirdID+Terr

The band ID only model had the best AIC and BIC, and the the model with both random effects had the best adjusted R^2 . All smoothers were significant, except for the territory smoother in the model that fit both random effects.

Probabilites used to generate simulation 22

Simulation 1

Probability of mate fidelity: 0.97

Probability of pair dispersal: 0.005

Probability of single male dispersal: 0.05

Probability of single female dispersal: 0.01

NULL

This run was generated to make it difficult to detect the territory effect.